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Development of a methodology for constructing GHG emission reduction curves incorporating the uncertainty associated with the main mitigation variables

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Summary

The overall objective of the study is to develop a methodology for the elaboration of greenhouse gas abatement curves for the energy demand sector, incorporating the uncertainty associated with the variables involved in the estimation of marginal abatement costs.

Subsequently, this methodology will be applied in a case study to the electricity generation sector and three demand sectors in six Latin American countries.

JEL Codes: L94, Q41, Q52, Q54
Keywords: Climate change, Abatement, Marginal cost curve
Uncertainty, Risk, Mitigation, Greenhouse Gases

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1 INTRODUCTION

This document was prepared by professionals Jacques Clerc, Manuel Díaz and Bruno Campos and corresponds to the final technical report of the project **"Development of a methodology for the construction of GHG emission reduction curves incorporating the uncertainty associated with the main mitigation variables",** developed for the Inter-American Development Bank.

This section initially presents a global context of the study and then presents, in a general way, the contents of the chapters that make up this document.

1.1 CONTEXTUALIZATION OF THE STUDY

Climate change as a global problem poses challenges that may eventually become demands. International market regulations use instruments that progressively attempt to internalize the external costs associated with global warming from consumption and production activities. For example, carbon markets (both compliance and voluntary) are growing, and it is expected that some developed countries will have to certify (or even establish standards regarding) the amount of greenhouse gas emissions generated in the production of each unit of good (carbon footprint) in the coming years, among other initiatives. Furthermore, these same markets are progressively incorporating low-carbon production into their preferences.

Discussions regarding climate change mitigation and adaptation are well-known. Various studies are conducted at the national and international levels, estimating greenhouse gas (GHG) emission reductions, costs associated with mitigation measures, and even co-benefits through counterfactual analysis. More recently, efforts have also focused on evaluating the requirements and costs associated with adaptation. Furthermore, attempts are being made to design financing mechanisms for mitigation and adaptation strategies.

Despite its relatively small contribution to global emissions, Latin America must also contribute to greenhouse gas (GHG) mitigation. The countries of the region must plan their GHG emissions reduction strategies in advance to achieve their targets and honor their commitments, but at the lowest possible social cost.

Therefore, this study aims to develop and apply a methodology for abatement curves, but first reviews background information on energy consumption, GHG emissions, and mitigation targets from different countries in the region. The purpose of this review is to identify similarities and unique characteristics among the various experiences.

While this study primarily aims to analyze the electricity generation sector, the review of consumption data contributes to the selection of additional sectors in which Chile will be used as a case study within the Latin American context. The selected sectors are (1) Pulp and Paper Industry; (2) Copper Mining; and (3) Transportation. These sectors were chosen considering

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Note that these are also relevant in other countries of the Region1 . The methodology for constructing abatement curves will be applied in the Chilean case, but the foundations will be laid so that they can also be used by other countries in the region and in sectors with similar challenges to those in Chile.

Methodologically, this paper reviews how to construct a marginal abatement curve. An abatement curve shows the emission reduction impacts of a measure and its relative costs, allowing for a ranking of different mitigation measures. This information enables various stakeholders to determine the timeline for implementing measures and to assess the associated efforts. Furthermore, from a private sector perspective, the curve allows for inferences about the cost of implementing a project that is more advantageous than inaction.

The main contribution of this study to others conducted in the country is to extend the usual methodology for constructing a marginal abatement curve to incorporate the inherent uncertainty of the variables associated with the costs, benefits, and emissions of each measure. The importance of explicitly incorporating this uncertainty lies in enriching the decision-maker's knowledge.

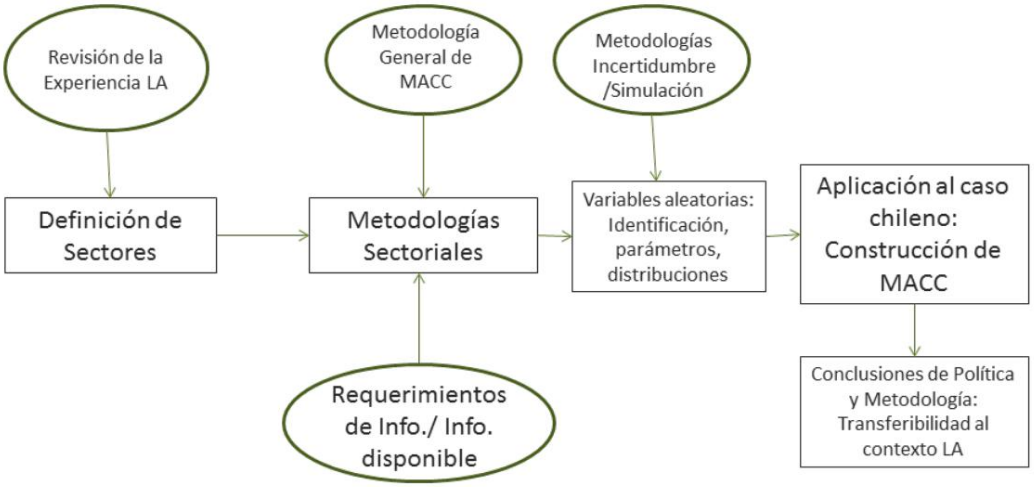
In this way, within a cost-benefit framework, present value metrics (or some equivalent annual cost) per ton of reduction will be established, and their probability distribution will be determined. This will be done using standard simulation methodologies. Incorporating uncertainty also helps to organize and clarify the assumptions that allow for cost calculations for each abatement option.

The following figure summarizes the main stages and products of the study. Based on a review of Latin American experience, sectors to be studied in the Chilean case are defined. In this regard, for each sector, and based on the general methodology, specific methodologies for evaluating mitigation measures are defined. These methodologies are enhanced by incorporating randomness into the evaluation by identifying and characterizing various relevant stochastic variables for subsequent simulation. Finally, the methodology is applied to the Chilean case to construct a marginal abatement curve for the selected sectors, incorporating uncertainty into the analysis. This curve allows for policy conclusions to be drawn, and, moreover, methodological recommendations for application in other contexts can be inferred from its construction process.

[1] Several countries have a paper and pulp industry. Copper mining, also very prevalent in Peru,
It can be likened to other mining activities in the region. Transportation is a challenge for the main cities in all countries of the region.

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Figure 1: Study Summary



Source: Own elaboration.

1.2 REPORT CONTENTS

The overall objective of this study is to develop a methodology for constructing greenhouse gas abatement curves for the energy demand sector, incorporating the uncertainty associated with the variables involved in estimating marginal abatement costs. Subsequently, this methodology will be applied in a case study of the electricity generation sector and three demand sectors. The chapters of this report are a necessary step toward achieving the objective of developing a specific methodology and implementing it.

In particular, Chapter 2 presents the definition of the concepts of uncertainty and risk, as well as showing why it is important to incorporate uncertainty into decision-making and choices.
For this purpose, the alternatives of using scenarios (simple sensitivity analysis) and simulation are considered.

Chapter 3 defines and briefly presents the fundamentals of constructing a marginal cost abatement curve. To this end, it provides a general explanation of the most well-known methodology developed by McKinsey & Co. (2009).

Chapter 4 presents a description of the energy consumption of a group of six Latin American countries. It also includes the latest available information from national communications on climate change regarding greenhouse gas (GHG) emissions in the energy sector. The objective of this chapter is to identify key sectors in terms of their GHG emissions for the evaluation of mitigation measures.

Chapter 5 defines and presents the general methodological framework for how mitigation measures should be analyzed, providing an approach to energy modeling and modeling approaches for gathering the information required to build the model.

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It is important to highlight the use of different approaches used for gathering information and assigning its uncertainty, which further enriches the methodological analysis for replicating this exercise in other Latin American countries.

Chapter 6 presents the methodology for developing greenhouse gas abatement curves in the power generation sector. This includes applying the methodology to characterize stochastic variables in the electricity generation sector and developing the abatement curve with uncertainty for the electricity generation sector in Chile's Central Interconnected System, including a sensitivity analysis of the variables.
considered in the model.

Chapter 7 presents an analysis of the availability of information and methodologies for developing mitigation options for the transportation, copper mining, and pulp and paper industries, focusing on the Chilean case. This chapter includes an inventory of relevant sources for evaluating mitigation measures and an analysis of weaknesses in the available information. Part of this chapter involves determining potential values for the relevant variables. This includes considering penetrations, potential impacts, costs, and other relevant technical aspects, while maintaining an uncertainty-based approach to the main key variables and presenting these variables with the uncertainty to be considered in the exercise.

Chapter 8 constructs the draft curves and presents the results, making a brief analysis of their particularities.

Finally, in chapter 9, comments and conclusions are given regarding the application of this methodology and the results of this exercise.

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2. UNCERTAINTY IN DECISION MAKING

The uncertainty surrounding economic variables and physical phenomena considerably complicates the economic analysis of problems. Since it is not always possible to make decisions based on an objective evaluation of scientific data, it is sometimes necessary to resort to more subjective methods to compensate for the lack of information. However, decision-making under conditions of uncertainty is not entirely subjective; it can frame the available knowledge, take into account the degree of uncertainty, and estimate the extent of the lack of knowledge (ignorance). Analysis under conditions of high uncertainty should be geared toward selecting the best strategy based on an impartial evaluation of the available quantitative and qualitative data. [Group Economic Experts 1991]

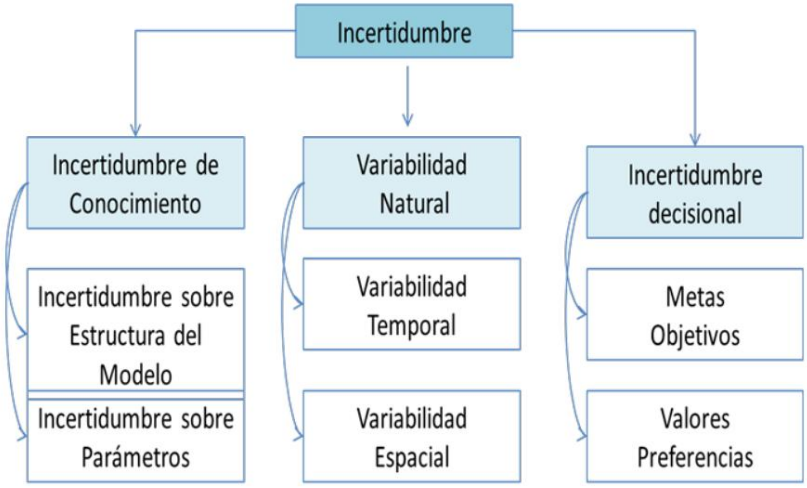
The central purpose of decision-making, and of policy formulation and analysis, is to identify the relevant factors and sources of disagreement in a problem, so as to anticipate the unexpected. Consciously addressing uncertainty compels us to think more carefully about these issues and about the eventualities and contingencies of the problem [Morgan and Henrion 1990].

This section defines the concepts of uncertainty and risk. It also explains why it is important to incorporate uncertainty into decision-making and outlines the alternatives for doing so.

2.1 **UNCERTAINTY**

Uncertainty affects the deterministic model by incorporating randomness in input variables.
Errors in model results can be due to natural variability (e.g., hydrology), as well as errors in the input data, model parameters, or the model itself. The following figure shows one possible classification of uncertainty types.

Figure 2: A Classification of Types of Uncertainty



Fuente: Water Resources Systems Planning and Management –UNESCO.2005.

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The uncertainty associated with the adequate representation of the process under study in the model is usually known as structural uncertainty.

An alternative classification, instead of comparing natural variability versus uncertainty of knowledge, distinguishes between sources of uncertainty. For example, information uncertainties, uncertainty associated with models, and numerical errors.

Annex A.1 discusses in more detail the sources of uncertainty, ways of classifying it, and the importance of taking it into account in decision-making.

2.2 **RISK**

There are countless definitions of risk. The most common definition describes risk as the possibility (probability) of an adverse event occurring. In contrast, the concept of uncertainty relates to indeterminacy, lack of definition, or ignorance. Uncertainty does not involve the value judgment of defining the adverse event.

While some definitions of risk focus on the probability of an event occurring, others incorporate both this probability and the consequences of that event. For example, even though the probability of a strong earthquake occurring on any given day may be low, the earthquake event is considered high risk due to its impacts.

In a decision-making context, uncertainty and doubt regarding the alternatives must be reduced in order to make a reasonable choice among them. The risk lies in not making the right decision in terms of the associated losses (e.g., monetary).

In summary, incorporating uncertainty into decision-making greatly enriches the analysis, providing improved information and analysis compared to deterministically developed evaluations or models, and guidelines on where to best invest resources to improve the quality of information.

2.3 **THE CRITERION OF AVERAGE VALUES**

A model that directly applies the “most probable” or “average” values has significant limitations. The probability that the result of the variable of interest will match the estimate using average values is practically zero. But more importantly, it ignores the risk of error.2 The focus should not be on the average but on the size, frequency, and causes of the variations [Fung, 2010].

[2] This section is based on content available at http://www.strategy-at-risk.com/2011/03/01/plans-based-on-average-assumptions/

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A simple example shows how ignoring uncertainty and using average values can lead to incorrect decisions. An agent has the option to invest 5 and their expected return is 10, so they obtain an expected net profit of 5. If the agent only considers this criterion, then
invest.

If the expected return is constructed from a binomial distribution where with probability 0.01 it obtains 1000 and with probability 0.99 it obtains 0. With probability 0.99 the agent obtains a net income of -5.
The decision to invest no longer seems so wise for any risk-averse agent.

This simple exercise reveals how ignorance of uncertainty carries a high risk of making the wrong decision.

2.4 **ALTERNATIVE TREATMENT OPTIONS FOR UNCERTAINTY**

To improve decision-making, uncertainty must be addressed. There are several ways to manage it. This document distinguishes between two of the most commonly used options: Scenario Planning/Sensitivity Analysis and Uncertainty Analysis/Simulation.

Use of Scenarios/Sensitivity Analysis

Sensitivity analysis using scenarios is the simplest and most widely used alternative. It explores and quantifies the impact of potential errors in a model's input data on the output. New assumptions about the future define scenarios comparable to the average case. Specifically, the values of the input variables are changed, and the calculation that yields the output variable is repeated. The main advantage of sensitivity analysis is its simplicity. The disadvantage is that it does not provide information regarding the probability of each scenario occurring.

In addition to the average value scenario, best-case and worst-case scenarios are often constructed, taking the lowest or best possible value for each variable. However, it is not necessarily true that evaluating the worst-case scenario for each variable will yield the worst possible result in the model output .

Despite the limitations of this approach, its use allows for exploratory analysis. Sensitivity analysis also allows for identifying how output values change in response to modifications in input values. Therefore, given the inability to develop analyses

[3] To illustrate the point, see http://www.strategy-at-risk.com/2011/03/01/plans-based-on-average-assumptions/
It is shown that the joint distribution of the sum of two independent normal random variables X+Y does not coincide with the sum of the distributions of X and Y for values that deviate from the expected values of the variables.

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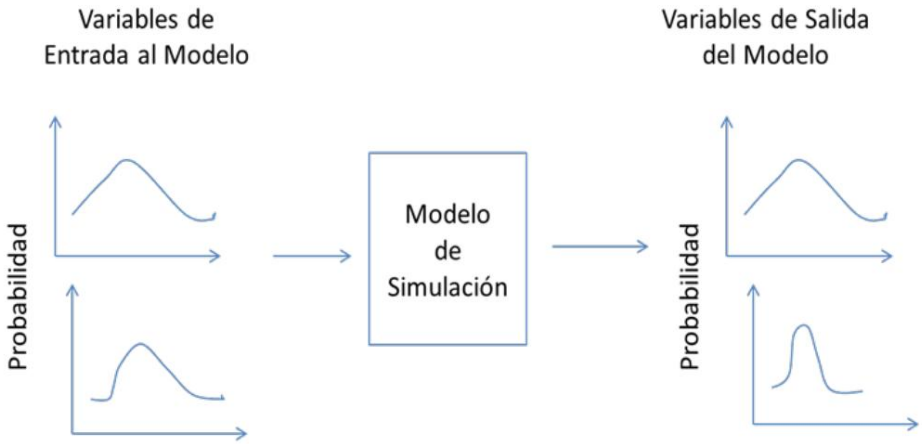
In cases of formal uncertainty, it is always advisable to sensitize the main variables through the use of scenarios.

Uncertainty Analysis/Simulation

Uncertainty analysis through simulation is a much more realistic alternative for dealing with uncertainty. This computational method allows for the quantitative treatment of uncertainty. Simulation provides decision-makers with a wide range of possible outcomes, their probabilities of occurrence, and reveals extreme cases.

Simulation allows us to build models of the possible outcomes associated with different values that certain variables can take. Each variable that involves uncertainty is then associated with a probability distribution. By using distributions, the variables can be assigned different weights to different events. The values of the different random variables are fed into a model (transformation function) to obtain the distribution of output variables (results). The following figure outlines the uncertainty analysis performed in a simulation.

Figure 3: Simulation Models



Fuente: Replicado de Water Resources Systems Planning and Management –UNESCO.2005.

A simulation allows us to describe the range of possible outcomes and associate some measure of probability with them. In a risk analysis context, it also allows us to estimate the probability that an output variable will exceed a critical value . ⁴

Simulation Example: Uncertainty Analysis in Net Present Value (NPV) of a Project

Uncertainty analysis in estimating the NPV of an investment project is a specific case of simulation. The project's benefit Vt is calculated for periods t=1,…,T. Generally,

[4] It also allows for formal inference, such as estimating the mean, variance, and fitting a distribution. others.

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The profit function is a complex function where both revenues Rt and costs Ct depend on several uncertain variables. If there are N factors that affect revenues and costs, the value of the project in period t is:

= (1, ′ .) = (1, ′ .) ÿ (1, ′ .)

Where 1, ′ . They denote the variables that determine income and costs.

Thus, the Net Present Value is expressed as:

= (1, ′ ,) = ÿ =1 (1 +) ′ 0

Where I0 represents investment costs (CAPEX).

The simulation model assumes that the variables < are those that most affect revenue and/or costs. Thus, the model provides simulations = 1, … , where B is the number of simulation runs. The simulated present value values are: 0 = 1, …,

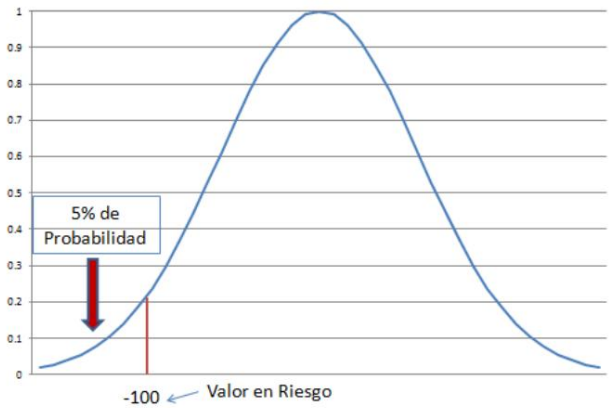
= (1, … ,), = 1, … ,

Where = 1 , …, ÿ ÿ (1, … ,)

In interpreting the results, in addition to the distributions of the output variables, it is relevant to consider a measure of volatility risk. For example, if a firm has the option of investing in two projects with the same NPV (and using the same discount rate), it is reasonable to choose the project with lower cash flow volatility first.

In this context, a "Value at Risk" or VaR metric for NPV can be useful. VaR represents the maximum amount that could be lost for a defined confidence level; this maximum amount has an associated probability of being lost. A VaR of 95% is commonly used. The following figure shows an example where, with 95% confidence, the loss is at most 100.

Figure 4: Example of “Value at Risk”



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Source: Own elaboration

Further details are provided in the annexes on the following points:

- Concepts of Uncertainty and Risk (Alternative classifications of uncertainty and key considerations associated with decision-making are discussed in greater depth). Annex A.1.
- Modeling Uncertainty Propagation (Formalizes mathematically the process of simulation). Annex A.2.
- Sampling Techniques (Important for achieving computational efficiency in runs of simulation). Annexes A.3 and A.4.
- Choice of Input Variable Distributions (Shows the alternative distribution assignments for the input variables that will determine the outputs). Appendix A.5.
- Correlation between Input Variables (Relevant methodological consideration for (improve consistency of estimates). Annex A.6.

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3 MARGINAL COST CURVE OF REDUCTION

Chile has committed to reducing its greenhouse gas emissions. To do so, it must determine a reduction strategy that allows it to meet its commitments without affecting economic growth. In practice, it must decide which set of mitigation measures to implement for this purpose.

One tool that is used in this cost-benefit (or cost-effectiveness) context is the marginal abatement cost curve (MACC), which allows for comparing the costs and emission reduction potential of different mitigation measures.

3.1 DEFINITION OF A MARGINAL COST CURVE FOR REDUCTION

A marginal cost abatement curve is defined as a graph that shows the cost, usually in $/ton of CO2, associated with the last unit (marginal cost) of emission abatement for different amounts of emissions reduced. Therefore, an unconstrained CO2 baseline must be defined in order to evaluate the marginal cost of abatement relative to the baseline.
The MAC curve allows you to analyze the cost of the last unit of CO2 abated for a defined reduction level, while also presenting the total abatement cost through the integral of the mitigation cost curve.

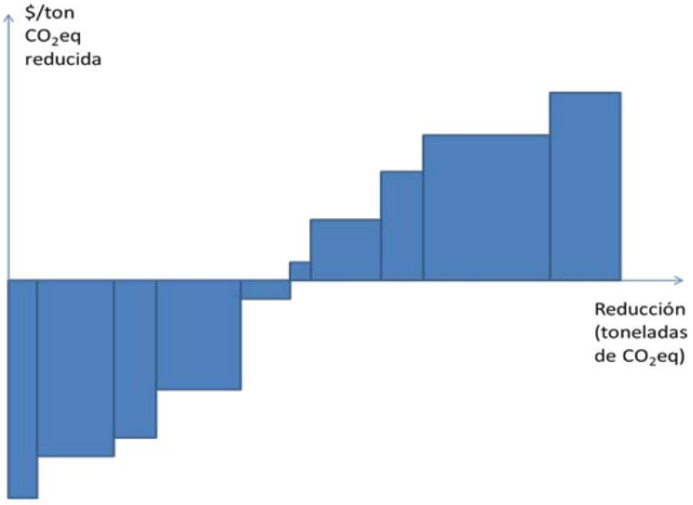
The abatement curve is a graphical representation of the abatement costs of a portfolio of mitigation options alongside their mitigation potential. It displays the different options ordered in ascending order according to their unit abatement cost. In this way, the curve presents, in a concise manner, the relevant information for evaluating and comparing the effects of different measures, which can also be grouped by sector.

The axes of the curve combine the abatement cost represented by the available technical measures and their relative impact. This relative impact corresponds to the potential reduction in the volume of greenhouse gas emissions for each measure. The reduction is measured counterfactually (the measure is compared to the BAU *("business-as-usual"* or reference case).

The following figure presents the shape of a typical abatement cost curve. Each bar/rectangle is associated with a measure (at the level of aggregation defined by the modeler). The width of each bar is interpreted as the emissions reduction *potential* associated with a given measure. This potential reflects the total installed capacity of abatement assets for that measure in the year of analysis, regardless of when that capacity was built. The height of each bar represents the average cost of avoiding one metric ton of CO2e in the year of analysis for each opportunity. This cost reflects the total active capacity of that opportunity; thus, it is the weighted average across sub-opportunities, regions, and years.

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Figure 5: Main dimensions of the abatement cost curve



Source: Own elaboration

To ensure comparability across sectors and sources, all emissions and sinks have been measured in a common manner, using CO2 equivalent measures in metric tons (tCO2e).
In general, for each measure a cost indicator is provided that shows how much it is possible to reduce in a year.

The order of merit of the abatement measures is based on the measures with the lowest costs
(in $ per tCO2e). Despite being called the marginal cost abatement curve, the term “marginal” is due to the increasing order of the measures. For each measure, in the construction of an expert abatement curve, an average cost is associated that is independent of the amount of emissions reduced.

According to the methodology used, MAC curves can be divided into expert opinion-based curves and model-derived curves [Kesicki F. 2010 (a)]. Expert opinion-based curves assess the cost and mitigation potential of each abatement measure based on expert consultation, while model-derived curves are based on calculations from energy or other models. Countries such as the United States and those in Europe have computable general equilibrium models that have typically measured the impact of raising the carbon cost.

In this study, we will focus on the **expert opinion-based approach,** also known as the technology cost curve. This approach is based on assumptions developed by experts for establishing the CO2 emissions baseline, the emissions reduction potential, and the corresponding cost of individual measures. Subsequently, the measures are explicitly ranked from least to most expensive, representing the cost of achieving the desired levels.

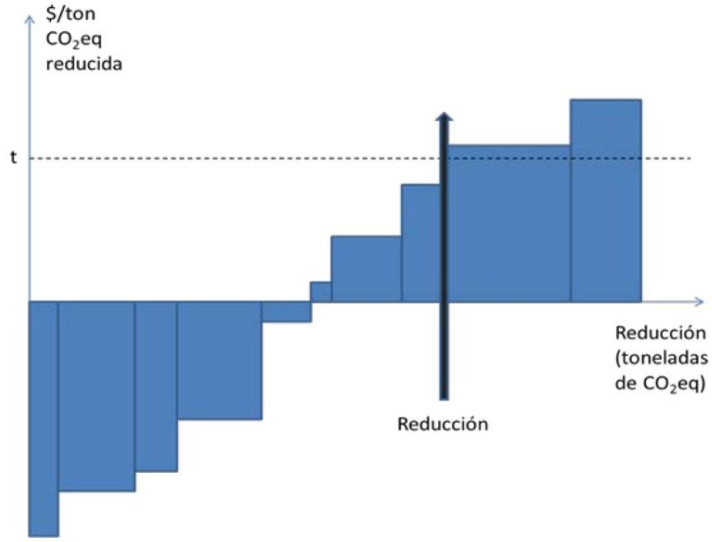
increasing emissions reductions.

This type of curve also does not allow for the analysis of economic instruments such as taxes (price-based) or tradable permit systems (quantity-based). Abatement curves do provide a general idea of the amount to be reduced in relation to a given carbon price (such as the tax level t in the

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(see next figure). Alternatively, when defining a reduction target, it is possible to establish the price that would have to be assigned to carbon to achieve such a level of reduction.

Figure 6: Interpreting the Abatement Curve



Source: Own elaboration

The abatement curve can be constructed from a private or social perspective. From a private perspective, only the cost and benefit components that directly affect the project developer are included, and the developer uses this to define the required rate of return for their initiative. In the social perspective, a social discount rate is used, and it is possible to incorporate external cost components.

3.2 ADVANTAGES AND DISADVANTAGES OF FOLDING CURVES

The main advantage of expert opinion-based curves is their ease of understanding. In general, marginal costs and reduction potential can be clearly allocated to a mitigation option. Furthermore, the technological details can be quite extensive, depending on the
refinement of the study.

Curves based on expert opinion typically show the technological potential of mitigation measures. Abatement curves based on expert opinion consider each possible technology separately. However, they present a maximum reduction potential because they do not consider behavioral aspects, nor institutional or implementation barriers. Behavioral aspects, such as changes in demand due to energy price adjustments, and rebound effects produced by increased energy demand in the case of efficiency improvements, are explained by exogenous adjustments to reference demand. The failure to address market imperfections is one reason for representing negative abatement costs, that is, mitigation measures that can simultaneously save money. However,

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This saving can only be achieved once market distortions are overcome. For example, in the residential sector, these distortions can take the form of split incentives.5
Lack of information and significant upfront payments are associated with a long recovery period.

Certain curves address some of the aforementioned aspects by simplifying reality. While assigning a single cost level to a technology is not plausible, this simplification is often made in many curves. Furthermore, for many renewable energy sources, such as photovoltaics or wind power, more detail should be incorporated into how costs are measured, depending on the location of generation capacity and environmental conditions. Additionally, studies based on expert judgment consider only a selection of existing technologies, for example, according to the likelihood of implementation, which may exclude promising technologies for the future. For sectoral studies, such as those for the transport sector, a problem can arise when mitigation costs are implemented from different decision-makers' perspectives, meaning that an aggregate of mitigation costs across all sectors is not properly applied [Kesicki F. 2010 (b)].

Another disadvantage of cost curves based on expert judgment is the potential for inconsistencies in the baseline assumptions. This occurs, for example, with the assumptions in the benchmark scenario. The calculation of marginal cost and reduction potential is performed in comparison to a benchmark development. In this context, it is important to adjust the benchmark scenario to the extent that the most economical reduction options have already been implemented in order to avoid double counting.

The most significant problem is the failure to consider different types of interactions. One such interaction is the intertemporal interaction of emissions reductions. The trajectory of emissions mitigation—that is, the amount and path of emissions reduction before and after the point in time under consideration—has a significant impact on the reduction curve due to potential cost reductions resulting from technological learning. Furthermore, curves based on expert opinion are unable to adequately capture the interactions between reduction measures, the dependencies of the overall economy, and behavioral interdependencies. The consequences of increased electricity use in the transportation sector, generated within the electricity generation mix, cannot be integrated into an assessment based on individual measures.

3.3 ESTIMATION OF THE MARGINAL COST CURVE FOR REDUCTION

In the most common and famous construction, the abatement cost is measured through the annualized cost or equivalent annual cost (EAC) of different measures in a given year, for each tonne of carbon dioxide equivalent (CO_2e) avoided compared to the baseline or BAU (Business as usual) scenario.

[5] Split incentives describes the situation where a tenant has incentives to improve thermal insulation of their home to reduce their energy expenses, but it is the property owner who must incur these costs without receiving the benefits that come with them.

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This metric allows for comparison of the cost-effectiveness of mitigation options with different lifespans [McKinsey & Co. 2009].

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= ÿ (1+)) ÿ1 ÿ

With

Where:

: Equivalent annual cost.

Capital recovery factor.

: Net present value of all project costs.

Discount rate.

: Number of project periods, or useful life of the asset.

The net present value of costs must incorporate all the flows generated by the project throughout its useful life; these may include investment costs, capital costs, operating and maintenance expenses, transaction costs, taxes, externalities, etc.

Then, the abatement cost is given by the following formula, which represents the difference between the cost of reducing one ton of CO2, if a mitigation option is developed, versus the
BAU scenario.

= [$ÿ 2]

The difference in emissions between both scenarios is defined as the reduction potential of the mitigation option.

Annexes A.7, A.8 and A.9 allow for further exploration of the following topics respectively:

- Derived abatement curves from models.
- Cost-Benefit Analysis. Distinctions between private and social evaluation. Inclusion of effects external.
- Treatment of the Discount Rate. A key issue in cost-benefit analysis and especially debated in the context of climate change.

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4 ANALYSIS OF ENERGY CONSUMPTION SECTORS

The selection of sectors is based on the specific weight of potential sectors for study in Latin America. To this end, an energy analysis is conducted for a group of six Latin American countries, selected primarily for their importance in terms of energy consumption within the region.

The selected countries are:

- Argentina
- Brazil
- Chile
- Colombia
- Mexico
- Peru

The analysis is based on national energy balances, which describe the energy consumption of the different sectors of the economy, making it possible to identify, to the extent that the information is sufficiently disaggregated, the most relevant sectors in terms of energy consumption.

A second component of the analysis comes from the review of national communications for the United Nations Framework Convention on Climate Change. These allow for the consideration of other sources of greenhouse gases in the analysis, as well as providing an overview of national strategies in the sectors under study, which will guide potential mitigation measures relevant to the region.

The analysis is conducted only for the energy sector, excluding the transformation sector and its self-consumption. This exclusion ensures comparability between energy balances, as their structure differs from country to country.

The following is a concise summary of the main results of the analysis. Detailed characterization of the countries can be found in Annex A.10.

4.1 SUMMARY

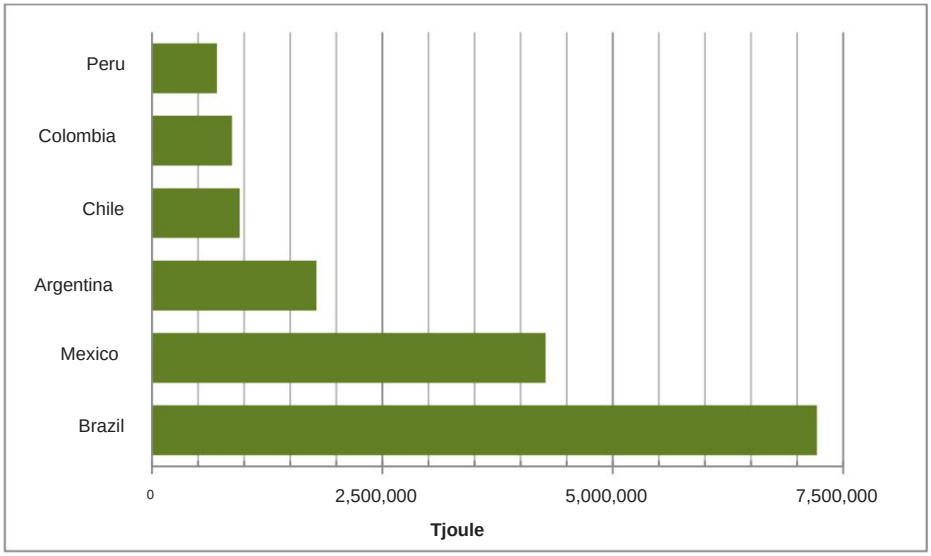
Analyzing the energy consumption of the countries studied allows for comparisons that better describe the energy behavior of the region.

First, a disparity in consumption is observed among the countries studied, mainly based on the difference in the size of their economies, populations, and territories. For example, Brazil has a final energy demand six times greater than Chile, as shown in Figure 6. A more interesting indicator in energy matters is energy intensity. This describes the energy units needed to generate one unit of product and allows for comparisons between countries. Figure 8 shows this indicator for the countries analyzed.

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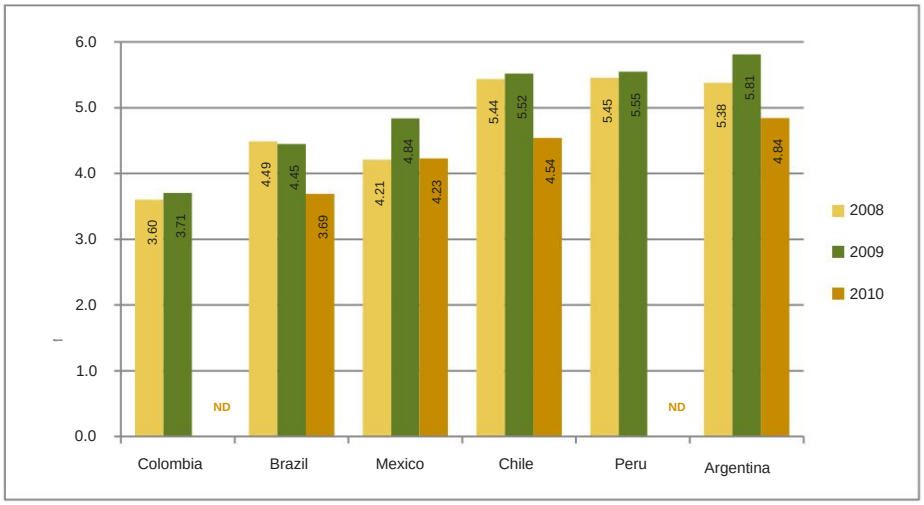
The lowest energy intensity is found in Colombia, followed by Brazil, suggesting a higher degree of efficiency in these economies. On the other hand, the least efficient economy among the countries analyzed is Argentina, reaching 4.84 Megajoules of final energy consumption per capita in 2010.
dollar of the Gross Domestic Product of that year.

Figure 7: Final energy demand 2009



Source: Prepared by the author based on the 2009 national energy balances.

Figure 8: Final consumption energy intensity (Mjoule/(2012)US$)



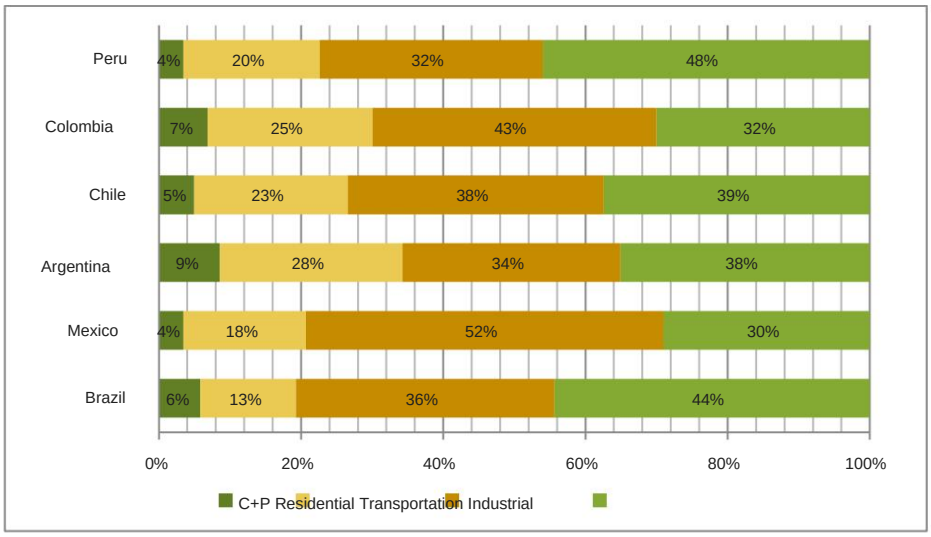
Source: Prepared by the author based on national energy balances 2008-2010 and the World Bank.

In most countries, the most significant sectors in terms of final energy demand are the industrial and transport sectors, the latter having clearly grown in recent years, which positions it as the main energy consumer in the short term, surpassing the

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industrial sector. An exception is the case of Argentina, where transportation is below the aggregate consumption of the commercial, residential, and public sectors. Figure 9 illustrates this result.

Figure 9: Distribution of final energy demand 2009



Source: Prepared by the author based on the 2009 national energy balances.

The transport sector is primarily dominated by road transport, which generally consumes around 90% of the sector's energy. However, the distribution of energy consumption in industry is much more heterogeneous among the countries of the region.

The following table shows the most important sectors for final energy demand.

Table 1: Main industrial sub-sectors in countries with available information.

Key sectors of the industry	Country	Average share of industrial energy consumption
Food and beverages	Brazil	27%
Cast Iron and Steel	Brazil	20%
Paper and pulp	Brazil	11%
Chemical	Brazil	9%
Copper mining	Chile	31%
Paper and Cellulose	Chile	20%
Other mining	Chile	7%
Food and Tobacco	Colombia	22%
Cement	Colombia	20%
Chemical	Colombia	12%
Paper and Printing	Colombia	10%
Iron and Steel	Colombia	9%
Iron and steel industry	Mexico	14%
Cement	Mexico	10%
Mining and Metallurgy	Peru	30%

Source: Prepared by the author based on the 2007-2010 national energy balances.

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It is observed that the most frequently appearing sectors, each appearing three times, are mining and paper, with the mining sector having a significantly larger share. Subsequently, the Iron and Steel, Food, Cement, and Chemical sectors each appear twice, with the latter having a larger share.
food sector.

In particular, the mining sector has a growing trend in energy consumption, this is a product of the increased energy intensity of the sector which is a consequence of the depletion of veins, or the exploitation of veins that are further away from the original processing centers, among other structural factors of the mining industry.

In conclusion, the most relevant sectors for analyzing mitigation measures, according to the data from the Latin American countries analyzed, are, firstly, the transportation sector, focusing on road transport. Next is the mining sector, primarily driven by the mining industries of Peru and Chile. The third sector is more difficult to select and is divided between the paper industry and the food industry.

The analysis of mitigation options is based on the importance of each sector in final energy demand, but it must also consider sector standardization. Sectors with heterogeneous processes and products generate many potential measures, but these lack significant added value in terms of GHG mitigation potential. This can be addressed through industry-wide measures, thus generating greater mitigation potential. For example, measures such as cogeneration, engine replacement, and energy recovery are applicable to most industrial subsectors, making them much more relevant and with a far greater potential for reducing greenhouse gases. Factors such as these suggest that the paper industry should be the third sector selected.

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5. GENERAL REVIEW OF METHODOLOGICAL APPROACHES

The following briefly presents the modeling approaches to be considered for quantifying the costs and effects of mitigation measures in the energy sector, and in particular in the transport and industry and mining sectors; sectors that will be investigated more specifically in chapter 7, and which have been chosen preliminarily, since, excluding the electricity generation sector, they are the most intensive in energy consumption and GHG emissions in the region.

5.1 APPROACHES TO MODELING MITIGATION MEASURES

Modeling mitigation options is crucial for the proper evaluation of different measures. Regulators generally use scenario studies to assess the potential effects of specific policy choices beforehand. In the case of mitigation measures, these scenarios are constructed using various methodological approaches. All models have both strengths and limitations.

There are alternatives for analyzing the relationships between energy consumption and economic variables such as production, prices, and emissions. In this sense, there are alternative methodological approaches for analyzing these relationships. According to UNFCCC (2008)6, the main categories of models are “top-down” and “bottom-up”.

Indeed, on the one hand, there are macroeconomic models, known as "top-down" models. Among these, computable general equilibrium models stand out, demonstrating significant consistency with economic theory. There are also econometric models that allow for the systematic incorporation of available historical data.
These approaches are useful for examining macroeconomic and fiscal impacts associated with GHG mitigation. They are especially useful for analyzing economic instruments such as carbon taxes or tradable permit systems.

On the other hand, there is the approach associated with microeconomic models known as "bottom-up" models. These models have a strong engineering foundation and, in many cases, specify the energy requirements of equipment and machinery to determine energy consumption and corresponding emissions. An emission factor can be associated with each process by fuel type, with further disaggregation by industry. This type of approach is useful for studying sectoral and technological impacts.

[6] UNFCCC. RESOURCE GUIDE MODULE 4: MEASURES TO MITIGATE CLIMATE CHANGE (2008).

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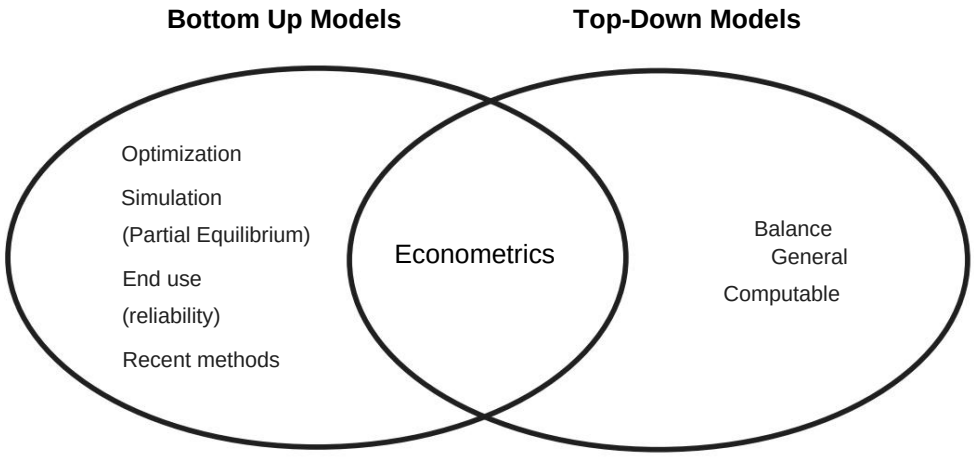
Table 2: Comparison between “Top-Down” and “Bottom-Up” models

Top-down	Bottom-Up
Use aggregated economic data	Use detailed data on fuels, technologies, and policies
Evaluate costs/benefits through impacts on production, income, GDP.	Evaluate the costs/benefits of individual technologies and policies
It generally assumes market efficiency.	It does not necessarily assume market efficiency; overcoming market barriers can result in cost-effective energy savings.
Capture feedback and cross-sector interactions	Capture interactions between projects and policies
This is not an appropriate approach for examining specific technology policies.	Used to evaluate costs and benefits of programs

Source: Based on UNFCCC 2005

The following figure presents the main types of models according to whether they are top-down or bottom-up. It can be seen that econometric methods can be either bottom-up or top-down depending on the case and application. Indeed, by way of example, it is possible to estimate energy consumption at a national or global level (top-down models) or at a sectoral level or for a small community (bottom-up).

Figure 10: Presentation of main categories of energy models according to "Top" classification Down" o "Bottom up"



Source: own elaboration.

According to UNFCCC (2008), the lack of top-down models, coupled with the need to evaluate individual projects, has led to the vast majority of evaluations being conducted using bottom-up approaches. This will be the approach considered in the development of this project.

5.2 INFORMATION **GATHERING TO FEED THE DETERMINISTIC ANALYSIS** AND **PROBABILISTIC**

The required information can generally be gathered in 3 ways, depending on the information available.

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• Gathering bibliographic information: The variables and their projections are obtained from a review of relevant literature. Appropriate methods and values for the calculations should be sought in existing documentation. For example, to project fuel prices (oil and coal prices), existing projections from organizations with recognized experience or relevance in the field of the variable being analyzed can be used, such as the International Energy Agency (IEA), the U.S. Department of Energy (DOE), among others.

A portfolio of mitigation measures, obtained from validated studies, can also be used to adequately consider the parameters required for calculations in a specific sector.

• Expert Opinion Analysis: It can often be very complex to accurately project a variable in a given year, but even more complex to determine the degree of dispersion of that forecast. A good alternative for obtaining this information is informed consultation with experts. A wide variety of sophisticated Bayesian evaluations and combinations of techniques have been used to analyze the information provided by experts. In practice, the Bayesian approach tends to be more difficult to apply, as it requires thorough assessments of the dependence between experts. Certainly, a simpler approach, compatible with some Bayesian formulations, is to combine the experts' distributions using a weighted average of the probability densities of the delivered value [Morgan and Henrion, 1990].

• Technical-economic modeling: This involves the use of statistical tools

Econometric methods for projecting variables require historical series or trend data that allow predicting the future behavior of the variables. Typically, these methods allow projections with a degree of variability, but only if the necessary prior information exists to obtain results with a significant degree of confidence.

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6 METHODOLOGICAL DEVELOPMENT AND RESULTS OF THE SECTOR ELECTRIC POWER GENERATION

In accordance with what was presented in section 3 regarding the fundamentals of the development of abatement curves and the incorporation of uncertainty in the analyses, the development methodology in the electricity generation sector is described below, in addition to its practical application in the electricity generation subsector in Chile.

The construction of an abatement curve generally involves identifying the measures to be evaluated and estimating their annualized costs and CO2e emissions in a given year. This allows for obtaining the abatement potential (horizontal axis of the graph) and the abatement cost per ton of CO2e reduced (vertical axis).

Chile's potential to reduce greenhouse gas emissions depends on a number of variables, such as the productive sector being analyzed, the energy consumption intensity (related to energy efficiency) of the sector, technological development, among other variables.

In the specific case of the Chilean energy sector, the potential is determined based on the capacity for technological advancement in the short term. This potential is then calculated through expert opinions from professionals at the Energy Center of the University of Chile, from which the reduction potential is derived.

Considering the above, in the development of the abatement curve for the energy sector,
They analyze the most cost-effective options for reducing electricity generation. The measures are presented considering their cost per ton of CO2e reduced and their reduction potential in the first year of operation, taking into account that construction can begin as early as 2012.

The methodology adopted for the development of this analysis comprises the following points:

- Review of international literature, in particular studies that have already assessed the potential of reduction of a series of measures in different productive sectors.
- Consult national experts7 regarding technologies, costs, emission levels, feasibility, applicability, acceptability and adaptation of technologies, etc.
- Establishing the cost of the measure, including its underlying costs and the potential penetration horizon. • The cost of the measures corresponds to the actual cost to society of adopting them.
  It considers investment, operation, and maintenance costs, and any savings in inputs or fuel. It does not consider some social costs or benefits, such as increased local pollutants or flooding of large areas resulting from reservoir construction.

[7] The experts are professionals from the Departments of Electrical Engineering, Mechanical Engineering and PRIEN of the University of Chile and the Ministry of Energy, among others.

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- Qualitative identification of other possible impacts (positive or negative) of the measures that could either facilitate or hinder its implementation.
- Calibration of results with respect to the costs of technologies from official bodies already in place results from the international literature.

Criteria have been defined for the selection and evaluation of the measures, so that only those feasible to apply in Chile within the evaluation timeframe are included. In this way, measures available only at an experimental level or with an incipient stage of development, measures not applicable in Chile, and unfounded judgments or opinions regarding certain technologies are not taken into account.

Although many studies only incorporate measures with costs below US$50 per ton of CO2e reduced, it is also possible to incorporate measures with costs above this range, according to their feasibility of implementation in the country.

The groups of measures considered, ordered from least to greatest importance with respect to reduction potential, are the following:

- New reservoir hydroelectric power plants.
- New geothermal power plants.
- New run-of-river mini hydroelectric power plants.
- New nuclear power plant.
- New wind farms.
- New biomass-fired thermal power plants.
- New coal-fired thermal power plants with CO2 capture system (CCS).
- New photovoltaic solar power plants (PV).
- New concentrated solar power (CSP) plants.
- New tidal power plants
- New natural gas power plants

In the case of costs, the main variables required are investment costs, operation, maintenance, useful life, investment periods, fuel prices involved, construction times, among others.

6.1 IDENTIFICATION OF THE MAIN VARIABLES SUBJECT TO UNCERTAINTY

It is possible to identify a wide range of variables that can be considered when conducting cost-benefit analyses in the energy sector. Some of these are known beforehand to have a significant effect on the evaluation, such as fuel prices. Others, however, have important effects but are often considered secondary or too complex to quantify, and therefore are not included in the evaluation, often underestimating the true costs of a project.

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The following table shows the variables that influence the cost of investment, operation and maintenance of an electricity generating plant, and which are also to a greater or lesser extent a source of uncertainty for the analysis, or a source of variability for the comparison between technologies.

Table 3: Relevant variables for the evaluation of the energy sector

Variables	
Investment cost	CO2 emission factor
Fixed cost	Discount rate
Cost of O&M	Fixed externalities
Transmission cost	ÿEmpleo
Fuel cost	Variable externalities
Fuel transport	ÿContaminación local
Plant factor	ÿEmission Date
Performance efficiency	Other environmental externalities
Construction time	Cost/Revenue Environmental Bonds
Operating life	Learning variables (Investment/efficiency)
Transaction time	Effect on the knot price
Transaction costs	Exchange rate

Source: Own elaboration

A generic analysis of technologies can distort the real uncertainty that some of these variables contribute to the problem. For example, the geographic location of a project strongly influences the transportation costs of inputs, the costs generated by variable externalities (how close the project is to an inhabited area), or the investment required in the transmission line. Therefore, these variables must be considered when evaluating specific projects, as they respond to, and can only be quantified as effects of, the geopolitical location of the project. Otherwise, the comparability of different technologies in cost-effectiveness analysis would be reduced.

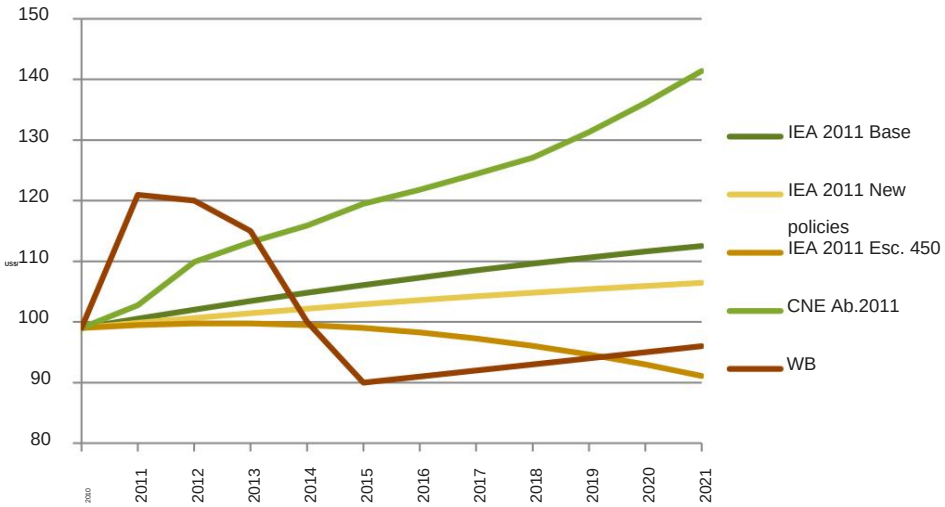
The variables considered in this evaluation are detailed below:

Fuel prices

Fuel prices are, a priori, a key variable in cost-effectiveness assessments, and at the same time, a significant source of uncertainty, given that economic results depend on fuel consumption for up to 40 years or more for thermal power plants, at prices that are highly erratic, even in the short term. This is illustrated in the following figure, which shows projections of the price of coal for electricity generation in 2021 published by the International Energy Agency (IEA), the World Bank (WB), and the Chilean National Energy Commission (CNE).

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Figure 11: Projection of the price of a ton of bituminous coal



Source: Own elaboration

Addressing this uncertainty is vital when deciding on an electricity generation project, since the project's economic results will differ greatly in a scenario where the price remains "low," favoring technologies that consume fossil fuels, versus a scenario of "high" prices where renewable energies would find themselves in a much more advantageous competitive position.

Cost of technologies

The investment cost of technologies is another key factor when evaluating a project. It represents the majority of the capital required for renewable energy projects, but for some new technologies, it is subject to constant innovation, which makes the price dynamic. In addition to this factor, costs differ between countries and geographical areas. For example, in 2010, PricewaterhouseCoopers estimated the investment cost for a CSP solar plant in Spain to be between €3,000 and €3,900/kW, and between €2,300 and €2,900/kW if it were installed in the Sahara .

O&M Costs

Operation and maintenance costs (excluding fuel consumption) are another source of uncertainty in the evaluation of power generation projects. These costs can be affected by the location and consequent distance of a plant from supply centers, by the efficiency achieved in the plant's operational processes, or by other variables inherent to each technology. In general, the values reported in the literature differ considerably for each technology; therefore, this variable is considered stochastic in this study.

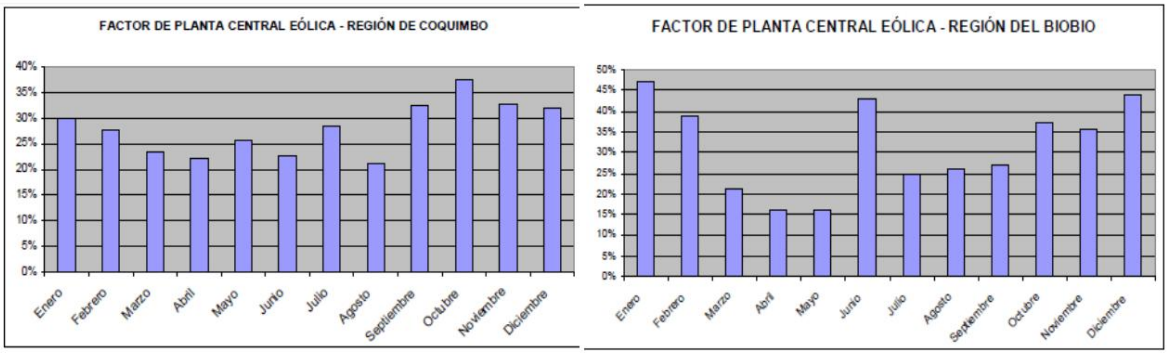
[8] "A roadmap to 2050 for Europe and North Africa", Price Waterhouse Coopers 2010

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Plant factor

The capacity factor is a variable that adds significant uncertainty to the evaluation, especially when dealing with energy sources that are dependent on weather patterns. Therefore, it becomes important to consider the capacity factor as a stochastic variable, since technologies like wind and solar exhibit a wider range of capacity factors, leading to significant variations in the cost of mitigating these technologies. As an example, the actual capacity factors obtained for two wind farms located in northern and central Chile, respectively, are presented.

Figure 12: Plant factor of wind power plants, 2010.



Source: Node price report April 2011, CNE.

The graphs show that there is not only a difference between the 2 plants, but also a difference throughout the year within each plant, reaching in the case of the Bío Bío region a difference of 200% in the plant factor between the "peak" months and those with the lowest plant factor.

Efficiency

Energy conversion efficiency influences the amount of fuel a power plant requires to generate one unit of energy. This can vary according to the calorific value of the fuel consumed and the specific parameters of the plant being evaluated. Looking at the specific consumption data from the April 2011 nodal price report for power generation plants installed in Chile's central interconnected system, values for coal-fired thermal power plants range from 0.35 to 0.4 tons of coal per MWh generated, which implies different levels of efficiency and, consequently, different fuel requirements.

Construction time

While different technologies have relatively standardized construction times, in practice there are factors that cause variations in these figures. Political, geographical, and social factors can affect these times, causing, if the deadlines are extended, cash flows to be delayed, resulting in a greater discounting of the flows shifted to the future.

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Emission factor

The CO2 emission factor differs between different installations of the same technology; it is affected by the same factors as efficiency. In general, the carbon content of the fuel used and the specific consumption of the thermoelectric plant will cause differences in the emission factor.

6.2 APPLICATION OF THE METHODOLOGY IN THE CHARACTERIZATION OF STOCHASTIC VARIABLES IN THE ELECTRIC POWER GENERATION SECTOR.

The construction of the stochastic variables of the model is detailed below, specifying the assumptions used and the sources of information.

Fuel prices

Fuel prices are modeled using medium-term and long-term estimates based on projections from expert organizations. The randomness of price variables is only considered for one year (2020) for the medium term and another year (2035) for the long term. Subsequently, price behavior in the intervening years will be a function of the aforementioned random variables. This design is chosen because the evaluation focuses on the effect of medium- and long-term fuel price trends, rather than on natural short-term market fluctuations.

In this case, the projections generated by the International Energy Agency (IEA) presented in the annual publication “World Energy Outlook 2011” were used. Projections are presented for a baseline scenario, one with the application of policies in accordance with climate change where CO2 reduction targets are projected for the main energy-consuming countries, and a scenario that considers with a 50% probability of limiting global warming to 2 degrees Celsius .

Additionally, projections from the World Bank and the National Energy Commission of Chile published in the October 2011 report on the nodal price of the central interconnected system are used.

Based on these projections, a random value for the fuel price in 2020 and 2035 is obtained. These prices are then interpolated between the base year and the value of the random variable in 2020. A convex trajectory was used for this interpolation to quickly cover short-term fluctuations in the fuel price, as shown in Figure 14. The interpolation between 2020 and 2035 was designed concavely to maintain the growth rate of the 2011-2020 interpolation and generate

[9] For more details on the scenarios, refer to the World Energy Outlook 2011 (IEA) document, chapter 1: “Defining the scenarios”

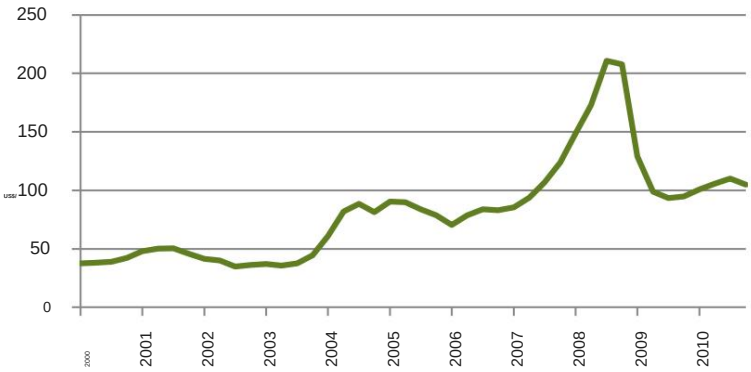
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a greater amplitude of the probability distribution of the value towards the final years of the projection

Coal price

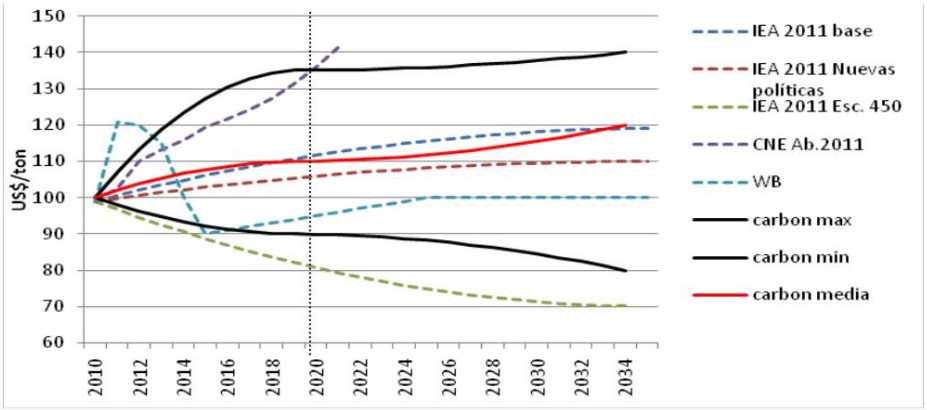
Figure 14 shows the projections observed in the various sources. The price limits and expected price per ton of coal (maximum, minimum, and average) used to define the random price variable for 2020 and 2035 are plotted on these projections. The IEA's scenario 450 projection was not fully considered because it is too optimistic compared to the behavior of recent years, as shown in Figure 13. For this reason, a 50% drop in the projection for 2020 and a 66% drop for 2035 are assumed. Since the distribution of these projections is unknown, a triangular distribution centered on the expected price and bounded by the maximum and minimum values defined in the projections is used. After 2035, a constant price is assumed.

Figure 13: Historical Price of Coal



Source: Prepared by the author based on data from the CNE

Figure 14: Bituminous coal price projections US$/ton



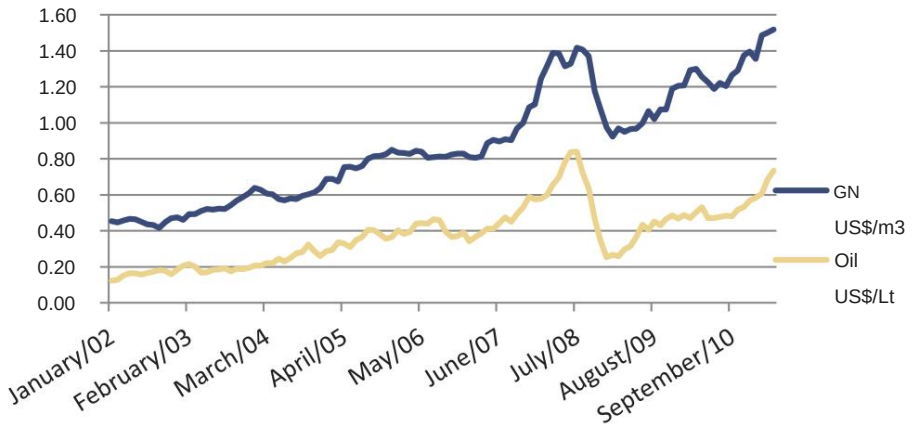
Source: Own elaboration

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Gas natural

The following figure shows the average price of natural gas in central Chile and the observed price of oil between 2002 and 2010, revealing a direct relationship between these variables. Based on this, an econometric prediction of the natural gas price is generated as a function of the price of a barrel of oil and natural gas price lags. See Appendix A.17.

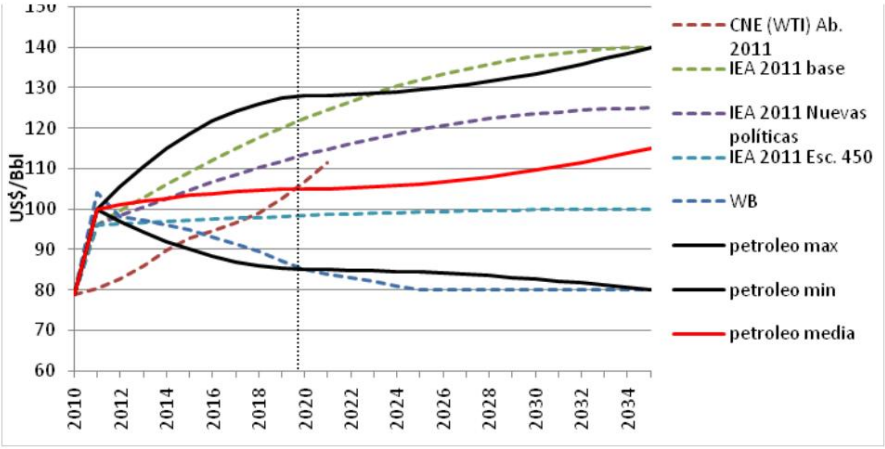
Figure 15: Historical prices of oil and natural gas



Source: Prepared by the author based on data from the CNE

Through econometric prediction, the parameters are obtained with which the price is projected according to the random variables for the price of the barrel of oil10, these are defined according to the limits and the expectation of the estimate to the year 2020 and 2035, obtained from the projections of the sources consulted indicated in the following graph.

Figure 16: Oil price projection US$/Bbl



Source: Own elaboration

[10] See econometric model in appendix A.17.

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Biomass

Projecting the price of biomass is more complex given the recent promotion of new uses for it as an energy source, but it is still an emerging market. The CNE's node price report (October 2011) was consulted, which shows a fuel cost for biomass-fired power plants of between US$35 and US$45/MWh. Although price stability is expected, a maximum price of up to US$80/MWh is modeled in a scenario where demand increases due to environmental trends away from fossil fuels.

Uranium

The demand for uranium is considered stable and relatively well-known, given the number of nuclear power plants installed worldwide and their consumption. Therefore, its projected price is also considered relatively stable at US$9.33/MWh, considering the complete uranium cycle.¹¹

The following table summarizes the values taken by the random variables.

Table 4: Parameters of random variables

Variable	Average value	Minimum	Maximum
Oil price 2020 US$/Bbl	105	85	128
Oil price 2035 US$/Bbl	115	80	140
Coal price 2020 US$/Bbl	110	90	135
Coal price 2035 US$/Bbl	120	80	140
Biomass price 2020 US$/MWh	47	40	60
Biomass price 2035 US$/MWh	55	40	80
Uranium Price US$/MWh	9,3	7,9	11,6

Source: Own elaboration

A positive correlation was also incorporated between the medium- and long-term variables for each fuel. This was done to reinforce the idea that the medium-term trend, up to 2020, necessarily has an effect on the long-term fuel price behavior (2020-2035). Thus, it would be expected that with high random variables for the 2020 price, the 2035 price would not fall within the lower range. While the correlation is unknown and arbitrary, it is adjusted to a value that allows for modeling the aforementioned effect; therefore, a correlation of 0.6 was used between the 2020 price of each fuel and its 2035 price.

Investment cost

To model the distribution of investment costs, literature on the evaluation of electricity generation projects was used, which presents investment cost ranges12. Additionally,

[11] "Projected Costs of Generating Electricity 2010", IEA y Nuclear Energy Agency NEA, Pag. 42.

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Experts from the Energy Center were consulted. Using this information, a range of probable prices and a mean value were constructed. The mean value was assigned according to expert opinion if it fell within the ranges reported in the literature, or as the value of the range that most closely matched the opinion. expert.

The assigned probability distributions are triangular since the distribution of possible values is unknown, but a "most probable" value, a maximum, and a minimum are known. The parameters are presented in Table 5.

Cost of O&M

To design the distribution of this variable, we used literature on the evaluation of electricity generation projects, which presents ranges of O&M costs. We also used the average variable cost of each power plant in the central interconnected system, obtained through the nodal price report. Based on this information, we determined possible O&M cost values, which were then modeled using a triangular distribution centered on the value with the highest density. The operation and maintenance cost of the Carbon Capture and Storage (CCS) system is more complex to quantify, given that the technology is still under development. To design a CCS range, we considered the information from the paper “Realistic Costs of Carbon Capture” [Al-Juaied 2009].

Table 5: Parameters of random variables

Technologies	Average cost value investment US$/KW	Min	Max	Average cost value O&M US$/MWh	Min	Max
Coal	2000	1800	2500	2,25	1	6
Mini Hydraulics	2800	2600	2800	5	4,25	6,97
Geothermal	3550	3000	5000	2	2	10
Biomass	2600	2500	3000	6,71	2,4	10
Nuclear	4000	3500	5500	15	9	20
Wind power	2100	1500	2400	7,7	7,7	10
Solar CSP	6200	5000	6800	20	10	20
Solar PV	4000	3500	4500	4,45	4,45	20
Tidal power	5000	4500	6000	17,5	17,5	28
CCS13	1900	1500	2100	100	80	120
Reservoir	2000	1900	2100	5	2,5	7
Gas CC	1000	800	1200	3,8	2,5	6

[12] “A Risky Proposition: The Financial Hazards of New Investments in Coal Plants”, Union of Concerned Scientists, Marzo 2011.

“Building a low-carbon economy – the UK’s contribution to tackling climate change”, Committee on Climate Change, December 2008.

[13] The investment cost of CCS technology refers to the additional value, over the cost of installing a conventional coal-fired thermal power plant.

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Source: Own elaboration

Plant Factor

To design the distribution of this variable, we consulted the literature on the evaluation of power generation projects, which presents plant factor ranges for each technology. We also sought expert opinion from the Energy Center. A triangular distribution was used, with the primary criterion for the mean being the value provided by expert opinion, if it fell within the ranges found in the literature. Otherwise, the closest value from the literature was used. The parameters are detailed in Table 6.

Efficiency

To design the distribution of this variable, we used literature on the evaluation of electricity generation projects, which presents efficiency ranges for each technology. We also compared this with information provided by experts. The variable was modeled with a triangular distribution whose parameters are presented in Table 6.

Construction time

To design the distribution of this variable, we use literature on the evaluation of power generation projects, which presents construction time estimates for different technologies. This is supplemented with expert information. In general, this information is fairly standardized, and the values differ very little from one publication to another. Using the information obtained, we design discrete distributions where the mean value has twice the probability of occurrence of the other probable values, which have the same probability of occurrence. The parameters are presented in the following table.

Table 6: Parameters of random variables

Technologies	Average value FP (%)	Min	Max	Value half efficiency a (%)	Min	Max	Value TC medium years	Min Max	
Coal	0,82	0,78	0,85	0,43	0,4	0,46	4		
Mini Hydra street	0,62	0,55	0,7	1	1	1	2		
Geoté rhemic	0,84	0,82	0,85	1	1	1	5		
Biome	0,8	0	0	0	0	0	4		

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work	5	,8	,85	.45	,4	,5		
Nucle with	0,85	0,8	0,9	0.34	0,34	0,36	7	
Wind power	0,3	0,25	0,35	1	1	1	1	
Solar CSP	0,45	0,35	0,5	1	1	1	2	
Solar PV	0,28	0,25	0,3	1	1	1	1	
Great wrap rice	0,28	0,25	0,3	1	1	1	2	
CCS*	0,82	0,8	0,85	0.43	0,43	0,46	4	
Embal with	0,54	0,5	0,75	1	1	1	4	
Gas CC	0,7	0,6	0,85	0.5	0,5	0,55	2	

Source: Own elaboration

Emission Factor

The emission factor was obtained through an analysis of the performance of coal-fired and natural gas-fired power plants. To this end, the emission factor of each plant was determined according to its specific fuel consumption (Ton/MWh). From the resulting set of emission factors, a range and a mean value were extracted for the stochastic emission factor. For simplicity, an emission factor of 0.108 tCO2/MWh is assumed for a coal-fired power plant with a CCS system.

Table 7: Parameters of the Emission Factor variable.

Technologies	Average value FE tCO2/MWh Min	Max	
Coal	0,95	0,9	1,05

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CCS*	0,108	0,108	0,108
Gas CC	0,45	0,35	0,55

Source: Own elaboration

6.3 DEVELOPMENT OF THE UNCERTAINTY-BASED REDUCTION CURVE FOR THE GENERATION SECTOR OF ELECTRIC POWER IN THE CENTRAL INTERCONNECTED SYSTEM OF CHILE.

This section presents the general methodology for estimating a reduction curve for the electricity sector. This analysis assumes that the baseload power plant uses pulverized coal technology. The reduction measures correspond to power plants with less CO2e-intensive technologies. For example, technologies associated with non-conventional renewable energy, hydroelectricity, and nuclear power are considered.

Initially, it is necessary to propose a portfolio of technically and economically feasible measures, relevant according to the contribution of GHGs to the total emissions of the sector involved.

The evaluation includes an analysis of the relevant variables of the measures, defining a statistical distribution of their behavior. Subsequently, a stochastic cost-effectiveness analysis of the measures is developed, generating a cost reduction curve that supports decision-making regarding which measures should be prioritized.

In the case of the national electricity generation sector, which includes the SIC, SING, Aysén, and Magallanes systems, generation levels will increase from 62.5 GWh in 2010 to 180 GWh in 2030. This implies an increase in capacity from 16,954 MW in 2010 to 40,000 MW in 2030, of which 52% corresponds to coal-fired power plants, compared to 30% in 2010. This will cause the sector's emissions to increase to 85 million TCO2e, raising the national electricity mix's coal intensity from 0.37 to 0.47 TCO2e per MWh. This is lower than the current value in developed countries such as Spain, Germany, Japan, and the USA, and even lower than the projected value for some of these countries. For example, the USA projects the carbon intensity of its electricity mix to 0.62 TCO2e per MWh in the year 2030 [PROGEA, 2008].

This increase in emissions resulting from the greater use of coal in the energy mix opens up significant opportunities for reducing GHG emissions by exploring alternative generation methods or mitigating these emissions (CCS). In this regard, this sector offers substantial reduction potential, which could reach a maximum of 37 million tons of CO2e by 2030. This represents the country's most significant GHG reduction potential.

The feasible areas to be considered to approach this reduction potential are Carbon Capture and Storage (CCS), Non-Conventional Renewable Energies (mainly mini hydroelectric plants, geothermal, wind power plants and biomass), nuclear energy and reservoir hydroelectric power.

Basic Model Description

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This section presents the general methodology for estimating an abatement curve for the electricity sector. This analysis assumes that the baseload power plant uses coal-fired technology. The abatement measures correspond to power plants using technologies that emit less than coal. For example, technologies associated with non-conventional renewable energy, hydroelectricity, and nuclear power are considered.

A key (and implicit) assumption in this type of modeling is that the unit abatement cost is independent of the reduction level for each technology. This allows us to estimate a unit cost and then determine the potential or bar width of the cost curve.

The following table shows the variables that are considered random in the model.

Table 8: Formalization of model variables.

Variable	Notation	Description
Unit investment cost		Units of US$/kW
Number of years of project construction		Also, the years in which the investment is distributed. The only variable with a discrete distribution.
Fraction of investment in year t		Applies during the construction years. = 0, ÿ ÿ 1. Must satisfy that ÿ = 1 $\sum_{=0}^{ÿ1}$. In our case is assumed = 1/ .
Plant factor		It represents the fraction of the year in which the technology is actually available for use.
Fuel price in year t		Prices must be in units of US$/ MWh. In some cases, they are estimated using econometric methods.14
Performance		Fraction that represents the efficiency of the process
Other operating costs	C	Units of US$/MWh
Technology-related emission factor ÿ	ÿ	Units of ton CO2eq/MWh.

Source: Own elaboration

The evaluation also takes into account the useful life (T) of each type of power plant and the discount rate (r). These two variables are not considered random for the purposes of the simulation15.

The first step involves estimating the net present value of the costs associated with each technology. A simple case is evaluated, assuming a generation of 1 MWh per year for each technology over the project's lifespan.

[14] For example, an econometric relationship is established between the price of crude oil and diesel. Given an outflow of Simulation of crude oil prices, this relationship allows obtaining estimates of diesel prices.

[15] In general, these exercises use different discount rates to adjust the results, but they do not... This includes it as a variable to be simulated.

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Each technology is associated with three main cost categories: Investment, fuel costs, and operating costs.

- Present Value of Investment Costs

-It is invested (US$/Kw) each year of construction = 0, ÿ ÿ 1.

The plant operates 365 × 24 × hours per year. Therefore, the previous cost can be expressed as 365 × 24 ×

now in units of US$/kWh, or, $\frac{1000ÿ}{365ÿ24ÿ}$ and US$/MWh.

-The present value of the investment costs turns out to be:

$$(\text{ from }) = \frac{1000}{365 ÿ 24 ÿ ÿ} \sum_{=0}^{ÿ1} \frac{}{(1 +)}$$

- Present Value of Fuel Costs.

-The present value of fuel costs is constructed from all the flows incurred during the useful life of the generation project = + ÿ 1.

-With a monetary amount, 1MWh of fuel can be purchased during the operating period t. However, since the efficiency of the generation process is less than 1, then the following are required to generate 1 MWh of electricity. Therefore, the NPV associated with fuel costs is:

$$(\quad) = ÿ \sum_{=}^{+ ÿ1} \frac{}{(1 +)}$$

- Present Value of Other Operating Costs

-Given that each year of operation incurs a cost of C (US$/MWh), the NPV is simply:

$$(\text{ from }) = ÿ \sum_{=}^{+ ÿ1} \frac{}{(1 +)}$$

In summary, the VPN associated with each of the technologies turns out to be:

$$= \frac{1000}{365 ÿ 24 ÿ ÿ} \sum_{=0}^{ÿ1} \frac{}{(1 +)} + ÿ \sum_{=}^{+ ÿ1} \frac{}{(1 +)} + ÿ \sum_{=}^{+ ÿ1} \frac{}{(1 +)}$$

To annualize, the equivalent constant annual amount K (with a period equal to the useful life of each project) is estimated as:

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$$= \ddot{y} \sum_{=0}^{\ddot{y}1} \frac{}{(1+)} = \frac{(1+)\ \ddot{y}\ 1}{(1+)^{\ddot{y}1}}$$

Identifying the Capital Recovery Factor (CRF):

$$= \frac{(1+)\,(1^{\ddot{y}1}}{+)\ \ddot{y}\ 1}$$

It remains = The latter has units of $/MWh.

The drawdown is estimated through a counterfactual analysis. This is how a unit drawdown is defined.

(tons reduced/MWh) as the difference between the emission factors of coal (c) and the technology

alfa: = ÿ ÿ .

Then the unit cost of abatement for each alpha technology is estimated as:

$$= \frac{}{}$$

In this way, the simulation model provides a distribution for each unit and annual abatement cost.

To construct the graph, the information is combined with that of the drawdown potential per measurement (which determines the bar width). In this study, uncertainty is not incorporated into the drawdown potential.

Potential for abatement

Constructing the curve requires a potential reduction for each mitigation option; in this particular case, we are referring to real options for incorporating electrical capacity from the evaluated technologies. These values were obtained through expert opinion from professionals at the Energy Center.

Table 9: Potential for reducing technologies

Technology	Potential annual reduction capacity (MMtonCO2)	
Reservoir	1000	4,38
Mini Hydraulics	300	1,50
Geothermal	500	3,41
Nuclear	1000	6,90
Wind power	600	1,46
Biomass	100	0,69
CCS*	300	1,76
Solar PV	200	0,48
Tidal power	10	0,02
Solar CSP	200	0,89

Source: Own elaboration

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Results

This section presents the results of the mitigation cost estimation model incorporating the uncertainty in the variables mentioned above and the consolidation of these results in the greenhouse gas abatement curve.

The economic evaluation is carried out with two discount rates, the first uses the social rate (6%) and the second corresponds to a discount rate closer to a private business (12%).

In the case of the assessment for the 6% rate, the following table presents the deterministic mitigation cost in the second column and the probabilistic values in their most probable value (Mode), along with the 5% and 95% confidence levels of the abatement cost, in the subsequent columns. For example, nuclear technology has unit costs less than US$2.15 with a 5% confidence level and less than US$24.66 with a 95% confidence level.

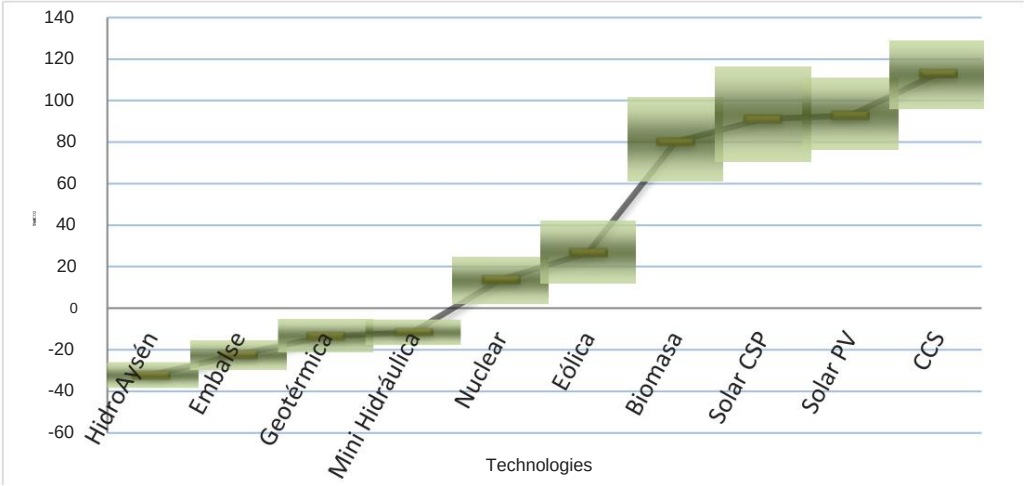
Table 10: Electricity generation technologies ordered by mitigation cost (r=6%).

Technology Cost	Mitigation US$/tonCO2	Fashion	Level of 5% confidence	Confidence level 95%
Reservoir	-$ - 19.1	-$ 18.1	21.2 -$ 28.0	-$ - 15.2
Geothermal	$ - -$ 10.8	-$ 13.2	-$ 20.6	$ - 4.0
Mini Hydraulics	$ 12.7	$ $ 75.7	10.9 -$ 16.9	$ 4.8
Nuclear	$ $ 95.8	$ 89.0	13.7 $ 3.5	$ 24.6
Wind power	$ 31.5		25.2 $ 14.0	$ 38.5
Biomass	$ $ 119.4	$	79.0 $ 61.6	$ 102.3
Solar CSP	$ 149.2	$	89.4 $ 71.7	$ 110.8
Solar PV	$		92.8 $ 76.2	$ 109.7
CCS	$		112.7 $ 97.1	$ 129.2
Tidal power	$		152.3 $ 134.0	$ 181.7
Gas CC	$ $391.7		371.1 $ 301.4	459.5
Diesel	$ $5,470.3	$5,795.7	$2,173.1 Source: Own elaboration	$ $ 13,899.6

Figures 17 and 18 present the main result, the abatement curve for the Chilean electricity generation sector, incorporating the uncertainty in the abatement cost of each applied measure (technology) (Figure 17), and the abatement potential (Figure 18). The solid line represents the expected values curve for this case, and the areas above this curve represent the uncertainty propagated through the analysis, reflecting the variability of the results shown in Table 10.

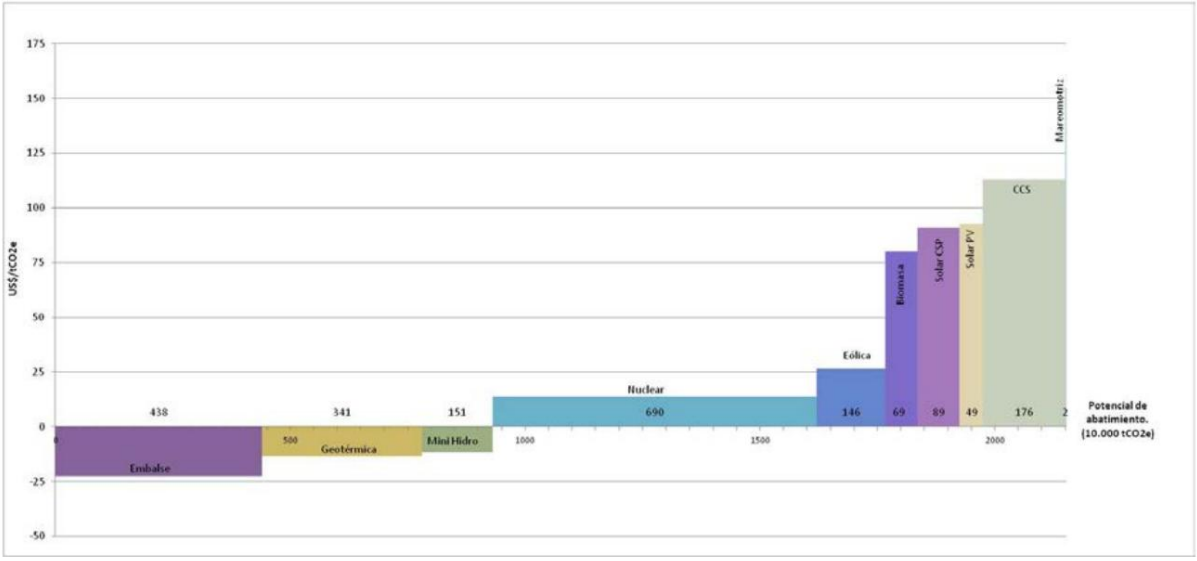
Figure 17: Abatement curve incorporating uncertainty (r=6%)

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Source: Own elaboration

Figure 18: Abatement curve incorporating average values (r=6%)



Source: Own elaboration

One of the key takeaways from these results is the shift in investment decisions that might occur in a decision-making process that considers uncertainty, compared to a deterministic one. For example, if the investment decision criterion is the Value at Risk (VaR) with a 95% probability, meaning the variables are ranked according to their probable cost with 95% confidence, this results in a change in the order between Geothermal and Mini Hydropower technologies, altering their priority in the deterministic scenario ($-18.12 vs. $-10.84).
compared to the VaR criterion (95%) ($-5.23 vs -$5.55), using the mitigation cost in US$/tonCO2 as an indicator.

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On the other hand, the following table presents the variability compared between pairs of technologies, checking in what percentage of cases, technology X is more expensive than technology Y.

This allows us to visualize in which cases certain technologies may behave differently from the deterministic value or the expected average value, given the incorporation of uncertainty.

Table 11: Variability of mitigation technologies

Geothermal>H. Embalse Minih	dro>Wind Geothermal>Nuclear	
98%	**69%**	**89%**

Source: Own elaboration

In the case of the evaluation for a discount rate of 12%, the following table presents the deterministic mitigation costs and the probabilistic values at their most likely value and the

confidence levels of 5% and 95% of the abatement cost.

Table 12: Electricity generation technologies ordered by mitigation cost (r=12%).

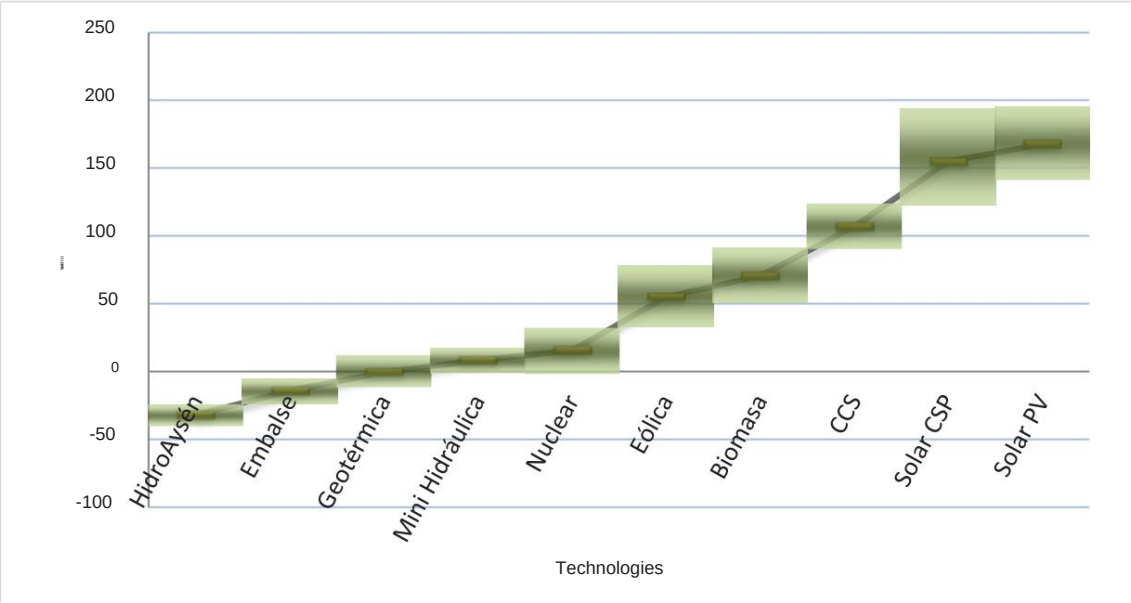
Technology Cost	Mitigation US$/tonCO2	Fashion	Level of 5% confidence	Confidence level 95%
Reservoir	-$ - 7.5	-$ 14.4 -$ -$ 0.6 -$	$ 24.2	-$ 5.4
Geothermal	$ 5.9	8.0 -$ $ 15.4 -$	$ 11.7	11.8
Mini Hydraulics	$ 9.8	54.7 $ $ 70.4 $ 115.8	1.4	17.3
Nuclear	$ 13.8	$ 106.7 $ 161.6 $	2.1	32.0
Wind power	$ 63.8	154.7 $ 166.6 $ 167.4	32.8	77.9
Biomass	$ 65.8	$ 231.4 $ 234.3 $	50.6	90.9
CCS	$ 463.7 $ 440.9 $ 5,384.9 $ 5,706.8		90.1	123.7
Solar CSP	$ $		122.8	193.4
Solar PV	$		141.1	195.4
Tidal power	$		198.4	272.6
Gas CC	$		344.9	553.6
Diesel	$		2,146.7	$ $ $ $ $ $ $ $ $ $ 13,662.1

Source: Own elaboration

Figures 19 and 20 present the abatement curve for the Chilean electricity generation sector, incorporating the uncertainty in the abatement cost of each applied measure (technology) and the mitigation potential. The solid line represents the deterministic curve (average expected values) for this case, and the areas above this curve represent the uncertainty propagated through the analysis, reflecting the variability of the results presented in Table 12.

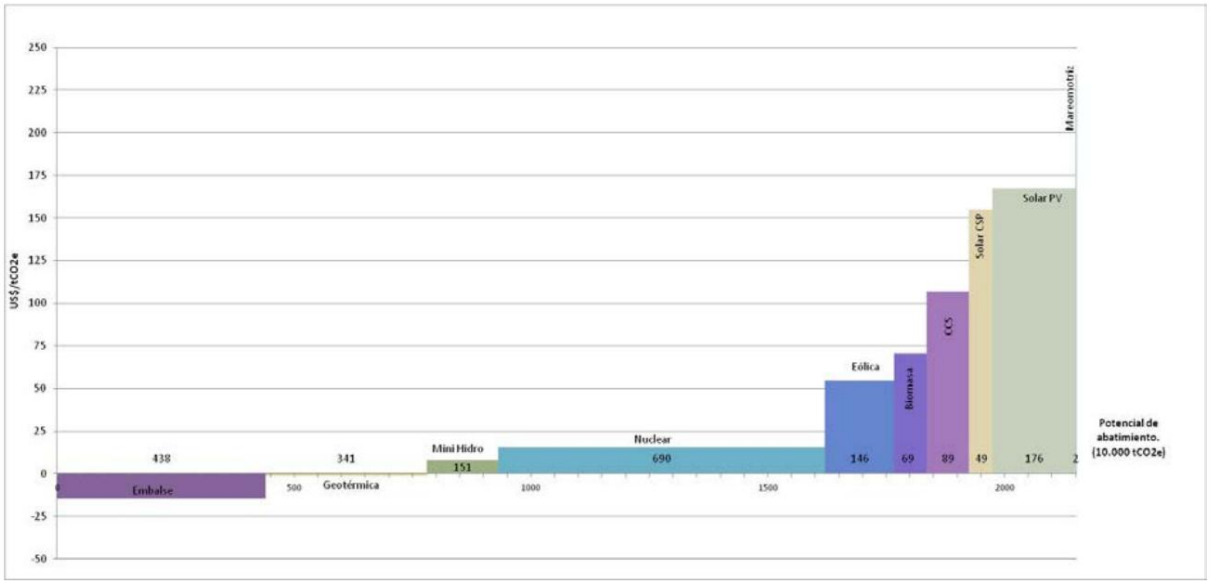
Figure 19: Abatement curve incorporating uncertainty (r=12%)

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Source: Own elaboration

Figure 20: Abatement curve incorporating average values (r=12%)



Source: Own elaboration

The following table presents the variability compared between pairs of technologies, showing the percentage of cases in which technology X is more expensive than technology Y. This allows us to visualize in which cases certain technologies may behave differently from the deterministic or expected average value, given the incorporation of uncertainty.

Table 13: Variability of mitigation technologies

Geothermal>H. Minihydro	**Embalse>Nuclear**	**Geotherma>Minihydro**	**Biomass>Wind**

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97%	**86%**	**77%**	**82%**

Source: Own elaboration

6.3.1 SENSITIVITY **ANALYSIS OF THE VARIABLES**

Annex A.16 presents the results regarding the sensitivity analysis of the variables involved in the estimation of emissions and reduction potentials for constructing the greenhouse gas mitigation curve.

One of the variables that most significantly contributes to the propagation of uncertainty to the final cost results is the technology investment value. The model is sensitive to variations in the type of technology (coal, gas, nuclear, solar, etc.), and this contributes to a greater or lesser degree of uncertainty. In the case of tidal and solar technologies, which are the least developed of those presented here, they accumulate the greatest amount of uncertainty in the analyses. This is largely due to the poor estimates of future costs for these technologies.

Another variable that is important in the uncertainty results is the price of fuels (oil, coal, etc.). It is also a variable of interest and propagates high variability due to the various (and different) cost projections and its direct correlation with the result of the different technologies.

An additional variable that accumulates and propagates significant uncertainty throughout the model is the technology's plant factor. The variability of this factor is very large and depends on the types of technologies, meteorological variables, industry investments, supply and demand, among other factors.

For more details on how the different variables behave in the final CO2 mitigation cost results, see Annex A.15.

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7 METHODOLOGICAL DEVELOPMENT AND RESULTS OF THE SECTORS OF ENERGY DEMAND .

The following exercise presents a modeling exercise for mitigation measures in the transportation, mining, and pulp and paper sectors. A group of measures for which national or international data are available has been selected. These measures will be described and characterized to determine their abatement costs, penetration potential, and uncertainty.

associated with them.

The preliminary findings show the potential of the measures studied and their relationship to the consumption of the corresponding sector. The calculation of the emissions fraction considers emissions from energy transformation avoided through electricity consumption, given that most of the measures imply greater efficiency in this energy source.

Table 14: Summary of mitigation potentials of the measures evaluated.

	Measurement	Potential	% of total sector 2020
Road Transport	Taxi Hybrid.	63.093	0,2%
	Hybrids	78.096	0,3%
	Ecodriving	37.180	0,1%
	Taxis Plug-in	97.832	0,3%
	Plug-in	87.190	0,3%
	Total	**363.391**	**1,3%**
Paper and Cellulose	Insulation	11.760	0,2%
	VSD	45.599	0,7%
	Cogeneration	347.900	5,2%
	Total	**405.259**	**6,0%**
Copper	BLIND	4016	0,0%
	MEPS	74.448	0,2%
	PRM	43.790	0,1%
	PVSD	21.247	0,1%
	Total	**143.501**	**0,4%**

Source: Own elaboration.

Additionally, a brief overview of the sector is presented to understand its structure and energy consumption.

7.1 MODELING AND ESTIMATION OF THE IMPACTS OF GAS MITIGATION MEASURES GREENHOUSE EFFECT IN THE TRANSPORT SECTOR

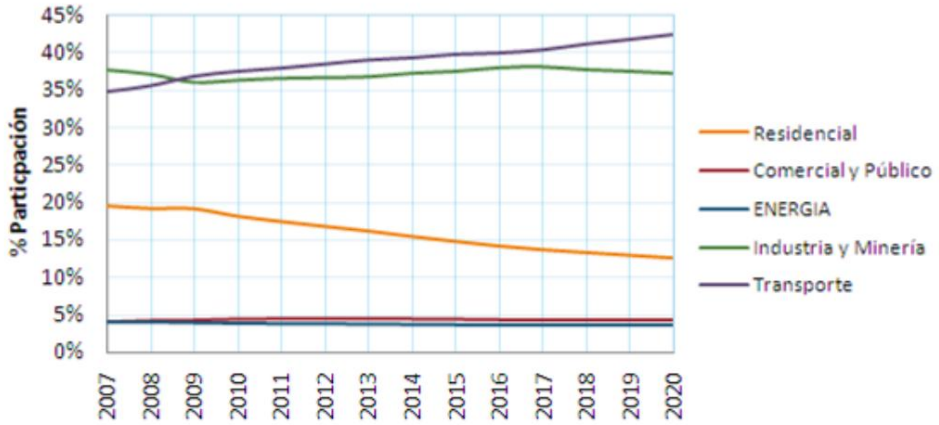
According to the IEA (2010), transportation is responsible for approximately 23% of global energy sector emissions. These emissions originate primarily from road transport, in addition to

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Maritime and air shipments are subject to international bunkering, which adds an additional difficulty for counting and assigning to countries and regions of the planet.

During 2009, the Transportation sector nearly overtook Industry and Mining as the largest energy consumer in Chile, accounting for 34.6% compared to 36.8% of the latter's final energy consumption, and projecting an even larger share in the short term, as shown in Figure 21. In this case, energy consumption is concentrated almost entirely (99%) in a single product: petroleum derivatives, which primarily include diesel and gasoline. There is also a smaller amount of electricity consumption associated with the Santiago Metro (urban electric train) and the railway, and gas consumption associated with a smaller fleet of light vehicles.

Figure 21: Sectoral participation in national energy demand.



Source: Prepared by the author based on “Implementation of the LEAP simulation tool for projecting long-term energy consumption scenarios and evaluating CO2 mitigation scenarios”, 2010. Ministry of Energy. PROGEA.

Table 15 presents the historical consumption of the transport sector by mode, from 2007 to 2010, extracted from the National Energy Balance of the CNE (National Energy Commission) for Chile.

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Table 15: Historical consumption by mode

	Oil Diesel	Oil Combustible	Gasoline Motor	Gasoline Aviation	Kerosene Aviation	Elect. Gas	Natural	Total	Total Annual
	Tcal	Tcal	Tcal	Tcal	Tcal	Tcal	Tcal	Tcal	Tcal
2007 Road 33,780 0			24.795	0	0	304	253	59.132	
Railway 208		0	0	0	0	47	0	255	86.853
Maritime 3,605		15.496	117	0	45	0	0	19.263	
Air	11	0	0	45	8.147	0	0	8.203	
2008 Road worker 35.006 2			25.682	0	2	321	144	61.157	
Railway 485		0	0	0	0	46	0	531	89.894
Maritime 3,616		14.988	49	0	0	0	0	18.653	
Air	7	0	0	47	9.499	0	0	9.553	
2009 Road worker 35.166 25			27.879	0	2	318	223	63.613	
Railway 467		0	0	0	0	45	0	512	86.027
Maritime 3,394		10.654	39	0	2	0	0	14.089	
Air	7	0	0	56	7.750	0	0	7.813	
2010 Road 37,870 809			31.482	0	0	351	197	70.709	
Railway 464		0	0	0	0	20	0	484	85.336
Maritime 1,593		5.091	0	0	0	4	0	6.688	
Air	7	0	0	84	7.364	0	0	7.455	

Source: National Energy Balance, 2010, CNE.

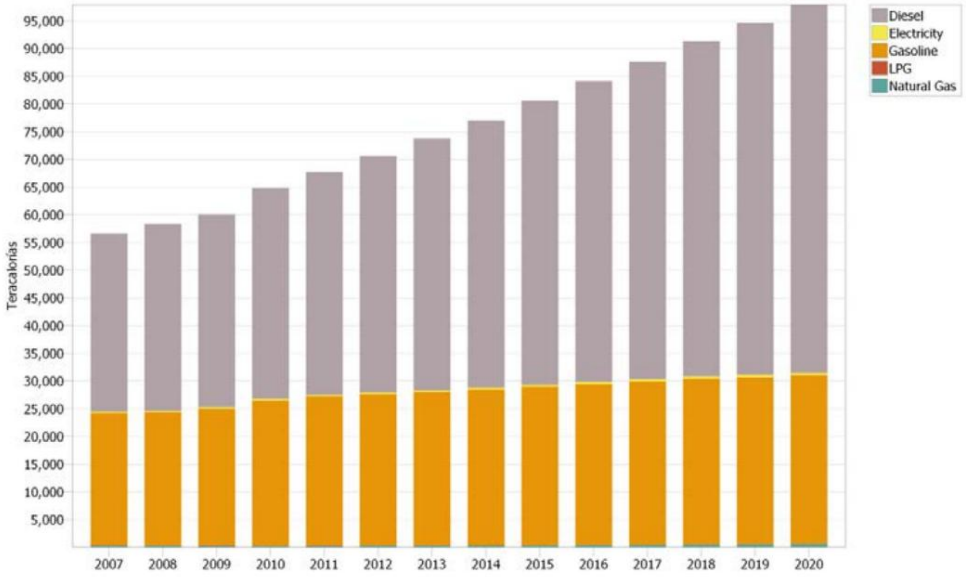
The road sector accounted for 68% of energy consumption in transportation in 2007 and 2008; however, during the last two years it has shown significant growth in its share.
reaching 74% in 2009 and 83% in 2010.

For the road transport sector, a slight upward trend in diesel consumption can be observed, while for gasoline this trend is more pronounced, with consumption increasing by 27% between 2007 and 2010. Overall, the road transport sector shows a growth of 19.5% during the last two years (2007-2010). On the other hand, the maritime transport sector shows decreasing consumption; specifically, consumption in 2010 fell by 47% compared to 2009.

The sector's consumption is projected to grow strongly. In fact, typical figures estimate average annual growth of 4-5% at least until 2020. The following figure presents an estimate of projected energy consumption for road transport up to 2020 by fuel. This projection is based on historical consumption data and does not currently consider variations or market effects of specific taxes applied to gasoline and diesel.

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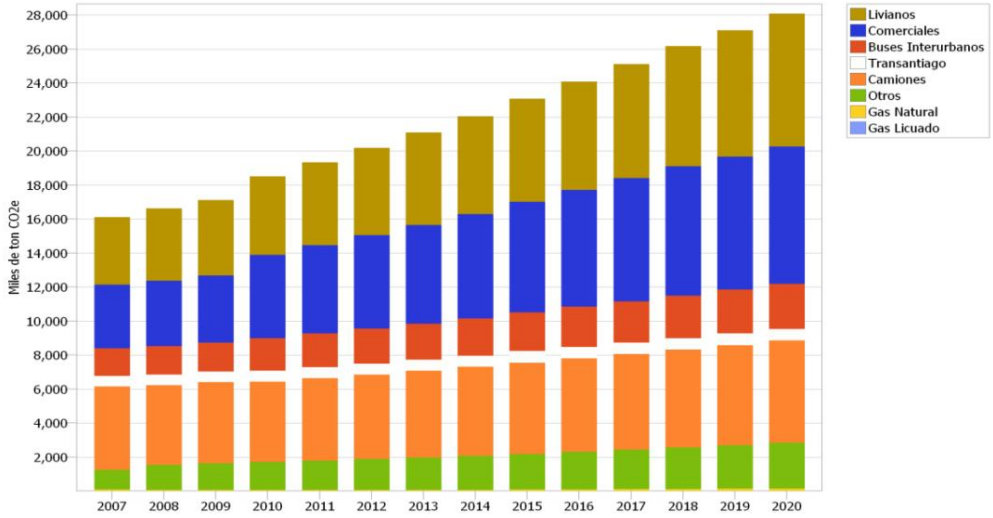
Figure 22: Projected energy consumption in Road Transport 2007 – 2020



Source: Prepared by the author based on LEAP 2011.

The following figure shows the projected emissions for the road transport sector. It indicates 28 million tons of CO2e for the year 2020.

Figure 23: Projected energy consumption in Road Transport 2007 – 2020



Source: Prepared by the author based on LEAP (2011).

7.1.1 **VARIABLES ASSOCIATED WITH** MITIGATION **MEASURES IN THE TRANSPORT SECTOR**

The transport sector is divided into road transport, air transport, and maritime transport.
Only road transport can be analyzed with a stock methodology, while the other two are modeled with an activity level methodology.

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The main variables for characterizing road transport are:

- Vehicle fleet and age (region)
- Standard associated with vehicles.
- Sales by region •

Variations in efficiency (consumption)

- Type of fuel used.
- Fuel consumption (final/useful) • Vehicle

categories (standards)

- Vehicle weight by category • Average load

by category • Average distance traveled

- Bike park
- Bicycle usage rate • Kilometers of

cycle paths

Other variables of interest include GDP growth rate, motorization, vehicle obsolescence, and population.

The main variables for characterizing the air and maritime transport sector are:

- Final/useful energy consumption according to energy type
- Type of trip (domestic/international)
- Average distance traveled

In addition, other variables of interest are required: GDP growth rates, exports, imports, and population.

Other transport sector (Metro, others).

- Consumption by energy type • Activity level

(km, passengers, other)

The opportunities for emissions reduction generally focus on trucks, passenger and commercial vehicles, and buses, given that road transport is the largest contributor to GHG emissions. These opportunities can be categorized as energy efficiency measures for trucks and buses, low-carbon technologies, reduced activity levels, and reduced carbon intensity in fuel. The following section details the GHG mitigation measures to be evaluated.

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7.1.2 **PRIVATE LIGHT VEHICLES AND HYBRID TAXIS**

Description

The measure involves requiring a certain proportion of new vehicles entering the vehicle fleet each year to have hybrid technology. Hybrid vehicles consist of two engines: one electric and one internal combustion. Compared to conventional vehicles, they significantly reduce the use of the internal combustion engine, resulting in lower fuel consumption and, consequently, lower emissions.

The driver for this measure corresponds to the number of hybrid vehicles entering the vehicle fleet annually. This number will be expressed as a percentage of the total number of new vehicles entering the fleet that year.

Additionally, the same measure is being evaluated for the replacement of shared taxis. This mode of transport has a constant fleet size, and it is estimated that from 2013 onwards, a gradual replacement of these vehicles will occur at a rate of 5% annually. This case is relevant given the increased activity level of taxis, which leads to a decrease in the cost per unit of mitigated CO2.

Background

The following studies have been considered where the effect of implementing this measure has been evaluated; the data obtained from these studies will serve as background to describe the parameters and variables of the evaluation.

1. Evaluation of Investment and Financial Flows of the Forestry, Agriculture, and Transport Sectors and Water Infrastructure Subsector, UNDP 2011
2. Cost and Potential Estimates for Reducing Greenhouse Gas Emissions Under Different Future Scenarios. Center for Global Change 2011.
3. Greenhouse Gas Emissions in Chile: Background for the development of a regulatory framework and evaluation of reduction instruments, PROGEA 2008
4. Thesis Title: “Stochastic analysis of the cost of an efficient portfolio of measures of mitigation of greenhouse gases to meet reduction targets in Chile”, Bruno Campos, 2011

Baseline

The baseline is considered to be the projection of the light vehicle fleet, which takes into account annual sales, a percentage of which will correspond to hybrid vehicles.

- The PROGEA 2008 report considers a baseline without hybrid vehicles.
- UNDP proposes a baseline penetration of 0.05% of sales in 2010, and 1.5% in 2015. year 2015, and 5.5% in the year 2020.
- CCG does not specify.

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• Campos 2011 proposes a baseline penetration of 0.05% of sales in 2010, a 0.5% in 2015, and 3% in 2020.

Penetration of the measure

• The PROGEA 2008 report considers a constant penetration of 1% of sales of light vehicles.

• UNDP proposes a penetration of the measure of 0.05% of sales in 2010, and 5.0% in 2019. year 2015, and 8.5% in the year 2020.

• CCG proposes a penetration of the measure of between 1 and 3% of sales in 2010 and between 1.5% and 4.5% in 2020.

• Campos 2011 proposes a penetration of the measure of 0.05% of sales in 2010, a 3% in 2015, and 10% in 2020.

Costs

Cost (US$)	PROGEA UNDP		CCG	Memory
Differential between a vehicle hybrid and a conventional vehicle 2011	Between 30% and 40%	5.000	7.500	5.000
Difference between a hybrid vehicle and a conventional vehicle 2020	Between 30% and 40%	5.000	4.000	3.000

*Fuel costs are calculated only once for all measures.

Other Variables

Variable	PROGEA UNDP		CCG	Memory
Activity Level (KM/year)	15.000	15.000	16.000	15.000
Taxi Activity Level (KM/year)	25.000	N/D	27.000	N/D
Conventional fuel efficiency (KM/Lt)	N/D	12	11,46	10
Hybrid Performance (KM/Lt)	N/D	19	15,62	19
Annual performance increase conventional (%)	N/D	1,39	1,39	1,39
Annual hybrid yield increase (%)	N/D	1,69	1,69	1,69
Useful life (years)	N/D	N/D	13	N/D

Mitigation potential

• The PROGEA 2008 report achieves a reduction in GHG emissions of 70,000 tons of CO2e per year 2020.

• UNDP achieves a reduction in GHG emissions of 107,000 tonnes CO2e in 2020.

• CCG achieves a reduction in GHG emissions of between 20,000 and 60,000 tons CO2e in 2020.

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- Campos 2011 achieves a reduction in GHG emissions of 245,000 tons CO2e in 2020. This value is very high compared to other references. This is due to two factors: the high level of penetration predicted for 2020 (7% above the baseline) and a lower performance used for conventional vehicles.

7.1.3 **PRIVATE LIGHT VEHICLES AND PLUG-IN HYBRID TAXIS**

Description

The measure involves requiring a certain proportion of new vehicles entering the fleet each year to have plug-in hybrid technology. Plug-in hybrid vehicles consist of two engines: an electric motor powered by the vehicle's energy recuperation system and directly through electrical injection, and an internal combustion engine. Compared to conventional vehicles, they significantly reduce the use of the internal combustion engine, resulting in lower fuel consumption and, consequently, lower emissions.

The driver for this measure corresponds to the number of plug-in hybrid vehicles entering the country.
annually to the vehicle fleet. This amount will be expressed as a percentage of the total number of new vehicles entering the fleet that year.

Additionally, the same measure is being evaluated for the replacement of shared taxis. This mode of transport has a constant fleet size, and it is estimated that from 2015 onwards, a gradual replacement of these vehicles will occur at a rate of 5% annually. This case is relevant given the increased activity level of taxis, which leads to a decrease in the cost per unit of mitigated CO2.

Background

The following studies have been considered where the effect of implementing this measure has been evaluated; the data obtained from these studies will serve as background to describe the parameters and variables
of the evaluation.

1. Evaluation of Investment and Financial Flows of the Forestry, Agriculture, and Transport Sectors and Water Infrastructure Subsector, UNDP 2011
2. Cost and Potential Estimates for Reducing Greenhouse Gas Emissions Under Different Future Scenarios. Center for Global Change 2011.
3. Thesis Title: “Stochastic analysis of the cost of an efficient portfolio of measures of "Mitigation of greenhouse gases to meet reduction targets in Chile," Bruno Campos, 2011.

Baseline

The baseline is considered to be the projection of the light vehicle fleet, which takes into account annual sales, a percentage of which will correspond to plug-in hybrid vehicles.

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- UNDP proposes a baseline penetration of 0.00% of sales in 2010, and 0.03% in 2013.
  year 2015, and 0.5% in the year 2020.
- CCG does not specify.
- Campos 2011 proposes a baseline penetration of 0.00% of sales in 2010, a
  0.03% in 2015, and 1.5% in 2020.

Penetration

- The UNDP study proposes a penetration of 0.05% of sales in 2010, and 1.5% in the following year.
  2015, and 5.5% in 2020.
- CCG does not specify.
- Campos 2011 proposes a penetration of 0.05% of sales in 2010, and 0.5% in 2011.
  2015, and 3% in 2020.

Costs

Cost (US$)	UNDP	CCG	US$ memory
Difference between a plug-in vehicle and a conventional vehicle 2011	30.000	15.000	30.000
Difference between a plug-in vehicle and a conventional vehicle 2020	30.000	-2.43% annual	15.000

*Fuel costs are calculated only once for all measurements.

Other Variables

Variable	UNDP	CCG	Memory
Activity Level (KM/year)	15.000	16.000	15.000
Taxi Activity Level (KM/year)	N/D	27.000	N/D
Conventional fuel efficiency (KM/Lt)	12	11,46	10
Electrical efficiency (KM/Lt)	19	17.25	19
Electrical consumption (Kwh/Km)	0,25	0,95*	0.25
Annual performance increase conventional	1,39	1,39	1,39
Annual performance increase of plug-in	1,69	1,69	1,69

*Data from 2007

Mitigation potential

- UNDP achieves a reduction in GHG emissions of 61,000 tons CO2e in 2020 without considering
  the measure regarding shared taxis.
- CCG achieves a unit GHG emissions reduction of between 17.3 tonnes CO2e in 2020.
- Campos 2011 does not present this information.

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7.1.4 **EFFICIENT DRIVING**

Description

The measure consists of training drivers of commercial buses and light fleet trucks in better driving practices (Eco Driving) reducing fuel consumption and therefore reducing CO2 emissions from vehicles.

Background

The following studies have been considered where the effect of implementing this measure has been evaluated; the data obtained from these studies will serve as background to describe the parameters and variables of the evaluation.

1. Evaluation of Investment and Financial Flows of the Forestry, Agriculture, and Transport Sectors and Water Infrastructure Subsector, UNDP 2011
2. Cost and Potential Estimates for Reducing Greenhouse Gas Emissions Under Different Future Scenarios. Center for Global Change 2011.
3. Greenhouse Gas Emissions in Chile: Background for the development of a regulatory framework and evaluation of reduction instruments, PROGEA 2008

Baseline

The baseline is considered to be the projection of training courses carried out as a percentage of the total number of drivers.

- UNDP considers a penetration from 0% to 13.34% by 2020.
- CCG considers a 10% penetration in the base year.
- The PROGEA 2008 report considers a 5% penetration rate in the base year.

Penetration

- UNDP proposes an additional 20% penetration of conductors above the baseline.
- CCG aims to achieve a penetration of 43% by 2020 (In 10 years).
- The PROGEA 2008 report considers a growing penetration of up to 15% of the drivers as of 2020.

Costs

Cost	UNDP	CCG	Prog
Training costs (US$)	126	148	N/D

*Fuel costs are calculated only once for all measures.

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Other Variables

Variable	UNDP	CCG	Prog
Activity Level of Buses (KM/year)	65.000	N/D	50.000
Truck Activity Level (KM/year)	70.000	27.000	60.000
Bus Performance (KM/Lt)	2,73	N/D	2,67
Truck Fuel Efficiency (KM/L)	3,5	8,33	4,84
Reduction in consumption through training	4%	From 4% to 0% in 5 years	10%

Mitigation potential

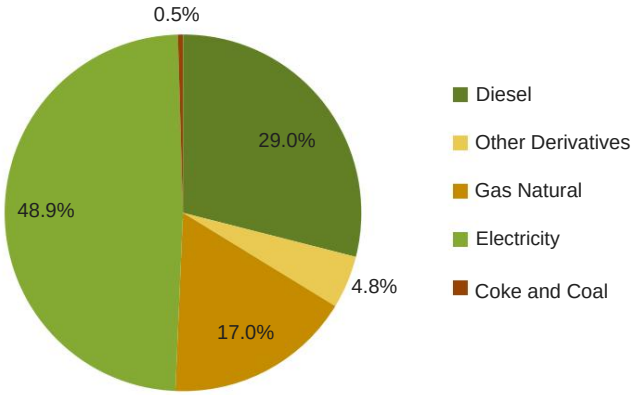
- UNDP achieves a reduction in GHG emissions of 66,000 tons CO2e in 2020 without considering the measure regarding shared taxis.
- CCG achieves a reduction in GHG emissions of 133,000 tonnes CO2e in 2020.
- Campos 2011 achieves a reduction of GHG emissions of 58,500 ton CO2e in the year 2020.

7.2 MODELING AND ESTIMATION OF THE IMPACTS OF GAS MITIGATION MEASURES

GREENHOUSE **EFFECT IN THE COPPER SECTOR**

Energy consumption in copper mining grew at an average annual rate of 6.1% between 2000 and 2010, increasing from 18,000 teracalories in 1990 to over 33,000 in 2010. According to the National Energy Balance (BNE), the main energy sources consumed in 2010 were electricity, diesel, and natural gas. The following figure illustrates the sector's energy consumption breakdown.

Figure 24: Composition of Energy Consumption of the Copper Sector-2010



Source: Prepared by the author using data from BNE 2010. Ministry of Energy.

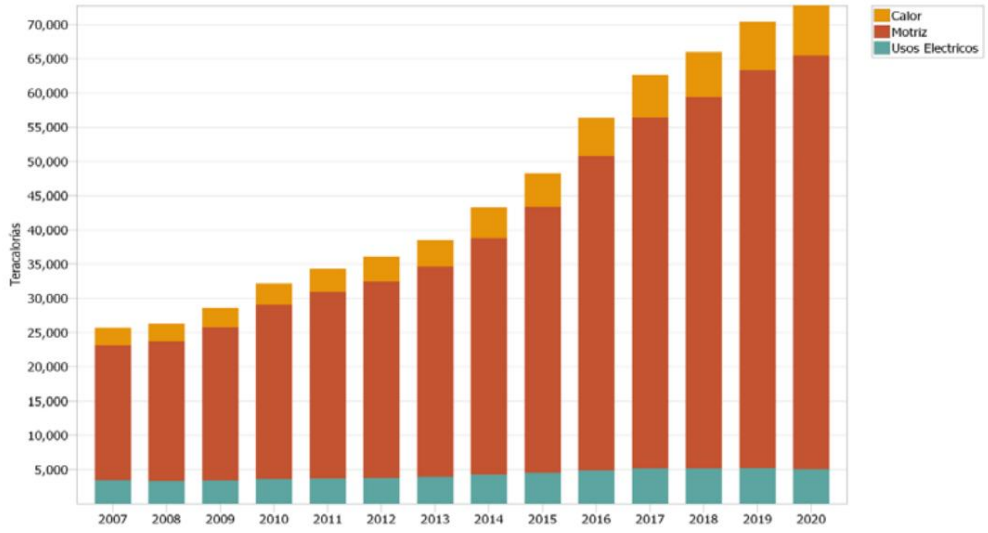
It is important to note that 2010 shows a significant difference compared to previous years in the importance of natural gas. During the years 2000-2009, the share of natural gas was consistently low.

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8%. The years 2007-2009 even fell below 3% due to a shortage of imports from Argentina. Therefore, 2010 shows a change that, if it continues, could be considered structural in the composition of consumption.

The sector's consumption is projected to grow strongly. In fact, typical figures estimate average annual growth of 6-8% at least until 2020. The following figure presents an estimate of projected copper energy consumption by use until 2020. Specifically, electricity consumption for 2020 is projected to reach 42 TWh.

Figure 25: Projected energy consumption by copper use in Chile 2007 – 2020



Source: Prepared by the author based on LEAP 2011

7.2.1 **DEFINITION OF ENERGY EFFICIENCY MEASURES**

In the copper sector, the main mitigation measures are related to energy efficiency in electricity consumption. Therefore, emission reductions will be indirect and linked to emission reductions in the electricity generation sector.

The measures are chosen as originating from the regulator and with specific goals. This reduces their complexity and makes modeling less difficult.

The measures to consider are:

1) Investment program for efficient engines (PIME).

[16] These estimates typically use the project portfolio, both existing and new, of the major mining companies as input. Cochilco periodically estimates copper production for the next ten years.

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It is assumed that the regulator sets a goal that 20% of new engines entering the fleet in 2013 and 2014 will be efficient (and where it is assumed that they would not be without the Program)17.

Only engines with power outputs less than 100HP18 are considered.

2) Engine Replacement Program (PRM).

The investment cost is the cost of the new engines minus the residual value. We can assume this value is in the order of 20-30% of the efficient motor value.

3) Motor Efficiency Standards (MEPS).

It is considered that in 2014 a regulation was established that requires that all engines (with power less than 100HP) entering the fleet be efficient.

4) Program to Promote Frequency Converters when new motors are introduced (i.e. efficient motor + VDF). (PVSD):

-2013: 20% of new engines with less than 100 HP will be efficient + VSD

-2014: 25% of new engines with less than 100 HP will be efficient + VSD

-2015: 30% of new engines with less than 100 HP will be efficient + VSD

It should be noted that this case presents the methodology with the most extensive data development and processing of available information, and it is used as a benchmark for the type of analysis that can be performed. This is not feasible for the other measures, given the lack of information and the complexity and resource demands involved. Therefore, they will be evaluated using somewhat simpler methods, taking advantage of the knowledge gained from these methodologies.

7.2.1.1 Methodology for Estimating Cost Effectiveness per Measure

To quantify the electricity and cost savings of the measures, the motor fleet is estimated up to the year 2020 by power category. The following table shows the total stock of motors by power range for copper mining for the years 2010, 2015, and 2020.

Table 16: Total Engine Fleet by Power Range (HP) 2010-2020

Power (HP) 2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020				
0 -5	53,782	57,215	60,012	63,766	71,355	77,772	83,793	89,806	91,154	93,724	96,092			
5-20	18,854	20,057	21,038	22,354	25,014	27,264	29,375	31,483	31,955	32,856	33,686			
20-50	10,080	10,724	11,248	11,952	13,374	14,577	15,705	16,833	17,085	17,567	18,011			
50-100	5,573	5,929	6,219	6,608	7,395	8,060	8,684	9,307	9,446	9,713	9,958			
100-200	3,472	3,694	3,875	4,117	4,607	5,021	5,410	5,798	5,885	6,051	6,204			
200-500	2,338	2,487	2,609	2,772	3,102	3,381	3,643	3,904	3,963	4,074	4,177			
500-1000	1,112	1,183	1,241	1,319	1,476	1,608	1,733	1,857	1,885	1,938	1,987			

[17] Strictly speaking, the goal would be to achieve an additional 20% of efficient engines.

[18] PRIEN 2006-2010 indicates that for power outputs above 100 HP, investments in efficient engines often yield results that are not cost-effective. Therefore, the programs have focused on power outputs within the range considered here.

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plus 1000	154	164	172	183	205	223	240	258	261	269	276

Source: Own elaboration.

Net park growth is also required, as shown in the following table.

Table 17: New Engine Entry 2011-2020

Power Range 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020										
0 -5	3,433 2,797 3,754 7,589 6,417 6,021 6,013 1,348 2,570 2,368									
5-20	1,203 981 1,316 2,660 2,249 2,111 2,108 472 901 830									
20-50	643 524 704 1,422 1,203 1,129 1,127 253 482 444									
50-100	356 290 389 786 665 624 623 140 266 245									
100-200	222 181 242 490 414 389 388 87								166 153	
200-500	149 122 163 330 279 262 261 59								112 103	
500-1000	71 58 78			157 133 125 124 28					53	49
but 1000	10 8		11	22	18	17	17	4	7	7

Source: Own elaboration.

Annex A.11 presents the methodology for estimating motors by power range.

The energy savings resulting from (1) using an efficient motor instead of a conventional one and (2) using an efficient motor with a variable frequency drive (VFD) instead of a conventional one are also estimated (a distinction is made here for pump motor and belt-driven applications). The following table presents these
annual savings.

Table 18: New Engine Entry 2011-2020

Range of Power (HP)	***Motor Efficient***	***Efficient Motor + VDF Pumps***	***Efficient Motor +VDF Straps***
5	1.2	4.0	1.7
5-20	3.2	19.4	8.3
20-50	5.7	50.9	21.8
50-100	8.8	93.7	40.1
100-200	4.8	185.6	79.5
200-500	16.0	431.1	184.8
500-1000	44.7	920.0	394.3
but 1000	297.9	6133.5	2628.6

Source: Own elaboration

These savings are valued at the free mononomic price of electricity or average market price, projected according to official variables and parameters used in the country's Node Price studies19.

[19] Reference node price report for projections APRIL 2011

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For each level of average power, average investment values are considered according to the following table:

Table 19: Engine Investment Values (US $)

Standard Medium Power		Efficient Efficient+VDF	
3	430	456	1,288
15	1,495	1,581	4,286
40	3,210	3,724	9,261
75	7,104	8,728	17,192

Source: Prepared by the author based on PRIEN 2007 and Greenlab UC 2011.

To estimate the investment values of a measure, the interest value is the difference between the value of an efficient engine versus a conventional one in cases where investment is made in an efficient engine. That is, this difference applies to the following measures: Investment program, and efficiency standards in engines.

Similarly, the difference between efficient motors considering VDF and conventional motors is applied to the VDF promotion measure.

Finally, in the case of early replacement, only the value of the efficient engine is used, but considering a residual value of 20% for the conventional engine (and therefore with respect to the value of the conventional engine).

7.2.1.2 Engine Fleet Estimation Methodology

The methodology used in the mining sector requires estimating the current fleet of engines and its growth. This methodology is a simplified version of that developed by PRIEN (2007, 2011).

According to PRIEN (2007), the electricity consumption of motors in the copper industry in 2004 reached almost 11,000 GWh. The following table shows the consumption and its composition according to the power range of the motors.

Table 20: Consumption of Motors in the Copper Industry - 2004.

Power (HP)	Average Power Consumption of Engines		Total
5	3	**415**	3.8%
5-20	15	**727**	6.7%
20-50	40	**1,036**	9.6%
50-100	75	**1,074**	9.9%
100-200	150	**1,339**	12.4%
200-500	350	**2,103**	19.4%
500-1000	750	**2,144**	19.8%
but 1000	5000	**1,982**	18.3%
Total	that	**10,821**	100.0%

Source: PRIEN (2007).

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According to PRIEN, for each power range, the number of engines that make up the fleet is given by:

$$= \frac{}{\ddot{y} \times 0.746 \times \times \ddot{y}}$$

Where:

: Number of engines.

ÿ ÿ: Average power of the power range considered (see previous table).

E: Annual electricity consumption by the motors (in kWh/year units).

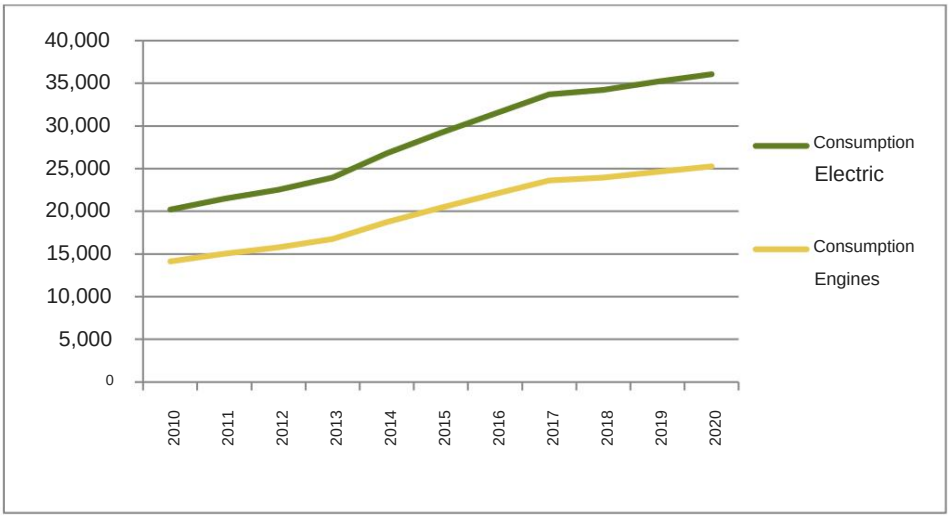
FC: Load factor. A value of 0.6 is considered as PRIEN (2007) does.

h: Hours of continuous use of the motors. As PRIEN (2007) does, this value is assumed to be 7500 hours.

To calculate the current motor fleet and make projections, we used the equation above. For the base case, we used the electricity consumption projection for the copper industry developed by PROGEA (2011) for the Ministry of Energy. We also used the precedent that approximately 70% of the electricity consumed in the sector is associated with the use of motors.

The following figure presents the projection of total energy consumption and motor consumption considered up to 2020.

Figure 26: Total Electricity and Motor Consumption for the Copper Industry 2010-2020.



Source: Own elaboration

To break down motor consumption by power range, it is assumed that the consumption structure remains unchanged and equal to that presented in the table.

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7.2.1.3 Estimation of Unit Savings

The following are the savings obtained by using an efficient motor over a conventional one and, (2) by using an efficient motor with a frequency converter instead of a conventional one.

- **Use of Efficient versus Conventional Motor**

The savings obtained by using an efficient engine versus a conventional one are estimated according to the following relationship:

$$= \times 0.642 \times \ddot{y} \times \times \ddot{y}1 \ddot{y} \qquad \frac{1}{2 \ddot{y}}$$

Where:

A: Annual Savings (in Mcal/year).

Q: Engine power.

: Engine efficiency (1 for conventional, 2 for efficient).

- **Use of Efficient Motor with Variable Frequency Drive (VFD) versus conventional.**

In this case, the savings, for each power range, are defined by:

$$= \frac{\times 0.642 \times \ddot{y} \times \times}{}$$

Where *Sr* corresponds to the percentage savings obtained by using frequency converters in a motor that is used r.

Two categories of uses are distinguished for which the savings of a motor vary. The first case is for pumps and fans, where a percentage savings of 35% is considered *(Spumps) ,* and another savings for air compressors, conveyor belts, and others, with a percentage savings of 15% *(Scomp).*

Savings are obtained for each power range using the average power.

[20] Both values are obtained from the study “Preliminary Estimate of the Potential for Efficiency in the Use of Electrical Energy to Supply the Central Interconnected System”, PRIEN 2008 correspond to 35 and 15% respectively.

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7.3 MODELING AND ESTIMATION OF THE IMPACTS OF GAS MITIGATION MEASURES

GREENHOUSE **EFFECT IN THE PAPER AND PULP SECTOR**

In Chile, there are currently twelve pulp mills operating a total of 17 production lines. Of these, 11 production lines manufacture Kraft pulp, and the rest produce pulp.
mechanics.

The country's pulp production reaches 5,500,000 tons, with an approximate energy consumption of 200 kWh for each ton of pulp produced [Minenergia, 2010].

This sector has experienced an average annual growth of 7.5% over the last 15 years and an average annual growth of 7.2% over the last 4 years. Its total energy consumption in 2010 was 13,404 teracalories. The main energy sources used in the pulp and paper industry that year were primarily firewood and electricity.

The evolution of industrial consumption is concentrated in the subsectors of Miscellaneous Industries, Pulp and Paper, and Steel. Of these, Pulp and Paper shows the greatest dynamism, while Steel does not show significant increases during the analyzed period. The large increase in consumption in the pulp sector may be due to the start of operations at Nueva Aldea (Celulosa Arauco) in 2006, producing 85,600 tons, and the expansion of Santa Fe (CMPC) in 2007, which increased its production by 780,000 tons.²¹ Furthermore, exports of Pulp and Paper increased from US$1.65 billion to US$2.616 billion FOB (approximately 60%) between 2005 and 2009.²²

According to the National Energy Balance, the following table shows energy consumption for the last four years. The significant reduction in firewood consumption in 2010, related to the implementation of cogeneration systems, is evident.

Table 21: Energy consumption in the Paper and Pulp Sector (in Teracalories)

TCal	Oil Diesel	Oils Liquefied Fuels	Gas	Electricity Coal		Gas Natural Other	Firewood and	Total Energy
2007	53	2,332	165	4,586	4	904	11,376	19,420
2008	104	2,260	190	4,597	0	18	12,390	19,559
2009	106	2,426	142	4,465	48	545	12,161	19,894
2010	119	1,199	58	3,767	0	976	7,285	13,404

Source: National Energy Balance

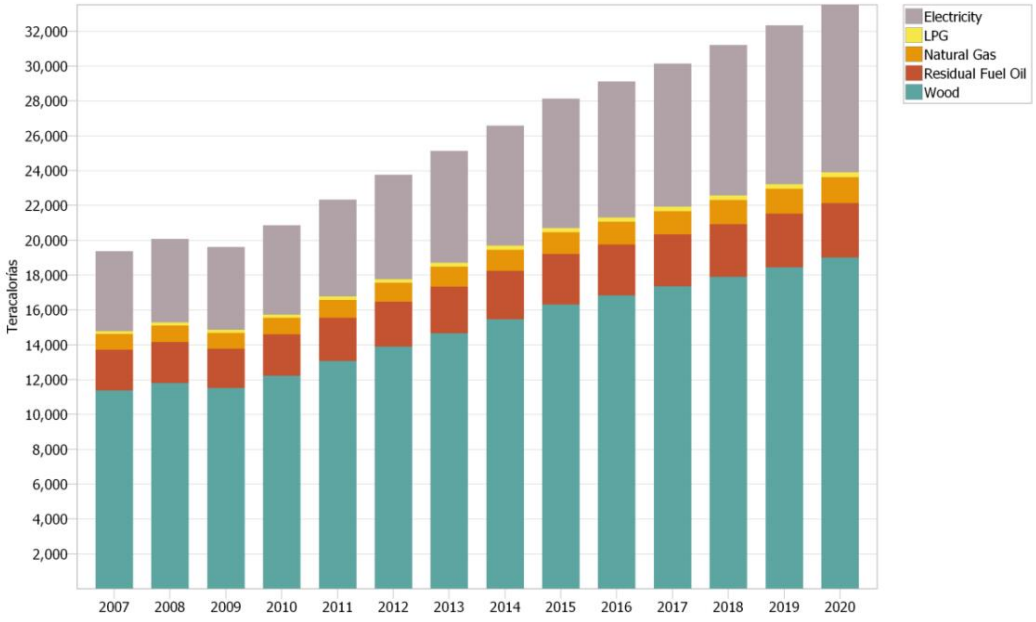
The energy consumed in industrial processes is primarily used as steam. Furthermore, the sector's largest share of electricity consumption is for motors, traction, and power.

21Fuente: “Latin American Markets”.
22Source: Central Bank of Chile.

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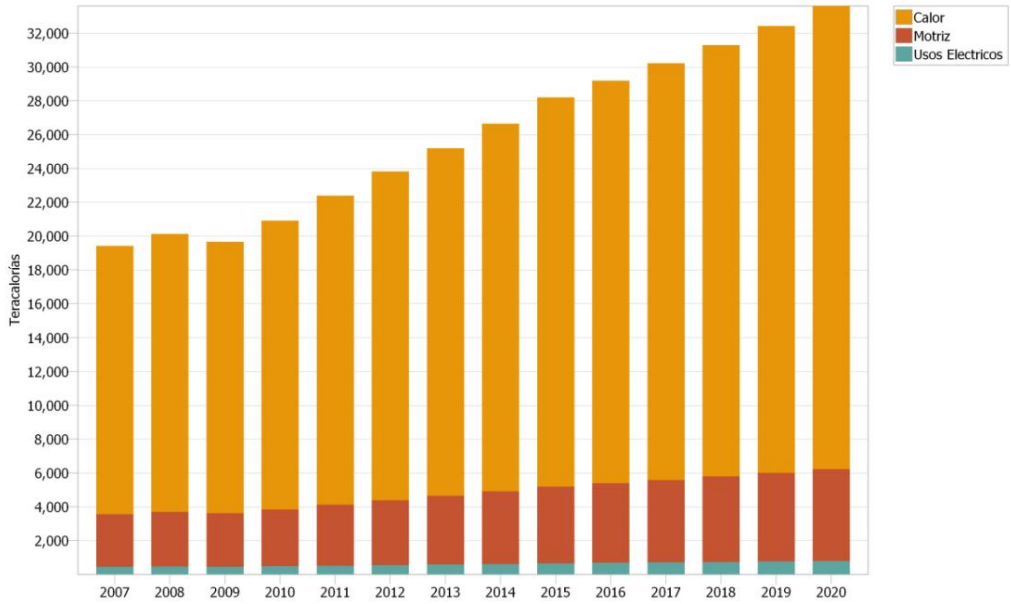
Consumption in the sector is projected to grow strongly. In fact, typical figures estimate average annual growth of between 6-7% between 2010 and 2015, and between 3-4% between 2015 and 2020.
The following figures present an estimate of projected energy consumption for the pulp and paper industry up to 2020, broken down by fuel and use. Specifically, the sector's expected electricity consumption by 2020 is projected to reach 11.2 TWh.

Figure 27: Projected energy consumption, Chilean paper and pulp industry 2007 – 2020



Source: Prepared by the author based on LEAP (2011).

Figure 28: Projected energy consumption by use, Chilean paper and pulp industry 2007 – 2020



Source: Prepared by the author based on LEAP 2011.

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Regarding the main sources of greenhouse gas emissions, basically three types of emission sources can be distinguished.

First, the consumption of fossil fuels required in the various stages of the production process, such as: boilers or gas turbines for power production, recovery boilers, incinerators, lime kilns and cooking, waste gas scrubbing or dryers powered by fossil fuels.

Secondly, the electricity consumption for the operation of the company, due to the automation of the process and technological development in general.

Finally, emissions can be generated from organic process waste, either during wastewater treatment or disposal in a landfill.

The mitigation measures that are mainly mentioned in the literature are grouped into the following:

- Improve the energy efficiency of the process
- Use of cleaner fuels and reuse of biomass generated by the process
- Use of cogeneration to meet heat and electricity requirements.

Improve the energy efficiency of the process

This group of measures includes:

- Installation of fuel economizers in boilers.
- Changing and adjusting burners in boilers.
- Improvement of vapor insulation and condensate lines.
- Optimization of the cellulose manufacturing process.
- Optimization of evaporation, burning and causticization.
- Optimization of the pasta preparation process.
- Improvement of the overall efficiency of the factory.

Use of cleaner fuels and reuse of biomass generated by the process

This group of measures includes:

- Emissions from biomass are considered neutral due to their biogenic origin
- The industry has great potential for the reuse of biomass as fuel, since it generates waste of organic origin.
- A double benefit obtained through the use of biomass: the reuse of a waste product and the replacement of a fossil fuel.

Use of cogeneration to meet heat and electricity requirements.

This group of measures includes:

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- Combined heat and electricity production to reduce the loss of both
- Reduction in fuel use and consequently in emissions to the atmosphere.
- This is a well-known and developed technology that can be applied to factories
  Existing and new. No significant problems have been detected in its use.
- Energy requirements and the heat/electricity ratio in the paper and manufacturing industries
  Cardboard is the most appropriate material for cogeneration.
- Emissions per unit of heat or electricity are considerably reduced as
  result of greater thermal efficiency.
- Overall thermal efficiencies can reach 93%, thus reducing the carbon dioxide emission rate by approximately 50%,
  compared to conventional systems.

The following is an evaluation of three measures that will be used to estimate the cost of mitigating GHG reduction opportunities in this sector.

In this industry, specific consumption in Chile decreased from 0.6 to 0.3 Tep/ton between 1990 and 2004. Internationally, yields of around 0.2 Tep/ton have been achieved. It is estimated that measures of this type could lead to improved yields, reaching consumption levels of around 0.2 Tep/ton. Over 10 years (by 2021), the potential annual reduction would reach 2.2% (PRIEN, 2008).

In the pulp and paper industry, the real opportunity lies in achieving effective energy integration and optimizing the use of process water (which also translates into reduced energy consumption). The introduction of cogeneration systems, one of the main measures being evaluated, depends on the heat supply and demand curves, which must first be optimized before designing the cogeneration system (i.e., its size). In the case of pulp, cogeneration is almost always feasible and profitable, but in principle, it is not beneficial (the project's profitability is lower). What should be done are energy integration development projects rather than just cogeneration projects. Water use is also an area that is not well addressed, and significant energy savings can be achieved through this approach (PRIEN, 2008).

7.3.1 **EFFICIENT COGENERATION SYSTEMS**

Description

The measure being evaluated is the implementation of efficient cogeneration systems for the production of steam, hot water, and electricity by burning liquors, bark, and other wood waste23.

[23] CANMET Energy Technology Center – Varennes

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Background

The following studies have been considered where the effect of implementing this measure has been evaluated. The data obtained from these studies will serve as background information for describing the parameters and evaluation variables.

1. Available and Emerging Technologies for Reducing Greenhouse Gas Emissions from the Pulp and Paper Manufacturing Industry. EPA (2010).
2. PPEE (2010). “Market Study of Energy Efficiency in Chile”, Country Program of Energy Efficiency. September 2010
3. CP/RAC (2012). Regional Activity Center for Cleaner Production (CP/RAC). www.cprac.org.

Baseline

The baseline used is the projected historical trend in energy consumption for the Pulp and Paper sector, obtained from the PROGEA study (2008). This projection does not consider any energy consumption reduction measures and only reflects market fluctuations. The data compiled in this study indicate that the industry's growth trend is expected to stagnate after 2015 due to the stagnation in forest planting.

Penetration of the measure

- In EPA (2010), a project to install a combined cycle gas turbine system was reviewed with the aim of ensuring the company's financial viability given the marked increase in energy prices. It corresponds to a 92 MW plant.

- The CP/RAC case considers a project with 365 GWh of energy produced by the cogeneration plant and 207 GWh of energy consumption, resulting in a surplus of 158 GWh that can be sold. The primary energy savings is 15%.

- In PPEE (2010), the installation of new cogeneration plants is evaluated over time, increasing the installed capacity by 80 MW by 2015 (4 plants of 20MW) and by an additional 100MW by 2020 (5 plants of 20MW).

Costs

Investment costs, as well as operation and maintenance costs, have been compiled according to the installed capacity of the implemented project. Additionally, one of the projects had a history regarding investment amortization.

Cost	**CP/RAC**	**PPEE**	**EPA**
Investment	30 million Euros 15.43 MMUSD		75 million US$
Installed Capacity (MW) 41.5		20	92
Operation and maintenance N/A		0,0148 US$ /kWhe 0,01	US$/kWhe
Amortization	between 8 and 10 years	N/D	N/D

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Mitigation potential

- The CP/RAC and EPA cases do not consider an emissions reduction, but this will be estimated using the SIC electricity generation emission factor. In the first case, the reduction is 365 GWh generated (158 GWh sold to the grid), and in the second, 75 GWh.
- In the PPEE study (2010), the reduction of emissions is linked to lower electricity consumption, even though fuel consumption increases on the industrial process side.
  These emissions are lower than the reductions from electricity (considering natural gas as the base fuel). This represents a unit emissions reduction of 0.169 tCO2e/kWh (electricity)-year. This translates into the following emissions reduction target for the additional generation of electricity.

Mitigation Potential	**CP/RAC**	**PPEE**	**EPA**
Electricity Generation (GWh)	356	142	75
GHG emissions (T CO2e)	174.440	69.580	36.750

7.3.2 **FREQUENCY CONVERTERS IN PUMPS**

Description

The measure involves installing variable frequency drives on potable water pumps and pumps requiring pressure control in paper manufacturing. In this case, the electrical energy consumed decreases by reducing the flow rate, instead of dissipating energy in a valve.

Pumps represent a significant portion of the electricity consumption of motor-driven systems in the pulp and paper industry. Pumps are used to pressurize and circulate water, process chemicals, and pulping mixtures, as part of the

Paper and pulp manufacturing process. Energy costs and operating and maintenance costs are the most significant components of the lifetime costs of a pumping system. Therefore, the design optimization of a new pumping system should focus on optimizing life cycle costs [EPA, 2010].

Background

The following studies have been considered where the effect of implementing this measure has been evaluated.
The data obtained from these studies will serve as background information to describe the parameters and variables of the evaluation.

1. Available and Emerging Technologies for Reducing Greenhouse Gas Emissions from the Pulp and Paper Manufacturing Industry. EPA (2010).
2. “End uses and energy conservation supply curve in the industrial and mining sector of Chile.” Prepared for the National Energy Efficiency Program, Ministry of Energy by the Federico Santa María Technical University, Energy Innovation Center.
   September 2010.

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Baseline

The baseline is considered to be the projection of the historical trend of energy consumption in the Paper and Pulp sector, obtained from the PROGEA study (2008).

Penetration of the measure

- In the case of the Minenergía study, the energy saved by adjusting the flow to the required level is 10%. The specific consumption is 1,034 kWh/ton, with an annual production of 5,500,000 tons/year. The study assigns a penetration percentage of 76%.
- In the case of the EPA study, savings of 72,000 MWh/Year are estimated.

Costs

The estimated unit cost for the 1,800 existing Variable Frequency Drives is US$1,000, implying an investment of US$2,880,000. An additional savings of US$6,824,214 is calculated.
year.

In the case of the EPA study, only the added values for investment and energy savings were obtained, which are summarized in the following table.

Cost	**EPA**	**Minenergía**
Investment	US$ 3.200.000	US$ 2.880.000
Saving	72,000 MWh/Year	US$ 6.824,214 per year

Mitigation potential

- Mining and energy. An estimated savings of 68,242 [MWh/year] is expected, which corresponds to 48% of the potential savings in the electricity sector.
- The EPA study shows a reduction of 72,000 MWh/Year.

Mitigation Potential	**Minenergía**	**EPA**
Electricity Generation (MWh)	68.242	72.000
GHG mitigation (T CO2e)	33.438	35.280

7.3.3 **IMPROVE INSULATION**

Description

The measure consists of improving the insulation of the steam and condensate lines, in order to reduce heat transfer between equipment/lines and the environment.

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Background

The following studies have been considered where the effect of implementing this measure has been evaluated. The data obtained from these studies will serve as background information for describing the parameters and evaluation variables.

1. Available and Emerging Technologies for Reducing Greenhouse Gas Emissions from the Pulp and Paper Manufacturing Industry. EPA (2010).
2. “Energy Efficiency Improvement and Cost Saving Opportunities for the Pulp and Paper Industry”. An ENERGY STAR® Guide for Energy and Plant Managers. Sponsored by the U.S. Environmental Protection Agency. Octubre2009.

Baseline

The baseline is considered to be the projection of the historical trend of energy consumption in the Paper and Pulp sector, obtained from the PROGEA study (2008).

Penetration of the measure

Two applied cases have been found in the literature, and it is estimated that the overall industry savings will be between 6 and 26%.

Costs

The costs obtained from the EPA study (2010) are summarized below.

Cost	EPA 1	EPA 2
Investment	US$ 69,280	US$ 25,000
Savings	US$ 138,000	US$ 80,000

Mitigation potential

According to the EPA study (2010), the total energy savings amount to approximately 63,000 MMBtu, with savings exceeding US$138,000. This implies an implementation cost of US$69,280, with a payback period of six months. In another scenario, which considers insulation repair, annual energy savings of US$80,000 are achieved at a repair cost of around US$25,000. The payback period for insulation repair is four months.

Mitigation Potential	EPA 1	EPA 2
Electricity Generation (MWh)	18.460	10.695
GHG emissions (T CO2e/year)	9.045	5.240

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7.4 GENERIC METHODOLOGY FOR CALCULATING REDUCTION COSTS

Costs are evaluated as annualized payments for the cost of capital and operating and maintenance expenses. This means that a constant annual cash flow is considered over a fixed period of years equal to the project's duration or the asset's useful life. This cash flow represents the constant payment resulting from distributing the project's present value over the n years of the project/ asset's life.

Then the annualized cost is the sum of the annualized investments and operating expenses and annualized maintenance.

$$CAE = Annualized\ I + Manualized$$

For its part, the annualized investment corresponds to the product between the annualized value of the unit investment incurred in each year, and the penetration or number of investments that were made during the same year:

$$I_{annualized} = NPV(Ii \times Pi) \cdot FRC$$

$$FRC = \frac{r(1+r)^{n}}{(1+r)^{n}-1}$$

With

Where:

I_i : Unit investment value in year *i.*

FRC : Capital Recovery Factor. Weighting a cost by this factor yields the annualized cost.

Pi : Additional penetration of the unit stock or investment in year i.

r Discount rate.

n : Number of project periods, or useful life of the asset.

The calculation of annualized operating and maintenance expenses is similar to the case of investment, with the exception that each year of investment or execution generates different flows throughout the useful life of the project; in turn, these flows are a consequence of the accumulated stock of the mitigation action or measure.

$$OManualizado = VAN(FOMi) \cdot FRC$$

FOMi : Corresponds to all operation and maintenance flows triggered by an action executed in year i.

Abatement costs are defined as the incremental cost of measures compared to the baseline scenario, evaluated as monetary units per tonne of CO2 equivalent avoided, where available capital is not considered a constraint.

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Then, the abatement cost is given by the following formula, which represents the difference between the cost of reducing one ton of CO2, if a mitigation option is developed, versus the
BAU scenario.

$$= \frac{\qquad\qquad\qquad\qquad\qquad\qquad}{} \quad [\$ÿ \quad 2]$$

The difference in emissions between both scenarios is defined as the reduction potential of the mitigation option.

7.5 RANDOM VARIABLES

The random variables used in the different sectors of analysis are described below.

considering the ranges of the reference studies. The list excludes the description of the random variables developed in the first stage of this study associated exclusively with the transformation sector; these are all the technical and cost parameters of the different generation technologies, along with the price of oil, coal, and biomass for the years 2020 and

2035.

In general, correlations were not defined in the variables except for the prices of the same fuel in 2 different periods, this is the case of oil, coal and biomass where the value of the year 2020 has a correlation factor of 0.6 with the value for the year 2035, the latter in order to ensure that the projection ensures some degree of coherence in the medium and long term trend of the price of fuels.

Transport sector

The variables considered random in the sector are:

- The future price of the technologies, which has a downward trend driven primarily by the eventual drop in the price of lithium batteries and the generation of economies of scale. Based on the various projections from the analyzed studies, a triangular distribution is defined for these variables.

- Vehicle activity levels and performance, which, although measurable technical variables, must be extrapolated to the entire fleet, adding uncertainty to their actual average value. The parameters of the triangular distributions are obtained based on the values from the various studies.

- Reduction in consumption and training costs are variables that vary depending on the individual being trained, and, in the case of training costs, on factors such as group size, geographic region, etc. Random distributions are assumed for these variables, based on the values presented in relevant studies.

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Table 22: Random variables in the transport sector.

Transport Variables	Average value	Min	Max
Additional cost Hybrid 2020 US$	4000	3000	5000
Additional cost Plug-in 2020 US$	13500	12000	15000
Light Activity Level KM/year	15300	15000	16000
Activity Level Taxis KM/year	26000	25000	27000
Conventional yield KM/lt	11,46	10	12
Hybrid Performance KM/lt	17,5	15,62	18,5
Plug-in Performance KM/lt	28,5		
Plug-in Performance and 2020 KM/KWh	5	4	5,2
Reduction in consumption due to training %	4%	2,0%	5,0%
Activity Level Buses KM/year	60000	50000	65000
Activity Level Trucks KM/year	60000	55000	70000
Performance Buses KM/lt	2,7	2,3	2,9
Truck Performance (KM/lt)	4	2,5	4,5
Training Cost US$	137	120	154

Source: Own elaboration

Copper sector

The variables considered random in the sector are:

- The price of conventional and efficient engines; although there is a price estimate, this variable is subject to external factors that can distort the value, such as commodity prices, shipping distances, etc. Therefore, it is considered that prices can range from 10% lower than the estimated value to 20% higher, with a triangular distribution centered on zero dispersion.

The additional efficiency gained from technological replacement can vary depending on the efficiency of the conventional engine, potentially resulting in a greater gain than predicted. Furthermore, efficiency increase calculations are performed under specific, appropriate conditions that allow for maximizing the efficiency potential for the evaluation. Under other conditions, this efficiency gain may decrease considerably. The values presented in the following figure show the dispersion with respect to the expected efficiency gain of an engine. The minimum and maximum values indicate a higher probability that the actual efficiency will be lower (up to 50%) than the expected value, compared to a higher probability (20%) than the reference value.

- The operating time of the engines will depend on annual production variables, and therefore can be a significant source of dispersion. For this reason, it is included as a random variable to measure the effect of this factor on the uncertainty of the results.

  possible variability of usage time.

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Table 23: Random variables copper sector.

Copper variables	Average value	Min	Max
Engine price dispersion (%)	0	-10	20
Efficiency dispersion 5 (%)	0	-50	20
Efficiency dispersion 5 - 20 (%)	0	-50	20
Efficiency dispersion 20-50 (%)	0	-50	20
Efficiency dispersion 50 - 100 (%)	0	-50	20
Usage time	7500	6500	8000

Source: Own elaboration

Paper and pulp sector

The variables considered random in the sector are:

- Investment and operating and maintenance costs are considered random variables, given the dispersion of the specific values found in the studies. For this reason, the average of the values obtained from the studies, along with the maximum and minimum values, is used to generate a triangular random distribution.

- The level of cogeneration will depend on annual production variables, and therefore can be a significant source of dispersion. For this reason, the potential variability of usage time is included as a triangular random variable to measure the effect of this factor on the uncertainty of the results.

- Finally, the costs and potential savings of frequency converter and insulation measures are considered, according to the averages of the referenced studies; these values are distributed triangularly.

Table 24: Random variables in the paper and pulp sector.

Variables C&P	Average value	Min	Max
Cogeneration time (hrs/year)	7100	6200	7300
Cogeneration Investment (US$MM/MW)	0,8297	0,77	0,9
O&M cogeneration US$/KWh	12,4	10	14,8
Savings from variable frequency drives	10%	5%	15%
Cost of variable frequency drives	1.600	1.500	1.700
Insulation savings	16%	6%	24%
Insulation Cost US$/MWh saved	3,00	2,34	3,75

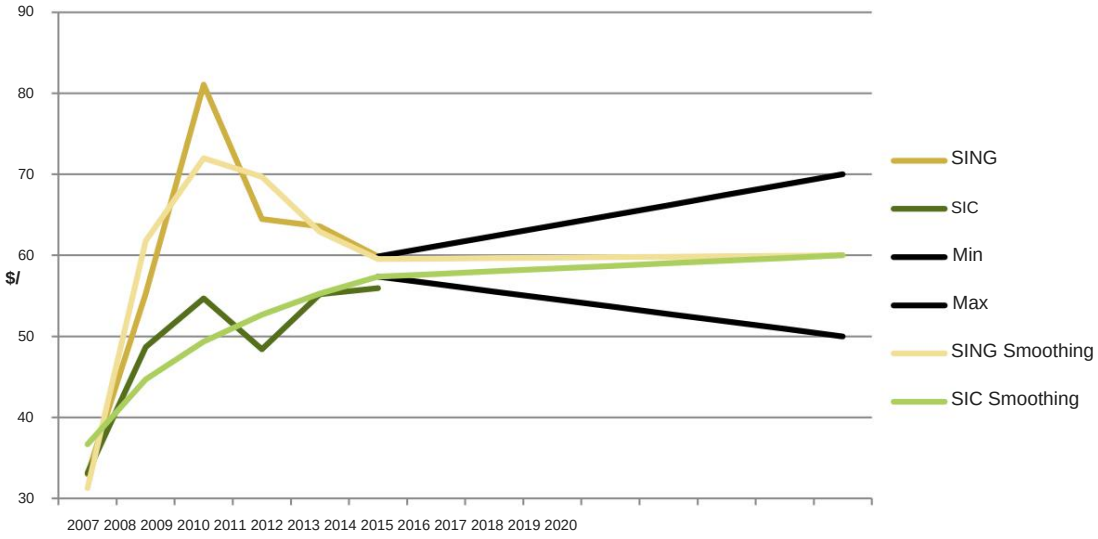
Source: Own elaboration

To evaluate these measures, the projected electricity price must be considered. For this purpose, the single-unit price is used, which is essentially the sum of the average spot price of electricity plus the quotient of the total amount of payments for available power divided by the amount of energy sold. This price is a good indicator of the average cost of energy. In our case, the annual average between 2007 and 2012 was used to generate a projection. The curve of actual data is smoothed and projected to the year 2020.

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with a single price for both Chilean electrical systems, the SIC and the SING: The following figure shows the projection of the average price with a maximum and minimum to the year 2020.

Figure 29: Projected electricity cost SIC and SING $/KWh



Source: Own elaboration

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8. CONSTRUCTION OF THE CURVE

To construct the curve, the measures are ordered from lowest to highest abatement cost, according to their average cost (the cost with 5% and 95% confidence is also presented), and are presented on the abatement potential axis, measured in Tons of CO2e24.

The table below shows the group of measures analyzed (including electricity generation measures) and the abatement curves, one that includes electricity generation measures and another that includes only those corresponding to the three energy demand sectors.

(transport, copper, and paper and pulp).

Table 25: Cost of abatement for mitigation measures in the electricity generation, transport, paper and pulp and copper sectors (r=6%).

Sector	Measurement	Moda (US$/tCO2)	5%	95% Potential (tCO2)	
Transport	Taxis Hibr.	-289,0	-323,6	-232,6	63.093
Transport	Hybrids	-276,0	-309,2	-224,0	78.096
P&C	VSD	-200,4	-236,1	-170,5	45.599
Copper	BLIND	-198,3	-228,0	-165,7	4.016
P&C	Insulation	-191,7	-224,3	-161,7	11.760
Copper	MEPS	-174,1	-205,2	-141,0	74.448
Copper	PRM	-168,8	-198,5	-136,0	43.790
Transport	Training	-143,0	-248,6	147,7	37.182
P&C	Cogeneration	-79,4	-114,2	-44,8	347.900
Copper	PVSD	-57,0	-98,8	-12,1	21.247
Transport	Taxi PI	-38,0	-83,5	-1,1	97.832
Reservoir **Transformation**		-21,2	-28,0	-15,2	8.760.000
Geothermal **Transformation**		-13,2	-20,6	-4,0	6.820.000
Mini Hydraulic **Transformation**		-10,9	-16,9	-4,8	3.000.000
Nuclear **Transformation**		13,7	3,5	24,6	6.900.000
Transport	Plug-in	22,0	-25,0	57,2	87.193
Wind **Transformation**		25,2	14,0	38,5	4.866.667
Biomass **Transformation**		79,0	61,6	102,3	1.380.000
Solar CSP **Transformation**		89,4	71,7	110,8	890.000
Solar PV **Transformation**		92,8	76,2	109,7	4.800.000
CCS **Transformation**		112,7	97,1	129,2	1.760.000
Tidal **Transformation**		152,3	134,0	181,7	200.000

Source: Own elaboration

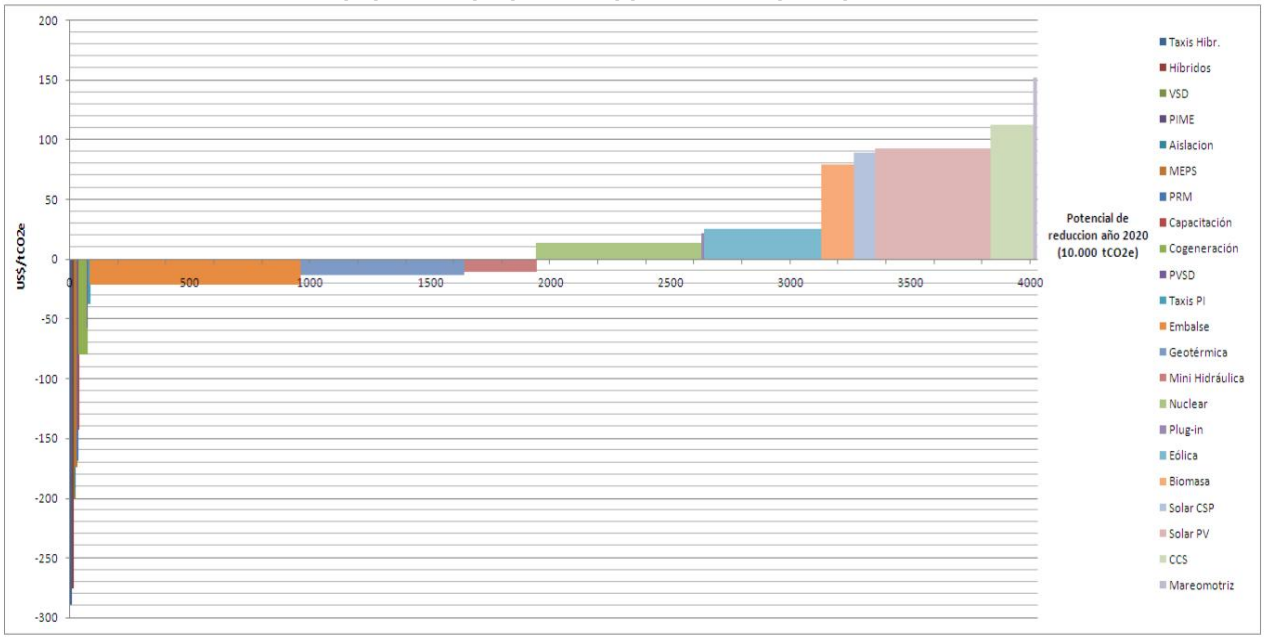
The table shows that there is a significant number of measures (with a 95% confidence level) with negative costs in all 4 sectors, which is a very important result.

[24] The final report of stage 1 of this study details the methodology for constructing the curve.

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taking into account the mitigation efforts that must be made by the countries where this methodology will be applied.

Figure 30: Abatement curve for mitigation measures in the electricity generation, transport, paper and pulp and copper sectors (r=6%).



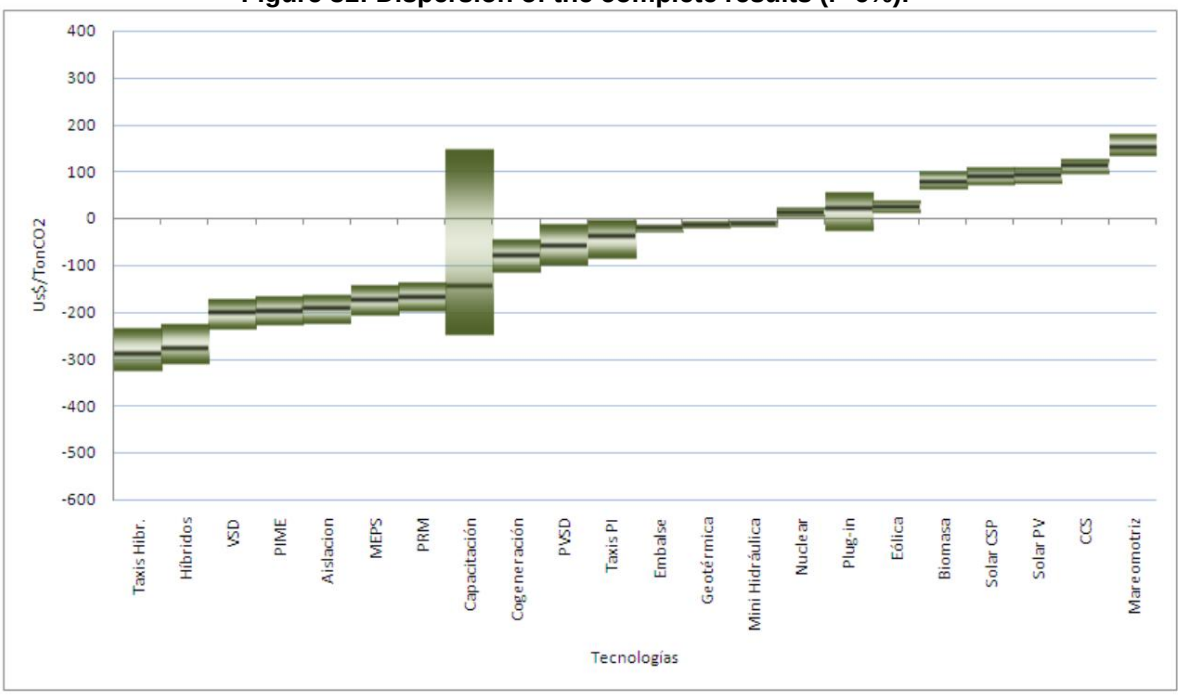
Source: Own elaboration

The figure above shows that the power generation sector has the greatest reduction potential among the sectors reviewed, followed by measures in the pulp and paper sector (particularly cogeneration). In this case, half of the reduction potential can be achieved with cost-saving measures.

The following figure shows this same curve, which includes the variation in the results (uncertainty) in the reduced tonne value for each measurement presented. Significant dispersion ranges can be observed in some of the measurements (even exceeding the actual measurement values). This reinforces the importance of conducting these analyses in this type of exercise.

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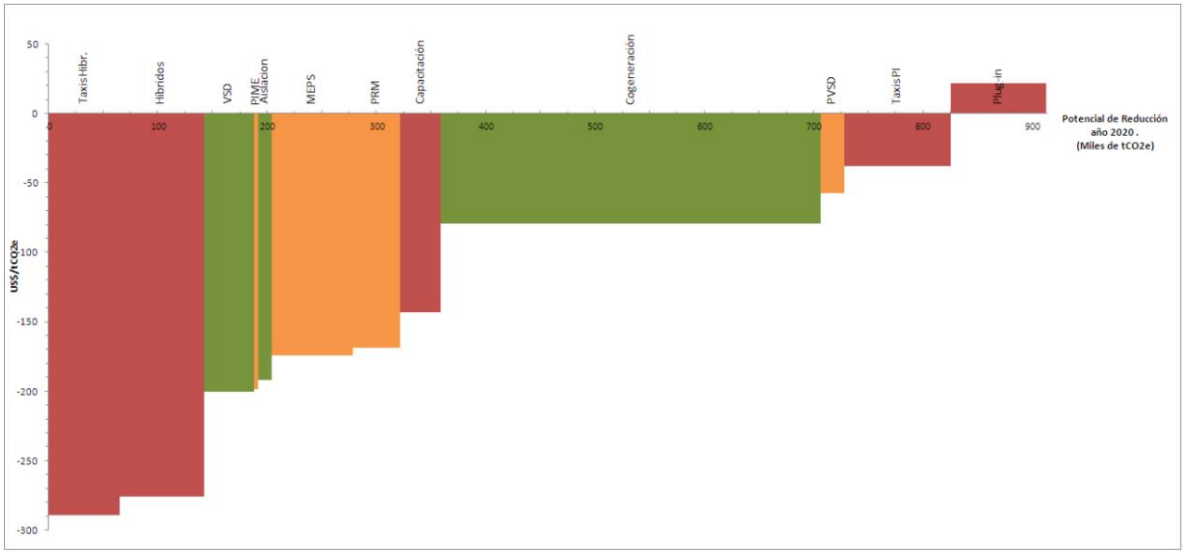
Figure 31: Dispersion of the complete results (r=6%).



Source: Own elaboration

The figure below shows that most sectors have measures with negative costs (an expected result, given that they are largely energy efficiency measures), and only the implementation of plug-in vehicles is a measure with positive costs. Almost 90% of the potential can be achieved with these "free" measures.

Figure 32: Decline curve for the transport, paper and pulp and copper sectors (6%).



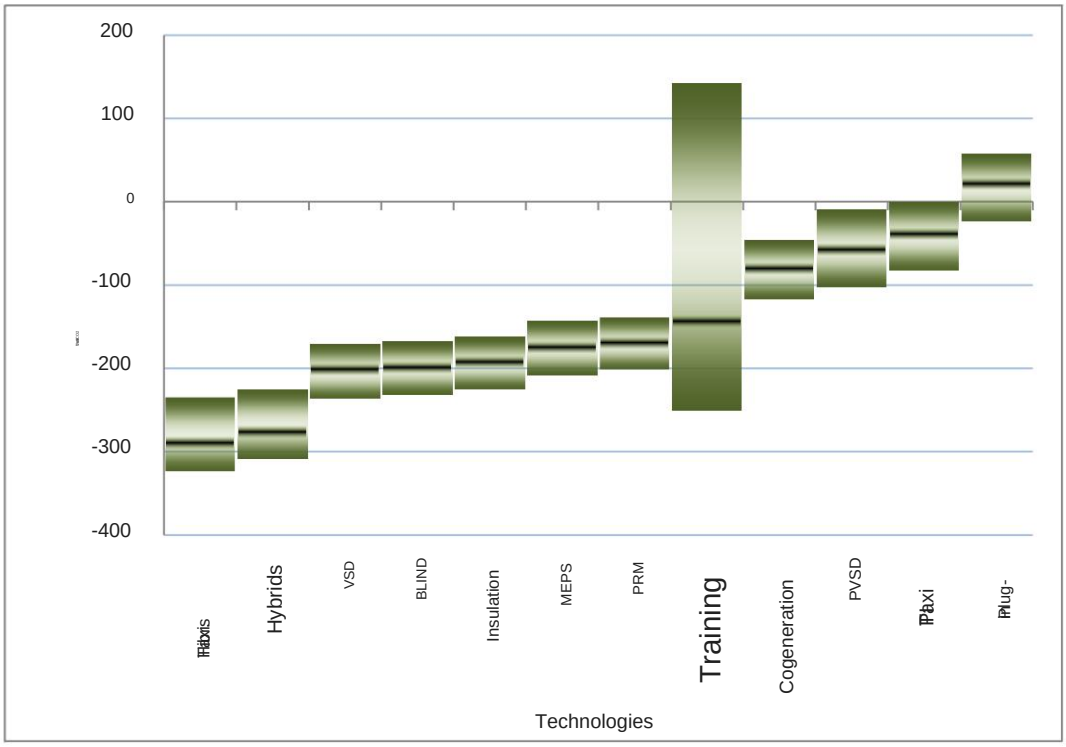
Source: Own elaboration

The following figure illustrates more clearly that measures such as training, cogeneration, plug-in vehicles, or copper frequency converters have a very high uncertainty

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high and that could change some decision by not knowing if it has a positive or negative cost or if the measure will have a real cost higher than the one that is supposed to be more expensive (post measure from left to right) given its high variability.

Figure 33: Dispersion of demand results (r=6%).



Source: Own elaboration

The table below presents one of the analyses mentioned above, which is the comparison between adjacent measures on the ascending abatement cost line. In this case, all evaluated measures are considered (including those for electricity generation), and those with a probability greater than 30% of being higher than the measure to their right on the curve were identified. This analysis becomes even more relevant when the measures are from different sectors (for example, transportation vs. mining), given the lack of correlation between sectors and their input variables.

Table 26: Comparative evaluation of measures with uncertain results

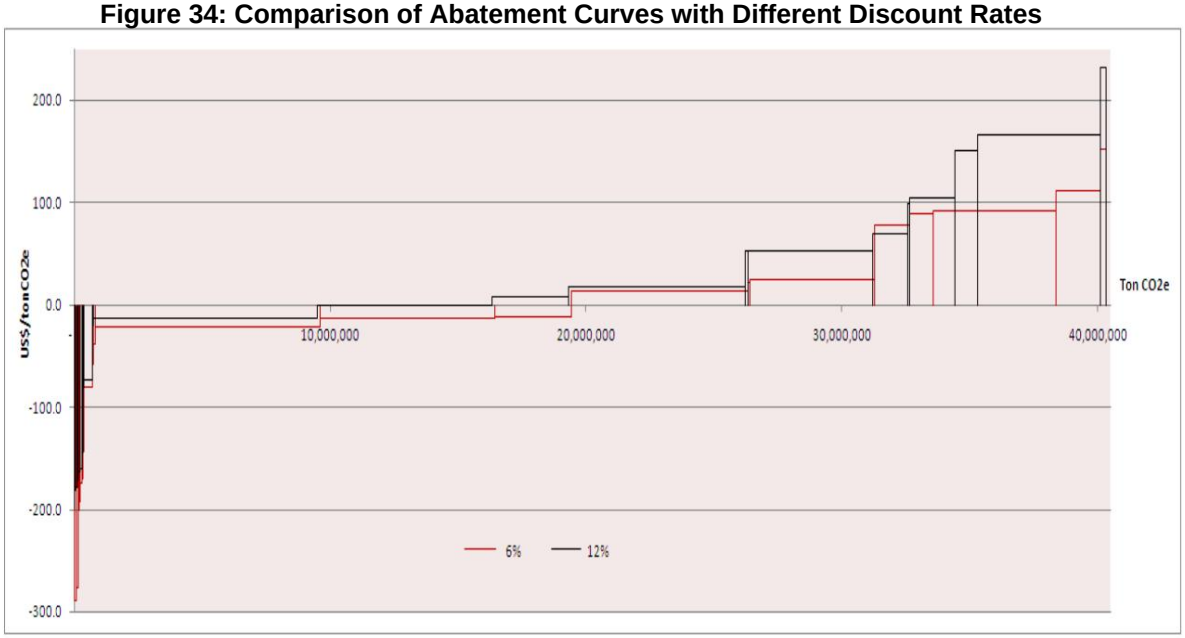
	VSD Pumps P&C > PIME	PIME > Isolation P&C	PVSD > Taxi PI	Geothermal energy >Minihidro	Nuclear > PI	PI > Wind power	Solar CSP > Solar PV
United Nations	45%	39%	35%	32%	48%	33%	43%

Source: Own elaboration

Finally, the marginal cost reduction curves for both discount rates used, the social rate of 6% and the private rate of 12%, are presented together.

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Figure 34: Comparison of Abatement Curves with Different Discount Rates



Source: Own elaboration

The latter is relevant because the results of the marginal cost abatement curves suggest that there is room for social and private benefits. The potential for increased efficiency implies that numerous barriers and market failures hinder the implementation of countless good initiatives.

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9 COMMENTS AND CONCLUSIONS

This study contributes methodologically to incorporating uncertainty into decision-making, specifically in the case of climate change mitigation. Incorporating uncertainty through simulation demonstrates the importance of including it in these types of analyses, which involve technical-economic and "environmental" variables (in this case, greenhouse gases). This is associated with improvements in the quality and quantity of information required for assessments and the costs involved in achieving a more reliable information base. The methodology proves highly relevant, and its application exists in various productive sectors in the country and in other countries in the region.

The results derived from this study enrich the deterministic analysis. In particular, it is possible to determine probability ranges for the cost levels of the different measures. The results show that the value obtained by ignoring uncertainty differs from the expected value of the unit abatement cost. This confirms that evaluating each parameter at its mean value does not yield the most probable value of the abatement cost function.

The project has been able to characterize mitigation options for the electricity generation sector and three specific demand sectors with relevance in their energy intensity and GHG emissions in Latin America, with the purpose of constructing an abatement curve,
including explicitly uncertainty.

Incorporating uncertainty allows for a reduction in the possibility of errors in policy decisions. This is because the probability of all random variables reaching their mean value is zero. Therefore, cost-effectiveness is highly uncertain, and this type of analysis allows for estimating limits on these costs. In general, the main sources of uncertainty are the cross-cutting variables of energy prices and the investment costs of electricity generation technologies.

The results obtained reveal some instances where deterministic analysis yields different values than probabilistic analysis, with a defined confidence level (95%), which would alter the investment decision. Indeed, as seen in the results section for the electricity generation sector, considering uncertainty makes the ranking of some options uncertain.

Considering a 95% confidence level, the ranking order changes for the mini-hydro options with geothermal energy and solar PV with solar CSP, confirming the relevance of the considerations of
uncertainty in decision making.

The copper and pulp and paper sectors demonstrate measures designed to achieve cost-effective results. This shows that energy efficiency savings translate into significant economic savings that justify the investment and the incorporation of tools to help overcome the barriers associated with their implementation.

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Given that the transport sector is one of the main emitters of greenhouse gases, the implementation of mitigation measures such as those proposed in this study is key to its sustainability and to meeting the commitments regarding climate change that are

acquiring the countries of the region. Unfortunately, the reduction potentials achieved are quite low and with considerable uncertainties.

This work provides an initial approach to greenhouse gas mitigation analysis using simulation methods. However, further study in several areas of the research would be desirable:

- In terms of enriching the model, there is the alternative of incorporating external costs to the extent that these costs can be quantified. For example, co-benefits derived from improved air quality in the electricity generation sector can be incorporated.

- Exploring the role of distributions in more depth is another important point. Specifically, it is relevant to seriously study how the final distribution of abatement costs changes if...
  The distributions change. For example, determine if there are significant differences if a uniform or triangular distribution is considered.

- In the same vein as the previous point, it is key to delve deeper into how to assign the form functional distributions and the key parameters that characterize them.

- It is important to analyze the role of correlation analysis. In the model, only the price trajectories of a single energy source are correlated. It is relevant to delve deeper into theoretical aspects before applying different levels of correlation between variables. Key questions arise: How do correlation considerations affect the results? Are these changes significant? Which correlations are the most significant?

- It is also interesting to incorporate trajectories for many of the random variables.
  Currently, variables such as investment costs, other operating costs, capacity factor, and efficiency show randomness, but only once (in the first year, not over a long period). It is relevant to consider movements where, for example, the capacity factor of a wind farm increases and approaches its potential.

- Finally, it is also interesting to be able to develop an analysis of the expected value of information, which allows us to quantify how much it costs to improve uncertainty in those cases where it is not possible to make decisions with the previously defined certainty criterion

Overall, the study results show that cost-effective mitigation projects are usually available.
A wide range of failures have been analyzed in the climate change literature, among which the following stand out: the low priority of agents (consumers and companies) to reduce energy costs and reduce emissions, lack of knowledge and problems with access to capital.

Finally, we understand that this methodology has a high degree of replicability in countries of the region, for which it is necessary to have information of a certain quality or expertise in the sector involved, so that the parameters, criteria and approaches necessary to incorporate the
uncertainty in the variables of interest.

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APPENDICES

A.1. UNCERTAINTY **ANALYSIS CONSIDERATIONS IN ECONOMIC EVALUATION**

One of the main problems that hinder decision-making in most everyday life situations is the presence of uncertainty in the variables that govern the processes.

Uncertainty analysis in economic evaluation is one of the most important issues to consider, as it allows us to determine how sensitive the analysis is to the inherent and large uncertainties in many of the model's key variables.

In the case of evaluating a mitigation measure, this requires integrating many types and sources of information known only with some level of precision. This is why directly using “most probable values” or “averages” has significant limitations. The probability that the result of the variable of interest will match the estimate using average values is practically zero. But more importantly, it ignores the risk of error.25 Therefore, one should not be concerned with the average but rather with the size, frequency, and causes of the variations [Fung, 2010].

Sensitivity analysis is an alternative that explores and quantifies the impact of potential errors in a model's input data on its output. New assumptions about the future define scenarios comparable to the average case. Specifically, the values of the input variables are changed, and the calculation that yields the output variable is repeated. The main advantage of sensitivity analysis is its simplicity. The disadvantage is that it does not provide information regarding the probability of occurrence of each scenario.

There are several methodologies for performing an uncertainty analysis, some of which are described below. continuation.

Why perform Uncertainty Analysis?: The problem of the Deterministic Approach

Today, deterministic approaches are recommended for analyzing decision-making problems. This approach relies on the use of algorithms, methodologies, and available data deemed appropriate for conducting an evaluation. This type of input data constitutes what is known as the "deterministic approach."

The main shortcomings of deterministic analyses are: i) The input variables of the models assume a simple fixed value, without considering the inherent variability of the exposure parameters. In general, the variables are not representative and are sometimes overly conservative. ii) The output is a simple value that does not estimate the propagation of uncertainty or consider the variability that the final result should have. iii)

[25] http://www.strategy-at-risk.com/2011/03/01/plans-based-on-average-assumptions/

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The effects of independent or correlated variables cannot be examined, since covariance is not studied in this type of analysis. Due to all these shortcomings, this approach has been heavily criticized [Salinas 1994].

A particularly challenging problem is that of deterministic sensitivity analysis, which involves estimating the extreme values (maximum and minimum) of a model's input variables and combining them to obtain the extreme cases of the resulting variable. The classical approach consists of modeling the "best case," based on estimates of the minimum input values, and the "worst case," based on estimates of the maximum input values. It concludes that the most probable value will be...
It is found between the two calculated extreme cases.

This method becomes extremely complex when applied to models with a large number of inputs, and where extreme values of the output variable have a very low probability of occurrence. For example, let's calculate the ratio between the "worst case" and the nominal value of a variable that depends on the product of five variables, whose range varies between 0 and 1, and with an estimated probability value of 0.1 for each. We have the following:

Factor for “worst case” = (1)5 = 1
Factor for “nominal case” = (0.1)ÿ = 10ÿÿ
Nominal ratio/worst = 10-5 /1 = 10-5

The ratio then represents a factor of 1 to 100,000, which is an extremely wide estimation range, making it impossible to shed any light on the likely extreme values of the estimate [Gratt, 1996]. Similar examples confirming the inconvenience of using the deterministic approach, and even more so, of using conservative (extreme) deterministic values, can be found in [Burmaster and Harris, 1993], [Burmaster, 1996], and [Burmaster and von Stackelberg, 1991].

Given the disadvantages and biases of the deterministic approach, we propose incorporating and quantifying uncertainty at each stage of the analysis process. Probability is certainly the best-known and most widely used formalism for quantifying uncertainty, and advances in commercial software facilitate its analysis and processing, allowing for the straightforward representation of random variables.

Decision theory has shown that it is important to consider the relevance of uncertainty in cases such as the following:

- When analyzing a problem in which people's attitudes toward risk are very important.
- When information from different sources with high uncertainty must be combined. • When a decision must be made about whether to use more resources to acquire additional information. In general, the greater the uncertainty, the greater the expected value of the additional information.

Some other advantages and disadvantages of sensitivity analysis are its ease of application and understanding; however, it only allows the analysis of variations of one parameter at a time.

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It does not use information such as the probability distributions of the parameter to be sensitized and does not provide probability distributions of the NPV and IRR profitability indicators.

On the other hand, the main advantages of random sampling simulation methods over deterministic approaches are the following:

1) Inputs defined as statistical distributions cover the full range of values possible.
2) The input parameters are varied randomly and simultaneously, allowing the propagation of uncertainty through the model.
3) The simulation generates statistically valid and fully characterized, covering the full range of possible values.

Thus, it is clear that random sampling methods allow for a more rigorous analysis than discrete simulation methods, enabling a thorough sensitivity and importance analysis of the variables. Furthermore, with careful analysis, it yields much more reliable results and a higher quality statistical presentation.

While the application of the simulation method offers many advantages, there are also associated disadvantages. The main disadvantages include:

1) The value of the simulation depends on the quality of the information associated with the input distribution parameters (therefore, careful characterization of the assigned distribution is required).

2) Any change to an input implies rerunning the entire model.
3) The covariance or degree of interdependence between the variables must be known.
4) More sources and documentation are needed to establish distributions, assumptions, and covarianzas.
5) The output of an analysis can lead to a false sense of certainty (overconfidence in the results).

Definition of Uncertainty: Nature and Sources

The uncertainty surrounding economic variables and physical phenomena considerably complicates the economic analysis of problems. Consequently, since it is not always possible to make decisions based on an objective evaluation of scientific data, it is sometimes necessary to resort to more subjective methods to compensate for the lack of complete information. However, decision-making under conditions of uncertainty is not entirely subjective; it can frame the available knowledge, take into account the degree of uncertainty of this knowledge, and estimate the extent of the lack of knowledge (ignorance). Analysis under conditions of high uncertainty should be geared toward selecting the best strategy based on an impartial evaluation of the available quantitative and qualitative data. [Group Economic Experts 1991]

The central purpose in decision-making, and in policy formulation and analysis, is to identify the relevant factors and sources of disagreement in a problem, in order to anticipate what

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Unexpected. The conscious treatment of uncertainty compels us to think more carefully about these issues and the eventualities and contingencies of the problem [Morgan and Henrion 1990]. Successfully addressing uncertainty involves more than simply describing the uncertainties in individual variables and their propagation throughout the model. Often, the difficulties in understanding the problem and the limitations and uncertainties in current models are more important in the analysis. Furthermore, developing models for decision-making involves defining a large number of variables, which generally carry some degree of associated uncertainty.

Types of Variables in Analysis

For a better understanding and analysis of the types of variables that can be considered in decision analysis models, [Morgan and Henrion, 1990] define the main types, along with their possibilities for representing uncertainty. Three of the most commonly used are presented below:

Empirical variables represent measurable properties in the modeling of real-world systems, and their uncertainty can be expressed as a probability distribution. It has been suggested in the literature that these are the only variables for which uncertainty can be appropriately represented probabilistically. In principle, all empirical variables are uncertain; there is never absolute certainty about the truth of any empirical proposition or about the exact value of any empirical quantity. No matter how high the precision, no experiment can measure a real quantity with zero error. However, for practical purposes, uncertainty is often negligible for many empirical variables, allowing them to be treated deterministically in the model.

Defined variables. Fundamental physical constants such as the gravitational constant, the charge of the electron, etc., are currently empirical constants, since they are measured and are inherently uncertain. However, some variables are certain by definition and are called "defined variables," for example, the number of days in January, or mathematical constants such as "pi."

Its only importance within this context is its differentiation from empirical variables.

Decision variables are variables over which the decision-maker has control. If a variable is a decision variable, then by definition it has no inherent value. It is up to the decision-maker to select its value. The process of developing and evaluating a model can be better focused if these decisions are made explicit.

Sources of Uncertainty in Empirical Variables

There have been many attempts to measure different types of uncertainty. Most of these have focused on the uncertainty of empirical variables, which generally constitute the majority of variables in models [Morgan and Henrion 1990]. Uncertainties in these variables can originate from various sources, so the appropriate method for characterizing and reducing them generally depends on the specific type.

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from source. Thus, according to the different types of sources from which it comes, the uncertainty in empirical variables can be classified as follows:

1) Statistical Variation. Imprecisions in measuring instruments and observation techniques will inevitably lead to variations from one observation to another. The resulting uncertainty depends on the magnitude of the variations between observations and the number of observations taken. Statistics provides a number of well-known techniques for quantifying this uncertainty, such as standard deviation, confidence intervals, and others.

2) Subjective Judgment. There is a consistent tendency to underestimate systematic error. This tendency has been found universally in almost all measurements of physical quantities that have been made. Perhaps this should not be surprising, given the difficulty in making subjective estimates about errors that are essentially unknown over time. When quantifying uncertainties, it is easier to underestimate the existence or effect of little-known sources of error than to overestimate them.

3) Variability. Many variables within a model are not constant over time and space. In general, they do not correspond to particular variables, but rather to populations of variables. However, these populations can be reasonably well specified and could be well described by frequency distributions. Frequency distributions are sometimes called probability distributions and have the same formal properties.

The uncertainty in a frequency distribution can be represented by the probability distribution and its main parameters, such as its mean, standard deviation, median, or other percentiles. Under these conditions, it is not difficult to handle the uncertainty of probabilistically represented frequency distributions, but dealing with the uncertainty in the probability distributions that might represent the variables is more problematic and complicates the analysis, as it requires a more complex and rigorous statistical estimation.

4) Inherent Randomness. Inherent randomness is distinguished from other types of uncertainty by being, in principle, irreducible. A quantity is said to be random if no design or model incorporating its variations is known. In this view, randomness, like probability, is a function of the evaluator's available knowledge. A quantity may legitimately be random for one person but deterministic for another who knows its generation process. Thus, numbers can be completely determined if the algorithm is known. Otherwise, they can be as random as they appear to us.

5) Disagreement. If data are difficult to obtain, it is complicated to quickly determine which values are relevant to the process. Despite this, policy decisions sometimes need to be made before disagreements are resolved. Thus, in practice, such disagreements among scientific experts are often a significant source of uncertainty in policy analysis.

6) Approximation. The uncertainty of an approximation arises because the model used is only a simplified version of the real-world system being modeled. Approximations are also introduced by the use of discrete probability distributions to represent continuous distributions, or by the finite number of iterations used in Monte Carlo simulation or other random sampling processes. Each approximation represents a compromise between the accuracy of the representation and the computational cost of the model.

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Uncertainty about the shape of the models

Up to now, uncertainty has been considered regarding the variables or parameters of a model, but not regarding the form or structure of the model itself. Uncertainty about the form of the

Modeling is generally more difficult than the uncertainty of the value of a quantity, but experienced analysts often argue that uncertainty about structure is more important and has a substantially greater effect on the results of analyses. However, there has been relatively little research on situations in which there is uncertainty or disagreement about which form of model to use [Morgan and Henrion 1990].

In any case, all the models are definitely false. Perhaps it's possible to say that one model is better than another, in the sense that it produces more reliable predictions, not that it's more likely; however, it's inappropriate to try to assign probabilities to the models.

Uncertainty about the form of the model often reflects disagreements among experts about the scientific basis or technical mechanism. When this leads to significant differences, as is often the case, it may be more appropriate to examine it parametrically, rather than attempting a probability distribution over alternative forms or the weight of different experts' opinions, as has been argued with respect to disagreements about empirical variables [Morgan and Henrion 1990].

The incorporation of uncertainty and its impact on decision-making

Uncertainty can be defined as the lack of precise knowledge or understanding of the causes that determine the behavior of a real system. It can also be defined as the lack of knowledge of the true value of an unknown quantity (Morgan and Henrion, 1990). This implies that the system's behavior cannot be described with certainty, and therefore a risk analysis must be performed, making decisions within a context of uncertainty.

In this context, the question arises as to why uncertainty is of interest to a risk evaluator.
Even when data exist, assumptions or inferences will be required because they may not be available for all aspects of the analysis. Furthermore, the available data may be of dubious quality or unreliable. Additionally, Morgan and Henrion (1990) pointed out other reasons:

- Risk Analysis should combine information from different sources and of different quality.
- A decision must be made about allocating resources – and how to do so – to acquire information.
  additional.
- There may be biased or inaccurate estimates.
- Important factors and/or potential sources of disagreement in decision-making can be identified
  decisions.

In summary, based on the risk analysis case, incorporating uncertainty into decision-making greatly enriches the analysis, providing stakeholders with improved information and analysis compared to deterministically developed assessments or models, and guidelines on where to best invest resources to improve the quality of information.

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A.2 . UNCERTAINTY PROPAGATION MODELING

In reality, we encounter highly complex systems. Real-world systems are often stochastic in nature and difficult to understand and/or control using traditional analytical tools. Analytical methods generally require numerous simplifications that allow us to capture only some of the elements of a system. Simulation allows us to handle complex systems more realistically. In this section, we follow Helton and Davis.²⁶

Uncertainty analysis can be understood as the study of functions of the type:

$$= ( )$$

Where the function f represents the model under study, the vector = (1, … ,) of input variables (“inputs”), and the vector of predictions = (1, … ,). The purpose of uncertainty analysis is to determine the uncertainty in the elements of y that result from the uncertainty in the elements of x. Sensitivity analysis turns out to be a simpler methodological approach in which it examines how changes in some of the components of the vector x impact the elements of the vector y.

To carry out the uncertainty analysis, it is necessary to characterize the uncertainty of the elements of x. In general, the uncertainty of these variables is characterized by different distribution (or density) functions:

$$1, 2, \dots ,$$

Where denotes the distribution associated with the k-th element of the vector x. Thus, what is known is the joint distribution of and what is of interest is to know the distribution of the statistic = ( ). In what follows, and for simplicity, it will be assumed that m=1.

The determination of the statistic of y is done by estimating the distribution function, the complementary distribution, or the probability density function.

Complementary distribution is defined as:

$$(>) = \ddot{y} [( )] ( )$$

Where d represents the probability density function corresponding to the distributions 1, 2, … , and dV represents the differential (which is denoted in this way because it is generally a multidimensional volume, of dimension n), and

[26] J.C. Helton, F.J. Davis “Latin hypercube Sampling and the Propagation of Uncertainty in Analyses of Complex Systems”. Reliability Engineering and System Safety 81 (2003) 23–69.

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$$[\ (\ )] = \ddot{y}_{0}^{1}\ (\ ) > \ (\ )\ \ddot{y}$$

Similarly, the distribution function is defined as:

$$(\ \ddot{y}\ ) = 1\ \ddot{y}\ (\ >\ ) = 1\ \ddot{y}\ \ddot{y}\ [\ (\ )]\ (\ )$$

Then, in formal terms, the simulation seeks an approximation for the value of this last integral.

One formal way to interpret sensitivity analysis is through analysis of variance. The variance is given by:

$$(\ ) = \ddot{y}[\ (\ )\ \ddot{y}\ (\ )]2\ (\ )$$

Where E(y) denotes the expected value of y, $(\ ) = \ddot{y}\ (\ )\ (\ )$

Thus, sensitivity analysis can be interpreted as the decomposition of the variance V(y) into the components of the individual elements of the vector x. The size of each component will provide a measure of the importance of each variable.

Monte Carlo simulation is one of the most common alternatives for putting into practice the estimation of the propagation of uncertainty of a system.27 Monte Carlo analysis is a probabilistic sampling procedure used to determine the relationship between the variables of $(\ >\ ) =$ input and output results. This relationship allows approximating the integral $\ddot{y}\ [\ (\ )]\ (\ )$ (uncertainty analysis) and also evaluating individual effects that some of the components of the vector x have on the output variable y (sensitivity analysis).

In a simulation, a sample of size S is generated that is consistent with the input distributions. $_{1, 2, \dots ,}$ and the joint distribution. The sample is represented by:

$$= \ddot{y}\ _{1, 2, \dots ,}\ \ddot{y}, = 1, \dots ,$$

There are several ways to perform the sampling, and the main ones will be presented in the next section. Regardless of the sampling method used, the procedure considers weights such that the element xi can be used together with $= 1, \dots ,$ wi to obtain the desired approximations28.

Once the sample is known, the evaluation of the function allows establishing a relationship between the input and output variables:

---

[27] Other alternatives include differential analysis (based on partial derivatives) and Fourier analysis between Others. 28

In the simplest case of random sampling, the weight is simply 1/S.

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[.], = 1, … ,

Where = ( ).

Thus, the key integral can be approximated as follows by virtue of the Law of the Great Numbers (Weak)[29] :

$$(>) = \ddot{y}\,[\,( )]\,( )\;\ddot{y}\;\ddot{y}(>) = \ddot{y} \qquad \sum_{=1}^{( )}$$

This expression provides the most complete representation of the uncertainty in and derived from the distributions of the input variables x.

Similarly, the expressions for the estimators of the expected value and variance are given by:

$$( )\;\ddot{y}\;\ddot{y}( ) = \ddot{y} \qquad \sum_{=1}$$

$$( )\;\ddot{y} \qquad ( ) = \ddot{y}[\,\ddot{y}( )\;\ddot{y}\,] \qquad \sum_{=1}$$

There are more formal alternatives for performing sensitivity analysis regarding the impact of the elements of x on y through the expression [. One possibility is to use regression methods that relate y to the xk. These relationships are of the form:

$$= \qquad {}^{0} + \ddot{y} \qquad \sum_{=1}$$

[29] In very simple terms, the Law of Large Numbers states that as the sample size (S) increases, the experimentally estimated probability gets closer and closer to the theoretical probability. In other words, the larger the sample, the better the population is characterized.

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A.3. SAMPLING TECHNIQUES

In simulation exercises, several sampling techniques are available. The most common alternatives are simple random sampling (sometimes also called Monte Carlo sampling) and Latin Hypercube sampling.

Simple random sampling is the most common, intuitive method, and the one that adheres to the rules of statistics (Law of Large Numbers). Random numbers are generated from the input vector of variables x. In general, simple random sampling is the conventional technique that uses pseudo-random numbers to sample values from a probability distribution. Because of this, the samples will have a greater chance of coming from the values with the highest probabilities.
Furthermore, if the tails of the distributions are significant, there is a limitation on the concentration of sampled values near the center of the distribution. This means that when S is small, there are few values in the tails and a high concentration in those ranges with the highest probability.

An alternative approach is Latin Hypercube LH30 sampling , an advanced random sampling technique within the intervals of the cumulative frequency distribution, which minimizes the number of samples required to represent the entire distribution. Latin Hypercube sampling does not concentrate samples near the center and is particularly useful when the tails of the distribution are significant [Gratt, 1996]. This method is often recommended by software developers and international agencies because it significantly increases computational efficiency.31

LH sampling uses a stratified sampling approach that improves the coverage of the sample space. Considering the graph of a distribution function Dj (Xj), stratification is achieved by dividing the vertical axis into S intervals of equal length (equally likely), where S also corresponds to the number of simulation runs. This is done using the inverse function of the Dj distribution.

$^{-1}$. These intervals divide the horizontal axis into S non-overlapping, equally probable intervals that are not necessarily of the same length.

The next step involves selecting a value in each of the S intervals of the vertical axis. A single x-value is obtained in each one. When these values are followed through the inverse Dj $^{-1}$.
of the S intervals of the horizontal axis. If X is an Sxn matrix (where S was the number of runs and n the dimension of the vector x) the j-th column contains the LH sample for the random variable Xj.
Finally, the values in each column of the matrix are sorted in a random process. This process of mixing the values allows us to emulate the result of simple random sampling.

One way sampling alternatives are compared is through variance estimates.
The expected value estimator is the same for all types of sampling:

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[30] See the appendix where the Latin Hypercube sampling method is explained in more detail.
[31] For example, and in different contexts, LH sampling is recommended by Price Waterhouse Coopers, the EPA, and it is an alternative available in most commercial simulation software.

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$$( ) \ddot{y} \ddot{y}( ) = \ddot{y} \quad =1 \quad = \frac{1}{} \quad \ddot{y} ( ) \quad =1 \quad =$$

For simple random sampling, the variance:

$$(\ddot{y}) = \frac{1}{} \quad ( )$$

Whereas for LH sampling the variance is expressed as for the case n=2:

$$(\ddot{y}) = \frac{1}{} \quad ( ) + \frac{\ddot{y}\ 1}{} \quad (1, 2)$$

Therefore, the variance obtained in LH sampling is smaller only when the covariance term is non-positive. This holds true for monotonic functions f.

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A.4 . LATIN HYPERCUBE SAMPLING THEORY (LHS)

Esta sección se basa en *"A User's Guide to LHS: Sandia's Latin Hypercube Sampling Software" Gregory D. Wyss and Kelly H. Jorgensen. Febrero 2008. Risk Assessment and Systems Modeling Department*
Sandia National Laboratories

LHS was developed to perform uncertainty assessment and sensitivity analysis in the following context. Considering a function Y that is a function of other variables X1, X2,…,Xk, we are interested in knowing how Y varies when the X's change according to a joint probability distribution.

A conventional approach to performing this type of analysis is Monte Carlo simulation. By obtaining a sample of the assumed joint density for the X's and evaluating each Y for each sample, the distribution of Y can be estimated along with other characteristics such as mean and variance.
The output of a program for n Monte Carlo iterations is a set of n variable vectors (where each vector is k-dimensional). Each vector can be evaluated by the function to generate n values of Y. This approach results in reasonable estimates of the distribution of Y if n is sufficiently large. However, this type of implementation requires significant computational resources.

A more economical alternative that allows for more precise estimates is a restricted Monte Carlo scheme. The LHS method developed by McKay, Conover, and Beckman (1979) is one such scheme and is used to generate samples of the X's. LHS performs sampling by selecting n different values for each of the variables X1, X2,…,Xk as follows. The range of each variable is divided into n non-overlapping intervals of equal probability. One value within each interval is randomly selected according to the probability density of the interval. The n values thus obtained for X1 are randomly paired with the n values of X2. These n pairs are randomly combined with the n values of X3 to form n triplets. Repeating this procedure yields n k-tuples. This is the LHS sample. It is convenient to consider this sample as an (n x k) matrix where row i contains specific values of the k variables for

to be used in the i-th run of the simulation.

To better understand how this method works, a simple example is described below for a sample of size n=5 with two input variables. It is assumed that the first variable X1 is normally distributed with mean ÿ and variance ÿ2 . The points that determine the partition are easily determined based on the parameters ÿ and ÿ2 .

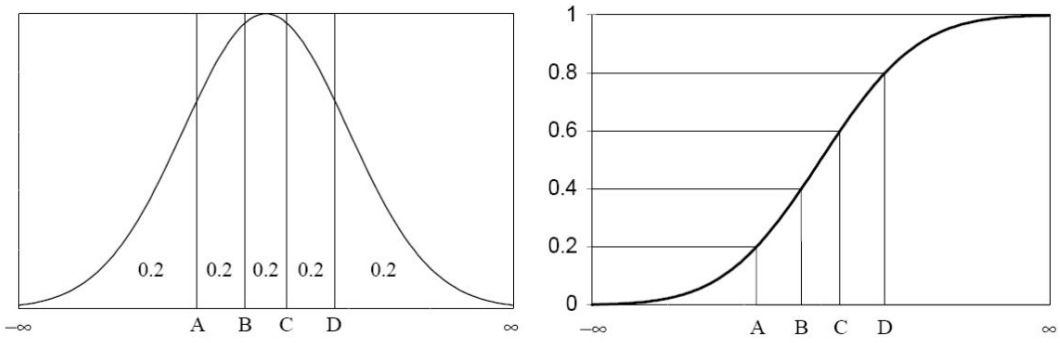
The following figure shows the intervals for both the probability density function and the distribution function. The intervals satisfy:

$$P(\ddot{y}\ddot{y} \ddot{y} X1 \ddot{y} A) = P(A \ddot{y} X1 \ddot{y} B) = P(B \ddot{y} X1 \ddot{y} C) = P(C \ddot{y} X1 \ddot{y} D) = P(D \ddot{y} X1 \ddot{y} \ddot{y}) = 0.2$$

Then each of the five intervals corresponds to a probability of 20%.

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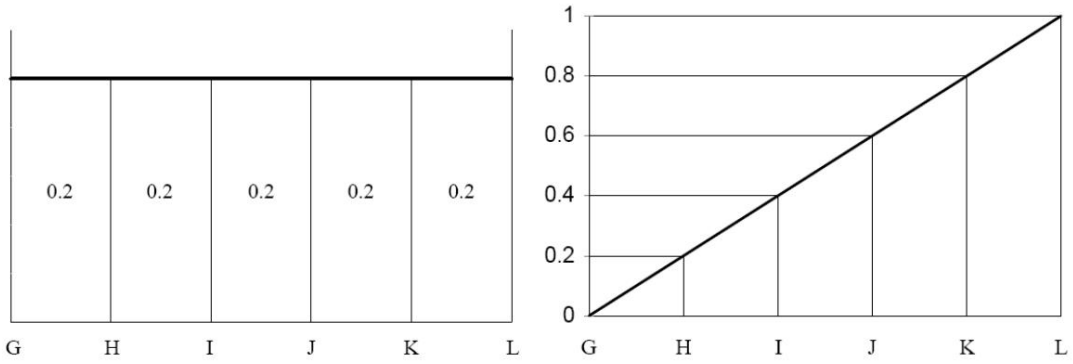
Figure 35 Intervals used in LHS of Size n=5 for Normal Variable



Source: Wyss and Jorgensen (2008)

For the second variable X2, it is assumed that it has a uniform distribution in the interval between G and L. The following figure shows the intervals for the probability density and the distribution function.

Figure 36 Intervals used in LHS of Size n=5 for Uniform Variable



Source: Wyss and Jorgensen (2008)

The next step requires obtaining specific values for X1 and X2 in each of the five respective intervals. This selection must be made randomly with respect to the density of each interval. That is, the selection should reflect the height of the density across the interval. For example, in the interval (-ÿ, A) for X1, values close to A will have a higher probability of selection than those values in the tail of the distribution extending to -ÿ.

Next, the selected values of X1 and X2 are randomly paired to form the five required two-dimensional vectors. In the original LHS concept, the pairing is done by associating a random permutation of the first n integers of each input variable. To illustrate this procedure, consider two permutations of the integers (1, 2, 3, 4, 5) as follows:

-Permutation Set 1: (3,1,5,2,4)

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-Permutation Set 2: (2,4,1,3,5)

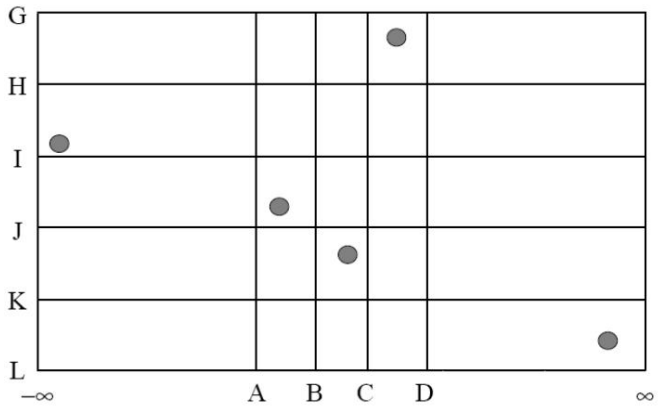
Using the respective position within each of the permutation sets as the interval numbers for X1 (Set 1) and X2 (Set 2), the following set of interval pairs can be formed:

Number of Iteration	Interval Used for X1	Interval Used para X2
1	3	2
2		4
3	1 5	1
4	2	3
5	4	5

Therefore, as an example, in the first iteration, the input vector is formed by selecting the specific value of X1 from the third interval (from B to C) and pairing it with the specific value of X2 selected from the second interval 2 (from H to I).

Once the values of each variable are obtained to form the five input vectors, a two-dimensional representation of the LHS sample can be realized as in the following figure. In general, a set of n sample points in a k-dimensional Euclidean space contains one point in each of the intervals for the k variables.

Figure 37 Two-Dimensional Representation of a Possible Sample of Size 5 Using X1 and X2



Source: Wyss and Jorgensen (2008)

Suppose we want an LHS sample of size n=5 from a normal distribution with ÿ=5 and ÿ2 =2.618. After defining the intervals of equal probability from this information, we need to randomly select an observation within each interval, but according to the distribution of the interval.
This is done by taking a uniform sample of the quantiles of the distribution and then inverting the distribution to obtain the distribution values that the remaining quintiles represent.

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To obtain specific values, n=5 numbers are randomly selected from the standard uniform distribution (U(0,1)). Let these values be denoted by Um, where m=1,2,3,4,5. These values are used to randomly select values within each of the n=5 intervals. To achieve this, each of the random values Um is scaled to obtain the corresponding cumulative probability, Pm, such that each Pm falls within the i-th interval. Then, for this interval with n=5,

$$PM = One + \frac{m - 1}{5}$$

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A.5. CHOICE OF INPUT VARIABLE DISTRIBUTIONS

In the development of uncertainty analysis, it is fundamental to characterize the uncertainty associated with
The input variables. Probability distribution functions often arise from the fundamental properties of the quantities they are intended to represent. For example, quantities formed from the sum of uncertain quantities tend to be normally distributed; quantities generated from the multiplication of uncertain quantities tend to be lognormal; and events that occur randomly over time lead to Poisson and exponential distributions. However, distributions are often selected empirically, based on those that provide a reasonable representation of the observed data. In other cases, the definition of the uncertainty of the input variables has relied on information provided by experts, as this is generally faster and less costly to obtain.

Often, there are differences of opinion regarding certain uncertain empirical quantities provided by various informed experts. In such cases, it is common to use the approach of combining opinions using relative weights. A wide variety of methods can be used to assign these relative weights. They can be derived from ratings made by the experts themselves regarding their own opinions or those of other experts, or they can be derived from the uncertainty expressed in the distributions themselves. They can also be assigned directly by the analyst based on information about the relative knowledge, reliability, and calibration of the data.

experts.

A decision-maker or analyst can consider each expert's opinion as given, supplemented by any evidence or judgment regarding the quality of the experts' opinions. This data and evidence can be used by the analyst to update their own opinion and arrive at a simple probability distribution, either by directly manipulating the relative weights or by applying the results to a model. The final distribution then represents the analysts' best judgment based on the experts' opinions.

A wide variety of sophisticated Bayesian evaluations and combinations of techniques have been used to analyze information provided by experts. In practice, the Bayesian approach tends to be more difficult to apply, as it requires thorough assessments of the dependence between experts. [Morgan & Henrion, 1990].

In practice, expert opinion is very useful given its great simplicity. Thus, some considerations are made when assigning probability distributions to variables for which little information is available. For example, if there is no historical data series or data that allows fitting a probability distribution with a certain level of confidence, simplifications such as the following can be used [Salinas 1994]:

- If nothing is known about the variance or factor under study, except its range, a uniform probability distribution.
- If only the range and mode of the variable or factor under study are known, a distribution is used triangular probability

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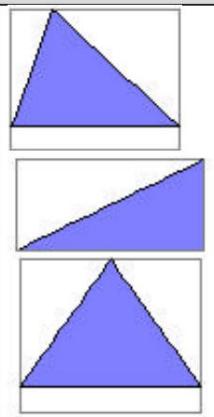
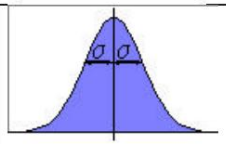
- If the values of a variable or a consensus data source appear to have an unrestricted range of possible values and are symmetrically distributed around the mode, a normal probability distribution is used.

- If the variable or factor under study assumes only positive values and the data suggest that the probability distribution might be skewed towards the lower end of the range, a lognormal probability distribution is used.
- Considerations are also made regarding extreme values in the probability distributions of the variables, which can commonly generate unrealistic values in the evaluation, so the input values or input variables of the model are truncated. For example, if the goal is to minimize the problem in asymptotic functions.

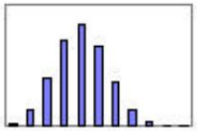
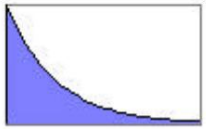
The following table shows common distribution functions and conditions under which their use is recommended.
use.

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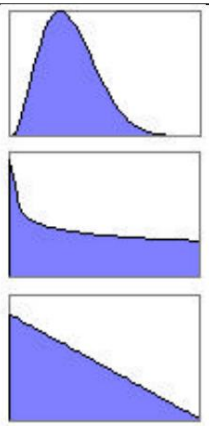
Table 27 Summary of Distribution Functions and Their Use in Uncertainty Analysis

Name (Parameters)	Examples of of distribution	Use	Appropriate uses	Examples
Triangular (Minimum, Most Likely, Maximum)		It is the The most commonly used maintaining a project have been modeled either because it is limited (finite to use. of the tails in situations where a simple the most probable value, but is given values close to the average.	distribution is not well-known and is not thought estimated as a minimum of M$40, with a more likely value of M$60 and a maximum of M$100. The actual cost could be tails) or because it is not symmetric. and intuitive understanding is of utmost importance by: Flexibility is a great advantage.	to be suitable for a normal distribution. The operating costs for like a triangular distribution. Note that it overestimates the size and the expected value of a triangular distribution is not = in + + on 3
Normal/ Gaussian (Mean, standard deviation)		Is a Error distribution. frequently used. This. Many natural variables fall This is due in part to the result in a normal distribution, as from the central limit theorem, the height of people. which states that the mean of a set of independent variables approximates a normal distribution is unknown, but it is known to be symmetrical around a mean value the extremes. +/- 1 +/- 2 Many distributions tend to normality within its limits, such as Poisson and Binomial	distribution. A random variable and that	Retail price inflation has been assumed to be 3% annually. However, it could be above or below this rate. The mean is 3%, and the standard deviation needs to be estimated. there is a greater probability of being near the mean than near ÿÿ of the medi ÿÿ of the medi ÿÿ of +/- 3 the me
Uniform (Maximum, Minimum)		It is used if the variable is the risk where there is little known maximum and minimum value. The position of a leak along a pipe, or the price at any given point in a commodity market such as gasoline, is bounded by a maximum and a minimum, and all values in between are highly sensitive. equiprobables.	Like other non-parametric distributions, this one has the advantage of being intuitively obvious, and highlights information about the distribution.	

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Name (Parameters)	Examples of of distribution	Use	Appropriate uses	Examples
Binomial (Number of events, probability of each event)		In each trial there are only two results (e.g., yes/no). This should be used if it is required to describe products, given that the probability of a defective item is 7%. The number of defective items will be 1(0) son independent: what happens in trials and a probability in an event does not affect the known occurrence. subsequent events. The probability is the same from one event to another.	the total number of defective items in a sample of 100 manufactured imparted by a binomial distribution with n=100 and p=0.07	
Poisson of (Occurrence rate)		The occurrence rate is described by this discrete distribution. If there is a measurement remains constant. The number of cases is not limited. The occurrences son independent.	system that allows for the reduction of payment each time an error occurs, and it is assumed that the incidence rate will be 20 times a year: the number of events that occur. The production in a given quarter is described by a Poisson distribution with a rate of 20/4=5 times/quarter	
Exponential of (Occurrence rate)		Describe the amount of time between events. It is only used to describe the time between a light bulb lasts for events. independent of the events, the light will be described by an exponential distribution if we	The time between (or until it happens) light bulb has an average lifespan of The occurrence rate is 5200 hours, the time that assume that the failure rate is prior. constant (i.e., the chances of them failing are the same throughout their useful life)	
Log normal (Mean, standard deviation)		This distribution is used for naturally occurring variables. The volume of gas in a gas tank frequently. The theorem The central limit states that if a quantity is the product of two or more variables chosen in a positively skewed way, the distribution is positively skewed. towards a log normal.	often follows a log normal distribution, being a product of their relationship to volume, pressure, and other naturally occurring variables. gas/liquid ratio, etc. Useful for the representation of will tend	

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Name (Parameters)	Examples of of distribution	Use	Appropriate uses	Examples
			quantities that vary by several orders of magnitude.	
Beta ÿÿÿÿÿÿÿÿ (Number of events, n, Number of events positive, r) ÿ1 = *r* + 1 ÿ2 = *n* - *r* + 1		probability of missing? This is used to have only a limited set of data and need to generate a probability distribution (n) from number of cases where no shot has been recorded. Use Beta(17,85). This also works for must generate a probability distribution of a number of (n) trials done with a series of there is some possibility of failure. primarily the place of the number of events to extrapolate the data obtained that occur. from a sample to the entire population.	determine the probability of an event. If you	If in 100 (n) shots from a firearm, 16 (r) miss, what is the a series of them. Note that this gives an estimate of the for the probability distribution of a limited set of data and recorded successes (r). This distribution is used for events or series of events, where

Source: Price Waterhouse Coopers (1999) and [Morgan and Henrion

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A.6. THE CORRELATION BETWEEN THE INPUT VARIABLES

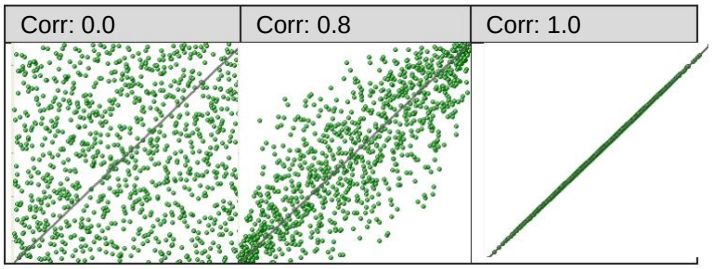
A common mistake in uncertainty propagation analysis is ignoring the correlation between input variables. If two variables are correlated, then the state of one gives us

Information about the possible occurrence of another. Not considering autocorrelation can lead to an underestimation of the overall level of uncertainty, and can also result in situations that could not occur in practice. For example, considering the base interest rate and the mortgage interest rate as independent variables could produce a nonsensical scenario where the base rate is higher than the mortgage interest rate.

Correlation is one of the most difficult aspects of risk quantification, and many definitions exist. The most common definitions of correlation are Pearson's and Spearman's rank correlation coefficient. Pearson's correlation coefficient (r) is a measure of the degree of linearity between two variables, or the amount of dispersion when one variable is plotted against another. In other words, a correlation coefficient of r equal to 1 means not only that the two variables move together, but that they move together linearly. It is quantified by the correlation coefficient r, which can vary continuously between -1 and +1, depending on the level of correlation.

If r is 1, it means the variables are perfectly positively correlated, meaning they move together. If r is -1, it means the variables are perfectly negatively correlated, meaning they always move in opposite directions. If r is 0, it means the variables are completely independent. The following figure graphically illustrates the concept of linear correlation.

Figure 38: Examples of rank correlation between variables.



Source: Own elaboration.

The disadvantage of Pearson's correlation coefficient is that it doesn't work if the relationship isn't linear. This problem is addressed by using Spearman's rank correlation. In rank correlation, both variables are ranked, and the Pearson correlation coefficient of the two rankings is compared. Rank correlation is used by most Monte Carlo simulation add-in software (e.g., Crystal Ball and @Risk).

There is an important distinction between correlation and dependence. An example of dependence is when one event can only occur if another happens. This should not be modeled as a 100% positive correlation, as this introduces the additional assumption that if the first occurs, the second will also occur.

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Second, it will definitely happen. In particular, the dependency should be modeled as an “IF - THEN” type decision through formulas or scenarios [PWC 1999].

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A.7. COST REDUCTION CURVES DERIVED FROM MODELS

The **model-based approach** seeks to calculate the cost and mitigation potential of the emissions through modeling. A number of energy models have been used for this purpose. method using a series of techniques. The most common way is to distinguish between top-down economic models and bottom-up engineering-oriented models. In both cases, the curves of Reductions are generated by summarizing the resulting CO2 price in "runs" with different limits strict emissions standards, or with the resulting emissions for different CO2 prices. This approach "Absolute emissions" means that the presentation of a model-based mitigation curve generally does not contain any technological details.

Bottom-up energy system models are, in most cases, models of
Partial equilibrium models represent only the energy sector, in contrast to top-down models, which cover the endogenous economic responses of the economy as a whole and are typically general equilibrium models. Bottom-up models are simulation or optimization models that calculate a partial equilibrium either through the minimization of
the system costs or by maximizing consumer surplus and the
producer. Compared to top-down models, bottom-up models contain more
Details of the technologies throughout the transition from primary energy to useful energy. Top-down models are based on substitution elasticities, mostly estimated based on historical rates and assuming that they are valid in the future [Kesicki F. 2010 (b)].

Since both types of energy models assume rational agents with cost-efficient behavior, it is not possible to represent cost-independent market distortions. As a result, these models do not show negative abatement costs. Furthermore, the model-based approach does not allow for representing technological details such as substitution possibilities, their different costs, and their technical characteristics.

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A.8. COMMENTS ON THE COST- BENEFIT ANALYSIS .

The proper execution of a cost-benefit analysis begins with defining the purpose of the evaluation and the scope of the system. For example, costs and benefits can be considered from the perspective of potential investors or from the perspective of a national economy. What about the impacts on other members of society or other countries? This is where the differences between a social cost-benefit analysis and a financial and economic cost-effectiveness analysis must be considered.

The analysis of the expected incremental costs and benefits of a project is called financial cost-benefit analysis when only the perspective of investors is considered.
of the project. This type of analysis considers expected market prices (or, financial prices) as a starting point, regardless of whether these prices include indirect subsidies or taxes. The reason is that the overall (financial) returns of a project for investors depend on the evolution of market prices for the resources (inputs) used. The weighted average return on investment that potential financiers expect, at a minimum, before accessing project financing, determines the financial discount rate at which incremental costs and benefits are converted to present values [Jansen JC 2006].

The analysis of the expected incremental costs and benefits of a project is called economic cost-benefit analysis when the impact on the entire national economy is considered. To measure a project's impact on the national economy, market prices must be adjusted to account for the effects of government intervention. Indirect subsidies received from the public sector must be added to market prices, and indirect taxes eliminated. This yields efficiency prices, or economic prices, which reflect the cost to the national economy of using the resources. In principle, economic analysis should correct for externalities, both positive and negative, on the well-being of individuals other than the project's investors.

In practice, positive or negative externalities are not easily quantifiable, and therefore are not typically included in economic cost-benefit analysis. The economic discount rate, that is, the discount rate applied in developing a cost-benefit analysis of a project, is often considered to be the expected risk-free interest rate (the effective interest rate on long-term government bonds) plus a risk premium that reflects the anticipated risk related to the future macroeconomic growth rate. Although the risk associated with a project's costs and benefits is not perfectly correlated with the risk of macroeconomic growth, in practice an "official" discount rate is applied to a wide variety of public investment proposals. To avoid crowding out private capital with public capital, when setting the economic discount rate, the trend should be given by the average cost of capital in the private sector. In contrast, in conventional economic cost-benefit analysis, the

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Concern for intergenerational wealth tends to play a minor role in setting commonly applied discount rates [Jansen JC 2006].

The analysis of the expected incremental costs and benefits of a project is called social cost-benefit analysis when it considers the impact on the broader society. Therefore, the boundaries of the analysis system must be clearly defined.

The social cost-benefit analysis of a project is based on the financial cost-benefit analysis, or private cost-benefit analysis. In this analysis, market prices are used, while impacts . However, they are used that do not have a financial relationship with the investor are not considered. Therefore, the social analysis requires the following adjustments:

- Market prices must be corrected by government interventions: the Indirect subsidies should be incorporated into the price, and indirect taxes eliminated.
- Externalities must be identified and their social value assessed to the greatest extent possible possible
- A social discount rate should be applied. That is, a discount rate appropriate for the social cost-benefit analysis of a project, rather than a financial discount rate that is only relevant to financiers.

It is worth noting that both financial and economic analysis provide highly relevant information for the public policy decision-making process. While social analysis allows for prioritizing different measures from a set of options from a social perspective, financial analysis can shed light on the support that private investors might potentially require to carry out such actions.

There is a wide range of externalities that are recommended to be included, at least qualitatively, in a social cost-benefit analysis. Among these, the following stand out [IDB 2010]:

- Macroeconomic and income impacts of the measures. Additional models are often required to capture inter-industry effects and the various feedback mechanisms. A common alternative is the use of computable general equilibrium models.

- Stimulation of the development of new technologies. More restrictive GHG reduction policies can stimulate technological development and innovation that reduce the demand for fossil fuels. In this way, it can lead to improved competitiveness for a country. Generally, technical knowledge, usually based on expert opinion, is modified exogenously.

- Impacts on employment. In general, it is desirable to estimate how a measure (or set of measures) impacts direct, indirect (in the supply chain), and secondary (in other industries) employment. Input-output and computable general equilibrium models are typically used to perform this type of analysis.

- Impacts on Local Pollution. GHG reduction policies and measures can have significant benefits on air quality. There are two main approaches to assessing the social value of the co-benefits of air pollution reduction. The first is to evaluate the cost of damage caused by polluted air (for example, in terms of health or through some estimate of willingness to pay for reduction).

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of a unit of contaminant). The second option is to use an approach of abatement costs avoided by achieving compliance with local standards.

- Impacts on Energy Security of Supply. GHG reduction policies and measures often have significant benefits in terms of long-term improvements in energy security of supply. In this context, energy security of supply is understood as “the availability of energy at all times in various forms, in sufficient quantities, and at affordable prices.” Estimating this externality in monetary terms is often difficult. However, in many countries, it is possible to estimate the reduction in expected failure costs of electrical systems using complex electrical models. • Costs of Public Programs. These costs include all government expenditures allocated to the implementation and monitoring of climate change programs.

- Other benefits and costs that are more difficult to quantify: Reduction of urban congestion, improvement in water supply, decrease in soil erosion, etc.

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A.9. SELECTION OF THE DISCOUNT RATE

It is common to assign a larger absolute value (positive or negative) to a given unit of benefit received today compared to one received in the future. Therefore, future benefits and
The costs associated with a greenhouse gas mitigation option must be
discounted at some discount rate to arrive at their present value. If a
Cost-benefit analysis in real terms, future values are expressed in dollars (or another currency) with the purchasing power of a given moment, for example, dollars of the year
2007. In that case, a real discount rate must be applied, that is, the rate of
The discount takes into account projected general price inflation. From now on, discussions will be based on real rates.

The choice of discount rate can strongly affect the relative attractiveness of an option
technological. Compare, for example, an operating-intensive generation option, such as a combined-cycle gas turbine, versus a capital-intensive generation option such as wind turbines. Assuming we use only a common discount rate, the choice
A "high" rate, in the range of 10%-12%, will positively affect operating expense-intensive technology, while the opposite choice, a "low" rate, in the range of 4-6%, will favor projects that are intensive in initial capital.

Choosing a "high" rate will generally favor fuel-based technologies.
fossils, discounting future fossil fuel expenses quite heavily, while
that the opposite will enhance the outcome of renewable options. This holds true even when
It attempts to account for the environmental externalities of fossil fuel use, including
greenhouse gas emissions and local pollutants (SOx, NOx, particulate matter, etc.)

It is a common practice of international organizations such as the International Agency for Development and Cooperation (IADC). Energy (IEA) and the World Bank use a discount rate to convert all values
Futures are priced at current values, regardless of the nature of the technology, the type of cost, or the benefit category. A high rate of around 8-10% is typically chosen.
compared to usual risk-free interest rates, which in OECD countries have
has been in the range of 1-3% (discounting inflation) in recent years. This practice is usually
justified by referring to the greater or lesser "opportunity cost of capital" in the
market.

In other words, a country must begin to invest its available capital resources in the
projects with the highest rate of return, followed by those with the next highest
performance, etc. This process must continue until all national capital resources are
spend. The return on the last unit of investment should match the OCC. One the rate of
Social discount expresses, among other things, a social premium associated with a marginal unit of
consumption on a marginal unit of future consumption in the context of opportunities
Productive investment and the functioning of local financial markets. Independent of

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This consideration, in practice, makes it difficult to determine the values of OCC and the social discount rate.
Establishing this practice in a solid manner, even from a theoretical perspective, is likely
and controversy [JANSEN JC 2006].

Some studies on long-term discount rates have also made it clear that a constant discount rate has only limited justification, and that recommendations for better practice are possible. Recent literature argues that discount rates vary over time and generally decrease as the evaluation horizon increases.

One of the most relevant works that deals with the topic of discounting is that of Arrow (1996).
According to Arrow, the discount rate is estimated as = + where d is the discount rate of consumption, is the pure time preference rate, is the marginal utility elasticity of consumption (in magnitude), and yg is the consumption growth rate. The first term indicates that present consumption is valued more than future consumption. The second term shows that the future consumer will be better off than the current consumer, so consumption must be shifted to the present.

In the context of Arrow, the Stern Review recommends a low discount rate.32 This is because considering high rates undervalues future generations. Using a discount rate of 1.4%, the Stern Review indicates that 1% of global output should be spent on efforts to curb climate change.

According to Nordhaus, the correct way to value future benefits is to consider how people actually value the future based on the opportunity cost of capital.
Nordhaus uses a rate of 5.5%, which yields considerably more conservative results regarding the necessary efforts. In fact, he indicates that only 0.1% of global output should be spent.

Hyperbolic discounting assumes a high discount rate in the near future and a low rate over longer periods. A hyperbolic rate allows the value function to be transformed into a hyperbola. This increases the value of future cash flows compared to traditional techniques.

[32] Citing empirical values, they take = 1, = 1.3 and = 0.1, which gives a rate of 1.4%.

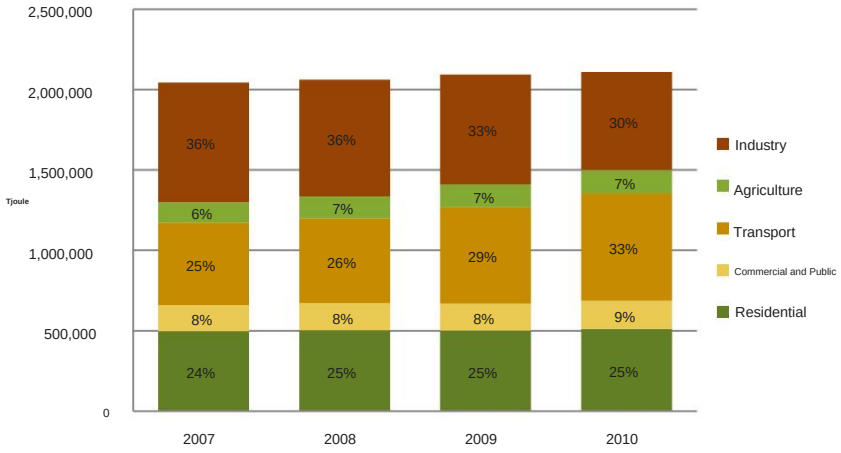
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A.10. ENERGY ANALYSIS OF COUNTRIES IN THE REGION.

A.10.1. ARGENTINA

Argentina's national energy balance has a lower level of disaggregation, distinguishing only the energy consumption of the commercial and public, residential, transportation, industrial, and agricultural sectors. The following figure shows energy consumption in these sectors and its percentage distribution for the period 2007-2010.

Figure 39: Sectoral energy consumption in Argentina 2007-2010 (TJoule)



Source: Prepared by the author based on the national energy balances of Argentina 2007-2010

The Republic of Argentina has the latest version of the 2nd National Communication to the United Nations Framework Convention on Climate Change published during the year 2007.
Currently, work is underway for the 3rd Communication.

The second communication includes the emissions inventory for the year 2000. Combustion emissions are presented below.

Table 28: CO2e emissions, Argentina 2000.

Sector	MMtCO2e	%
Generation	36,08	30,1%
Manufacture	15,30	12,8%
Transport	40,24	33,6%
Residential	17,25	14,4%
Commercial	3,17	2,6%
Agriculture	7,59	6,3%
Others	0,29	0,2%
Total combustion	119,89	100%

Source: Prepared by the author based on the second national communication of Argentina 2007.

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The table shows that a significant portion of the emissions are generated during the transformation process. These emissions, although generated during electricity generation, are produced in an effort to meet the demand for electricity, which is primarily driven by the manufacturing sector followed by the residential sector.

The following are some mitigation measures identified in the 2nd Argentine communication as potential or currently being implemented:

Table 29: Mitigation measures proposed in the 2nd Argentine national communication.

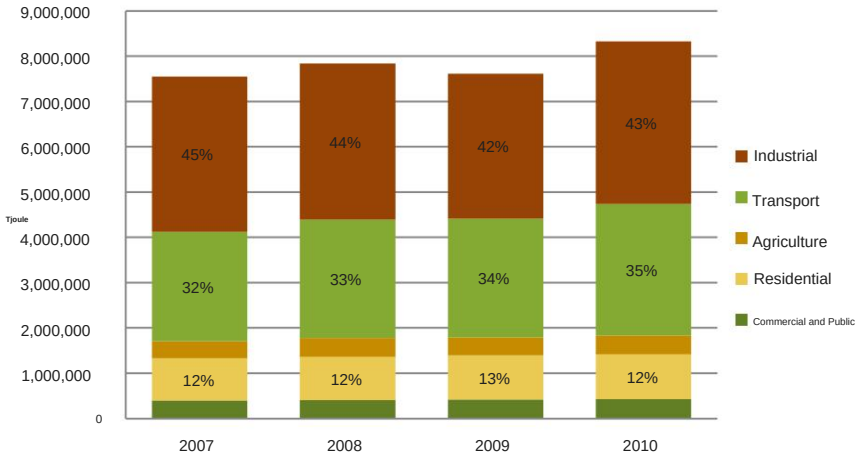
Measurement	Sector
Efficient public lighting	Public
Energy Efficiency Program (PURE) CPR	
CPR thermal conditioning	CPR
Incandescent light bulb replacement	CPR
Refrigerator replacement	CPR
Cogeneration systems	Industry
Fuel change	Transport
Good handling practices	Transport
Speed control	Transport
Modal shift from truck to rail	Transport

Source: Prepared by the author based on the second national communication of Argentina 2007.

A.10.2. BRAZIL

Brazil's national energy balance has a higher level of disaggregation for the industrial sector, allowing for the differentiation of energy consumption by subsector. The following figure shows energy consumption as a percentage of final energy demand, and its distribution during the period 2007-2010.

Figure 40: Sectoral energy consumption in Brazil 2007-2010 (TJoule)

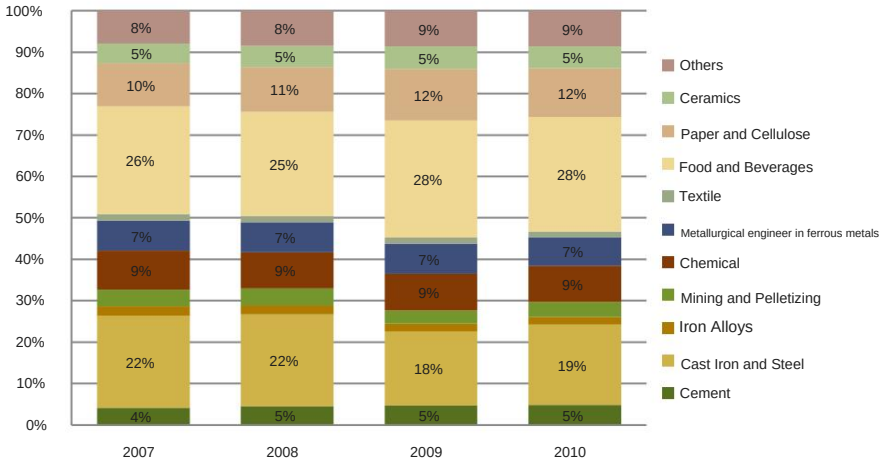


Source: Prepared by the author based on Brazil's national energy balances 2007-2010

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The following figure shows the breakdown of energy consumption in the industrial sector.

Figure 41: Energy consumption of industrial subsectors in Brazil 2007-2010



Source: Prepared by the author based on Brazil's national energy balances 2007-2010

Among the industrial subsectors, the food and beverage sector stands out with 28% of the industry's energy consumption during the years 2009-10, followed by the most important sectors: Cast Iron and Steel (19%), Paper and Pulp (12%), and the chemical sector (9%).

Brazil has the latest version of its 2nd National Communication to the Framework Convention on Human Rights.
United Nations on Climate Change, published in 2010 by the Ministry of Science and Technology.

The second communication includes the emissions inventory for the year 2005. Combustion emissions are presented below.

Table 30: CO2e emissions, Brazil 2005.

Year 2005	MtCO2	%
Energy subsector	48.601	16%
Industry	75.620	25%
Steel industry	16.467	5%
Chemical industry	15.446	5%
Other	43.707	15%
Transport	133.431	44%
Air	5.374	2%
Road worker	122.765	41%
Others	5.291	2%
Residential	15.484	5%
Agriculture	14.809	5%
Others	11.996	4%
Total	299.941	100%

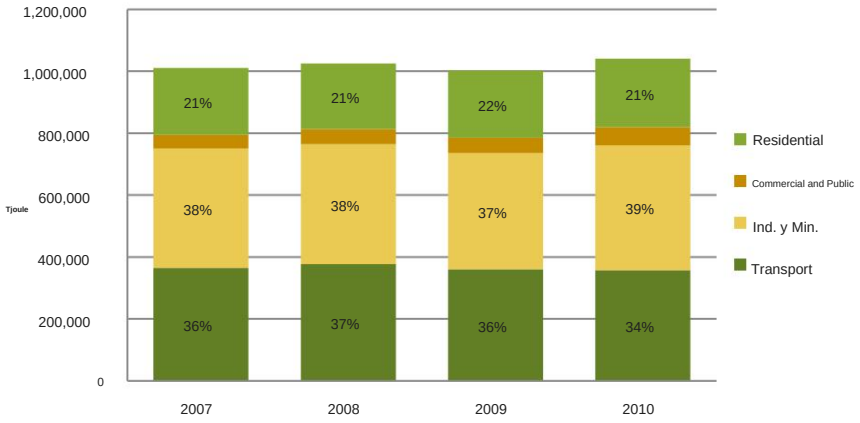
Source: Prepared by the author based on Brazil's second national communication 2010.

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A.10.3. CHILE

Chile's national energy balance has a higher level of disaggregation for the industrial sector, allowing for the differentiation of energy consumption by subsector. The following figure shows energy consumption as a percentage of final energy demand, and its distribution during the period 2007-2010.

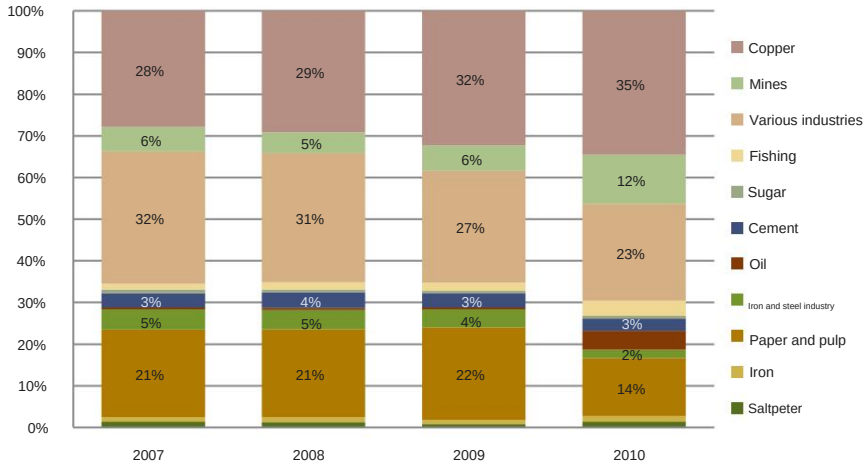
Figure 42: Sectoral energy consumption in Chile 2007-2010 (TJoule)



Source: Prepared by the author based on Chile's national energy balances 2007-2010

The following figure shows the breakdown of energy consumption in the industrial sector. Of particular note is the mining sector's energy consumption, which increased from 34% in 2007 to 47% in 2010, driven primarily by an increase in the energy intensity of the copper sector. Secondly, the pulp and paper industry stands out, having decreased from an average of 21% in 2007-2009 to 14% in 2010.

Figure 43: Energy consumption of industrial subsectors in Chile 2007-2010



Source: Prepared by the author based on Chile's national energy balances 2007-2010

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Chile's latest version is the 2nd National Communication to the Framework Convention on Human Rights.
United Nations on Climate Change, published in 2011 by the Ministry of Environment.

The second communication includes the emissions inventory for 2006. Combustion emissions are presented below.

Table 31: CO2e emissions, Chile 2006.

2006	MtCO2e	%
Energy industry	20.751	36%
Manufacturing, construction and mining	13.170	23%
Transport	17.062	30%
Public, residential and commercial	4.058	7%
Fishing	316	1%
Fugitive emissions	1.435	2%
Firewood and biogas	1.013	2%
Total	57.806	100%

Source: Prepared by the author based on the second national communication of Chile 2011.

The following are the mitigation measures identified in the 2nd Chilean communication as potential or underway:

Table 32: Mitigation measures proposed in the 2nd Chilean national communication.

Measurement	Sector
Energy efficiency labeling	CPR
Thermal regulations for housing	CPR
Minimum Energy Efficiency Standards (MEPS)	CPR
Advance replacement of efficient engines	Industry
Cogeneration	Industry
Efficient showers	Residential
Efficient residential lighting, Reduction	Residential
of standby losses, Efficient residential	Residential
cooling, Home insulation, Solar collectors,	Residential
Efficient commercial	Residential
cooling, Use of biofuels,	Residential
Efficient driving (Ecodriving)	Commercial
	Transport
	Transport
Modal shift due to the construction of new kilometers of Metro Aerodynamic	Transport
improvements Truck fleet	Transport
renewal Light vehicle fleet renewal Hybrid	Transport
vehicles for fleet renewal Energy efficiency in	Transport
copper Source: Prepared by the author based on the	Transport
second national communication of Chile	Copper

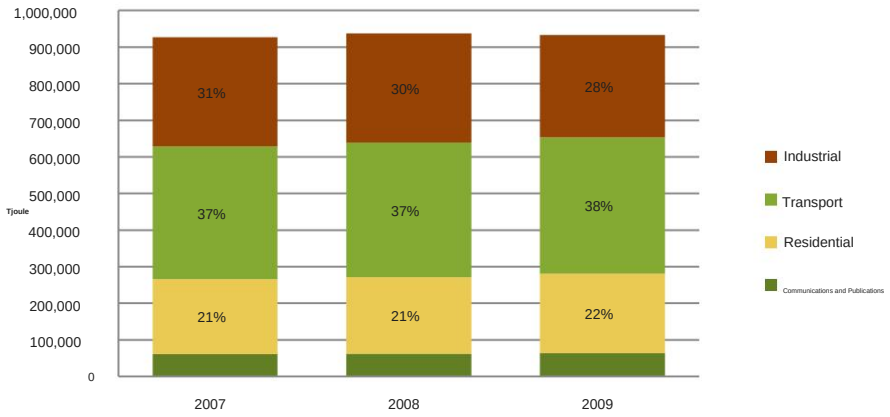
2011.

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A.10.4. COLOMBIA

Colombia's national energy balance provides a good level of disaggregation for the industrial sector, allowing for the differentiation of energy consumption across subsectors. The following figure shows energy consumption as a percentage of final energy demand, and its distribution during the period 2007-2009, according to the information available in the national energy balances.

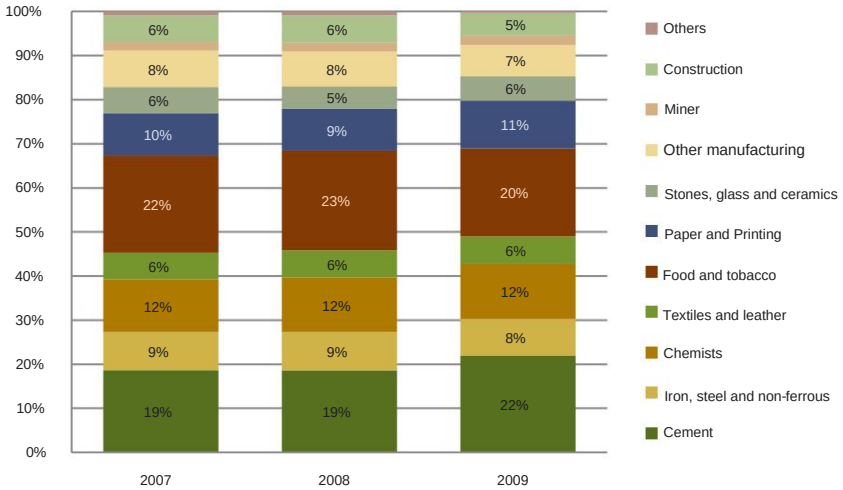
Figure 44: Sectoral energy consumption in Colombia 2007-2009 (TJoule)



Source: Prepared by the author based on the national energy balances of Colombia 2007-2009

The following figure shows the breakdown of energy consumption in the industrial sector.

Figure 45: Energy consumption of the industrial subsector in Colombia 2007-2009



Source: Prepared by the author based on the national energy balances of Colombia 2007-2010

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Similar to the Brazilian case, the food and tobacco sector is the most energy-intensive, along with the cement sector, followed by the chemical and paper and printing sectors; the energy demand of the Iron and Steel sector also stands out.

Colombia's latest version is the 2nd National Communication to the Framework Convention on Human Rights.
United Nations on Climate Change, published in 2010 by the Ministry of Environment, Housing and Territorial Development.

The second communication includes the emissions inventory for the year 2004. Combustion emissions are presented below.

Table 33: CO2e emissions, Colombia 2004.

Sector	MtonCO2e	%
Electricity Generation	16.845	23
Manufacturing and Construction Industries	13.940	19
Transport	19.987	27
CPR, Agricultural and others	6.494	9
Fugitive emissions	7.523	10
Coal mining	3.279	5
Oil and natural gas	4.243	6
Biomass burning	715	1
Total	**73.031**	**100**

Source: Prepared by the author based on Colombia's second national communication 2010.

The following are some mitigation measures outlined in the 2nd communication
Colombian:

Table 34: Mitigation measures proposed in the 2nd Colombian national communication

Measurement	Sector
Efficient residential and commercial lighting	CPR
Replacing coal boilers with natural gas	CPR
More efficient boilers	CPR
Efficient residential cooling	Residential
Solar collectors	Residential
Restructuring of urban public bus service	Transport
Electric light vehicles	Transport
Electric mass transit vehicles	Transport
Use of biofuels	Transport
Increased occupancy level of private vehicles	Transport
More efficient electric motors	Transversal

Source: Prepared by the author based on Colombia's second national communication 2010.

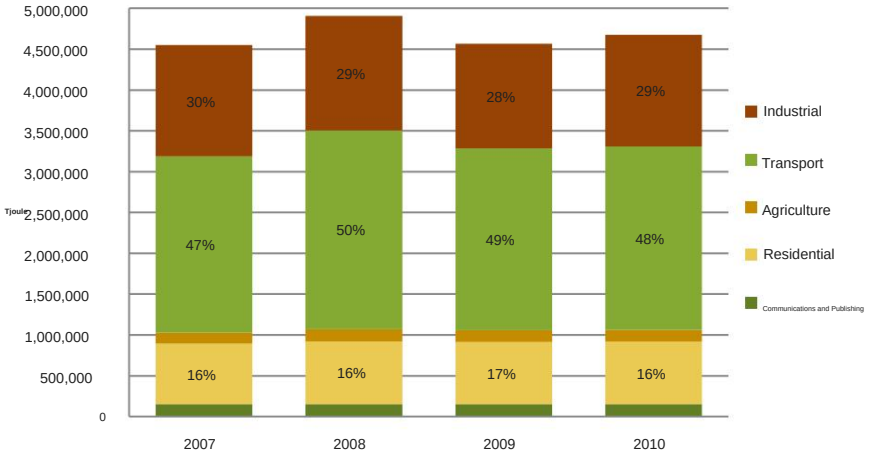
A.10.5. MEXICO

Mexico's national energy balance has a higher level of disaggregation for the industrial sector, allowing for the differentiation of energy consumption across subsectors. The following figure

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It shows the energy consumption of the final energy demand, and its percentage distribution in the period 2007-2010 according to the information available in the national energy balances.

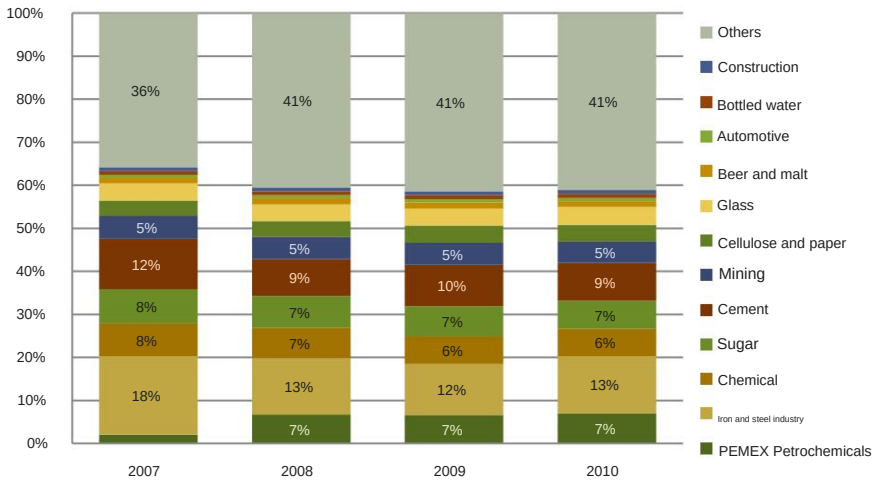
Figure 46: Sectoral energy consumption in Mexico 2007-2010 (TJoule)



Source: Prepared by the author based on Mexico's national energy balances 2007-2010

The following figure shows the breakdown of energy consumption in the industrial sector.

Figure 47: Energy consumption of industrial subsectors in Mexico 2007-2010



Source: Prepared by the author based on Mexico's national energy balances 2007-2010

Mexico shows a highly diversified industry, in which the sectors with the highest energy demand are the steel and cement sectors, with average shares of 13% and 10% respectively in the years evaluated.

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Mexico's latest version is the 4th National Communication to the Framework Convention on Human Rights.
United Nations on Climate Change, published in 2009 by the Ministry of Environment and Natural Resources.

The second communication includes the emissions inventory for 2006. Combustion emissions are presented below.

Table 35: CO2e emissions, Mexico 2006.

Sector	MtonCO2e	%
Energy generating industry	149.137	31
Manufacturing and construction industry	56.832	12
Transport	144.690	30
Other sectors	32.042	7
Fugitive emissions	47.395	10
Solid fuels	2.410	1
Oil and natural gas	44.984	9
Total	**477.490**	**100**

Source: Prepared by the author based on Mexico's fourth national communication 2009.

The following are some mitigation measures outlined in Mexico's 4th communication:

Table 36: Mitigation measures proposed in the 4th Mexican national communication.

Measurement	Sector
Efficient residential and commercial lighting	CPR
Environmental conditioning	CPR
Efficient residential cooling	CPR
Solar Collectors	Residential
Scrapping	Transport
More efficient electric motors	Industrial

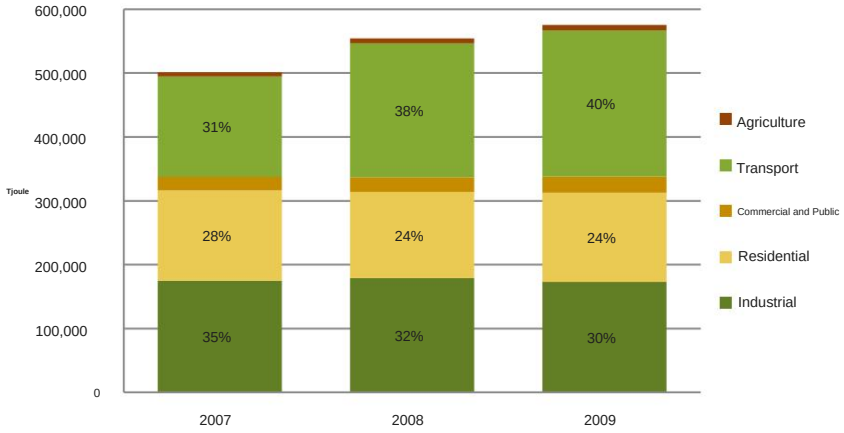
Source: Prepared by the author based on Mexico's second national communication 2009.

A.10.6. PERU

Peru's national energy balance has a lower level of disaggregation for the industrial sector, only distinguishing the energy consumption of non-metallurgical mining, fishing, and the rest of the sector as a whole. The following figure shows energy consumption as a percentage of final energy demand and its distribution during the period 2007-2009, according to the information available in the national energy balances.

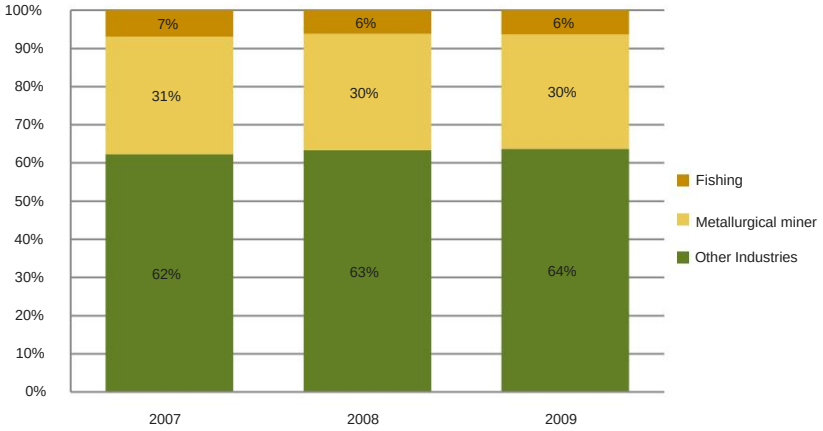
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Figure 48: Sectoral energy consumption in Peru 2007-2009 (TJoule)



Source: Prepared by the author based on the national energy balances of Peru 2007-2009

Figure 49: Energy consumption of industrial subsectors in Peru 2007-2009



Source: Prepared by the author based on the national energy balances of Peru 2007-2009

The Peruvian balance does not allow a detailed breakdown of the industrial sector, however the importance of the mining and metallurgical sector can be appreciated, with 30% of the energy demand of the industry.

Peru's latest version is the 4th National Communication to the Framework Convention on Human Rights.

United Nations on Climate Change, published in 2010 by the Ministry of the Environment.

The second communication includes the emissions inventory for the year 2000. Combustion emissions are presented below.

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Table 37: CO2e emissions, Peru 2000.

Sector	MtonCO2e	%
Energy industries	3.083	12
Manufacturing and Construction Industries	3.260	13
Transport	9.938	39
CPR and Agriculture	5.224	21
Fishing	2.127	8
Mining	1.357	5
Solid Fuels	4	0
Oil and Natural Gas	407	2
Total	**25.400**	**100**

Source: Prepared by the author based on Peru's second national communication 2010.

The following are some mitigation measures outlined in the 2nd communication
Peruvian:

Table 38: Mitigation measures proposed in the 2nd Peruvian national communication.

Measurement	Sector
Efficient residential and commercial lighting	CPR
Environmental conditioning	CPR
Efficient rural kitchens	Residential
Solar Collectors	Residential
Scrapping	Transport
Biofuels	Transport
Ecodriving	Transport
Hybrid vehicles	Transport
Efficient boilers	Industrial
Electric motors	Transversal

Source: Prepared by the author based on Peru's second national communication 2010.

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A.11. ENERGY EFFICIENCY ACTIONS IN THE INDUSTRIAL AND MINING SECTOR AND TRANSPORT

The following tables detail the different actions and programs studied by sector and subsector.

Table 39: Energy Efficiency Actions in the Industrial and Mining Sector

Subsector	Action	Action description	Expected Reduction	Fountain
Industries and various mines	Replacement of Motors for industrial and mining use (*)	a) Projects new. Engine installation efficient in new industries and tasks mining, or expansions of existing facilities. b) Engine replacement obsolete. c) Use of Variable Frequency Drives of VSD frequency in variable flow processes. **Supposed:** Penetration from 10% to 60% until 2016 in the form lineal	The three measures They generate savings of 208.7 GWh/year. This corresponds to 0.325% of total consumption. for the year 2016 (PRIEN 2008)	"Characterization of the current park of electric motors in Chile" (PRIEN 2006) and "Estimation of the potential for energy savings, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Industries and various mines	Labeled of Energy Efficiency in Motors and VSDs	Incorporate equipment and industrial systems of use generalized al National Program of Efficiency Labeling Energy	893 GWh for the first year of labeling expansion	PRIEN estimate 2007 and PPEE Performance Agreement-SEC 2007
Industries Several	Use of efficient electrical devices	The use of light fixtures efficient and the application of programs such as "Energy Star" has achieved a reduction in consumption of energy of for computers and office equipment.	It is possible to achieve savings of the order of 1% of annual consumption total energy	"Estimate of the energy saving potential, of through improvements in the energy efficiency of the various sectors (PRIEN 2008)
Industries Several	Use of heat wasted	Even in industries where a high level has been reached Despite the degree of energy conservation, there is still a high potential for heat recovery.	It is estimated that it is possible to achieve increases 2.5% efficiency annual for 15 years.	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct

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Subsector	Action	Action description	Expected Reduction	Fountain
				sectors (PRIEN 2008)
Industries Several	Recycling of and reuse of materials	Recycling of materials with high content Energy consumption has increased over the last fifteen years in most industrialized countries. For example, the rate of crude steel production, obtained by he Scrap processing of is at 43% globally. In developed countries, the rates of The recycling rates for non-ferrous metals are 30% for aluminum, 35% for copper, and 25% for zinc.	He recycling of materials can provide increases of efficiency of 0.9% annual	"Estimate of the energy saving potential, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Cellulose and Paper	Cogeneration	Cogeneration systems efficient to the production of steam, hot water, and electricity by means of burning liquors, barks, and other wood waste.	The consumption Specifics in Chile have decreased from 0.6 to 0.3 Tep/ ton in the period 1990-2004. Internationally, he reached returns of the order of 0,2 Tep/ton. It is estimated that the efficiency measures energy will involve the improvements in yields reaching consumption on the order of 0.2 Tep/ton. In 15 years (by 2021), the potential of reduction annual It would reach 2.2%.	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)

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Subsector	Action	Action description	Expected Reduction	Fountain
Iron and steel industry	Application of heat recovery technologies	Further efficiency gains can be achieved through he waste preheating; of better use of coal and oxygen; and coke recycling in ovens. Consideration should be given to that there are 2 ways to produce different steel with energy requirements: the reduction-oxidation process that receives iron ore in The blast furnaces. second is the process electric what is fueled by scrap metal steel, which is melted in electric ovens.	7% savings on Energy consumption in 2021 according to PRIEN (2008). Specific consumption in Chile is between 0.4 and 0.5 Tep/ton of steel produced, and internationally it is of 0.3 Tep/ton (CNE 2004	"Estimate of the energy saving potential, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Petrochemistry	Redesigns in the processes and a reduction in the chemical waste		Increases in the energy efficiency of 2.6% per year. (In EU countries – the intensity 15, energetic ha decreased by 40% between the years 1990 and 2004)	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Cement	Reductions in the energy use through modifications al process, specifically to the operation of ovens.	The new ovens of Dry processes are more efficient than wet processes because they do not require additional energy to with remove moisture. In a dry process, the gases from dry, emissions from the furnace They are used to pre-calcine the raw material and eliminate the presence of water. These processes can use 50% less energy than a conventional (wet). The industry national of ha	The consumption Specific values range between 0.07 and 0.09 Tep/ton. A consumption goal specific of 0.06 Tep/ton over the 15-year period, which would mean a savings accumulated to year 15 of 22%	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)

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Subsector	Action	Action description	Expected Reduction	Fountain
		incorporated is modifications.		
Sugar mill	Systems of cogeneration and heat recovery		Specific consumption can be reduced from 0.267 Tep/ton to 0.184 Tep/ton (specific consumption recorded in Belgium for the year 1999). Savings accumulated over 15 years would reach 31% of reported consumption in 2006.	"Estimate of the energy saving potential, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Fishing	Improved of the operation of propulsion engines in ships, of the heat cycles in the flour manufacturing of fish and the improvement of pumps and small engines		The annual savings potential over a 15-year period is 0.5%.	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Mining of copper	Replacement of Motors for industrial and mining use (*)	a) Projects new. Engine installation efficient in new industries and tasks mining, or expansions of existing facilities. b) Engine replacement obsolete. c) Use of Variable Frequency Drives of VSD frequency in variable flow processes. Penetration from 10% to 80% until 2016 in the form lineal	The three measures They generate savings of 622.5 GWh/year with regarding the projected scenario for the year 2016 (PRIEN 2008)	"Characterization of the current park of electric motors in Chile" (PRIEN 2006) and "Estimation of the potential for energy savings, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Mining of copper	Processes hydrometallurgical		Potential of improvement of 0.5% per year. Year 10	"Estimate of the energy saving potential, of through improvements in the energy efficiency of the various sectors (PRIEN 2008)

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Subsector	Action	Action description	Expected Reduction	Fountain
Mining of copper	Production of cathodes by means of pyrometallurgy		Potential of improvement of a 15% annually. Year 10	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Great Industry and other mining	Replacement of Motors for industrial and mining use (*)	a) Projects new. Engine installation efficient in new industries and tasks mining, or expansions of the existing facilities. b) Replacement of obsolete engines. c) Use of Variable Frequency Drives of VSD frequency in variable flow processes. Penetration from 10% to 80% until 2016 in the form lineal	The three measures They generate savings of 649.6 GWh/year compared to the projected scenario for the year 2016 (PRIEN 2008)	"Characterization of the current fleet of electric motors in Chile" (PRIEN 2006) and "Estimation of the potential for energy savings, through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Iron	Heat recovery, of use of efficient equipment and increased of the use of material recycling		The potential for improvement can be up to 4.1%.	"Estimate of the energy saving potential, of through energy efficiency improvements of the distinct sectors (PRIEN 2008)
Saltpeter	Heat recovery of		Savings of 2.7% per year	"Estimate of the energy saving potential, of through improvements in the energy efficiency of the various sectors (PRIEN 2008)
	Introduction of energy-efficient equipment efficient		Savings of 0.5% year	"Estimate of the energy saving potential, of through improvements of the

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Subsector	Action	Action description	Expected Reduction	Fountain
				energy efficiency of the different sectors (PRIEN 2008)

(*)For economic analysis, two types of motors are distinguished: standard motors and efficient or Premium motors. • Standard motors have an efficiency equal to or worse than that defined by NEMA Standard 12-10 of 1998. This efficiency also corresponds to the EFF2 designation of the European Labelling Scheme. • Efficient or NEMA Premium motors are those that comply with the NEMA Premium Standard (NEMA Standards MG 1-2003) and correspond, very closely, to the EFF1 designation of the European Labelling Scheme.

Source: Prepared by the author based on energy efficiency studies

Table 40: Energy Efficiency Actions in the Transport Sector

Subsector	Measurement	Description of the measure	Expected Reduction	Fountain
Sector transport road worker	Improvements in engines, vehicle weight, air resistance and maintenance.		Savings potential of a 12% (a savings growth rate of 0.85% per year) to 2021, which is equivalent to 15,396 GWh/ year.	Estimation of the potential for energy savings through improvements in the energy efficiency of different sectors (PRIEN 2008)
Subsector trucks	Age Change Maximum Truck Size	Two are studied cases: Ages maximums of 25 and 20 years old.	Reductions 113,900-152,400 (m3/year) for 25 and 20 year scenarios maximum age. Source: MTT Presentation (2007).	"Efficiency Analysis Energy in Transport Intercity Cargo" Ministry Presentation Transport and Telecommunications 2007
Sector transport air	Introduce new engine concepts		Energy savings of 20 to 30%	Estimation of the potential for energy savings through improvements in the energy efficiency of different sectors (PRIEN 2008)
Sector transport maritime	Energy efficiency of the engines maritime		5 to 10% savings fuels	Estimation of the potential for energy savings through improvements in the energy efficiency of different sectors (PRIEN 2008)

Source: Prepared by the author based on energy efficiency studies

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A.12. ESTIMATED GLOBAL POTENTIAL FOR **CO2 REDUCTIONS BY 2030**

Table 2 Estimated potential for CO_2 emission reduction in 2030

	Area[a]	2030 production (Mt)[b]		Mitigation potential (%)	Cost range, ($/tCO$_2$-eq)	Mitigation potential (MtCO$_2$-eq/year)	
		A1	B2			A1	B2
CO_2 emissions from processes and energy use							
Steel[c,d]	Global	1,554	1,578	15–40	<50	430–1,500	420–1,500
	OECD	436	388	15–40	<50	90–300	80–260
	EIT	176	193	25–40	<50	80–240	85–260
	Dev. Nat.	941	997	25–40	<50	260–970	250–940
Primary aluminum[e,f]	Global	49	43	15–25	<100	53–82	49–75
	OECD	12	12	15–25	<100	16–25	15–22
	EIT	6	6	15–25	<100	12–19	8–13
	Dev. Nat.	31	25	15–25	<100	25–38	26–40
Cement[g,h,i]	Global	5,524	4,418	11–40	<50	720–2,100	480–1,700
	OECD	596	553	11–40	<50	65–180	50–160
	EIT	313	219	11–40	<50	40–120	20–60
	Dev. Nat.	4,615	3,645	11–40	<50	610–1,800	410–1,500
Ethylene[j]	Global	329	218	20	<20	85	58
	OECD	138	147	20	<20	35	40
	EIT	19	11	20	<20	5	3
	Dev. Nat.	171	60	20	<20	45	15
Ammonia[k,l]	Global	199	195	25	<20	110	100
	OECD	20	18	25	<20	11	10
	EIT	19	22	25	<20	10	12
	Dev. Nat.	159	155	25	<20	87	80
Petroleum refining[m]	Global	4,838	4,697	10–20	Half <20	150–300	140–280
	OECD	2,220	2,123	10–20	Half <50	70–140	67–130
	EIT	412	415	10–20	Half <50	12–24	12–24
	Dev. Nat.	2,206	2,160	10–20	Half <50	68–140	65–130
Pulp and paper[n]	Global	1,226	927	5–40	<20	49–420	37–300
	OECD	657	536	5–40	<20	28–220	22–180
	EIT	62	42	5–40	<20	3–21	2–13
	Dev. Nat.	508	349	5–40	<20	18–180	13–110
Other industries, electricity conservation							
	Global				25% <20	1,100–1,300	410–540
	OECD				25% <50	140–210	65–140
	EIT				50% <100	340–350	71–85
	Dev. Nat.				-[d]	640–700	280–320

Table 2 (continued)

	Area[a]	2030 production (Mt)[b]		Mitigation potential (%)	Cost range, ($/tCO$_2$-eq)	Mitigation potential (MtCO$_2$-eq/year)	
		A1	B2			A1	B2
Total							
Sum[o,p,q]	Global					2,300–7,500	1,500–6,100
	OECD					400–1500	300–1,300
	EIT					500–900	200–600
	Dev. Nat.					1,500–4,600	1,000–3,800

Results are presented for selected energy-intensive industries and for three world regions. Impact of increased recycling is included in the potentials as (material) efficiency improvement. Note that it was impossible to distinguish fuel mix effects from efficiency changes. However, fuel mix effects are generally very small, except for the cement and pulp and paper industries

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A.13. INFORMATION AVAILABILITY ANALYSIS

This section analyzes the availability of information associated with each variable relevant to the construction of an end-use model. To this end, the institutions that might possess such information are first identified and subsequently interviewed. In particular, a distinction will be made between public information, private information, and confidential and non-existent information.

In order to conduct the information search, a group of institutions has been identified Public and private institutions that have participated in the implementation of EE measures and those that are potential holders of relevant information. The following table lists the institutions associated with each productive sector:

Table 41: Relevant institutions in the search for information

Transport sector	Industry and mining	Commercial	Public	Residential
ANAC	CPC	Table of Ef. Energy in Retail	MIDEPLAN	CHILECTRA
TRANSANTIAGO	SOFA	Cencosud, D&S, etc.	MOP	TRANSELEC
METRO	National Council of Cleaner Production	Inspects National Economic	Ministries	Superintendence of Electricity and Fuels
Ministry of Transport and Telecommunications	Mining Roundtable Energy Efficiency in Large-Scale Mining	PPEE (audits) energy)	Undersecretariat of Regional Development	CONAMA
SECTOR	COCHILCO	I HAVE	Intendancies and Governments Regional	SERNAC
CORFO	Ministry of Economy		Address of Budget	Municipalities
I HAVE	Ministry of Mining		Superintendence of Electricity and Fuels	CORFO
PPEE	CORFO		Chile Buys	CDEC
DGAC and JAC	SEC		Chilean Association of Municipalities	CNE
General Directorate of Maritime and Merchant Marine Territory	CONAMA		Municipalities	MIDEPLAN (CASEN and others)
	Public companies		CNE	MINVU
	CNE		PPEE	PPEE
	INE (ENIA)		I HAVE	I HAVE
	PPEE			

Source: Prepared by the author based on the “Strategic Plan 2007-2015 of the Chilean Country Energy Efficiency Program”, PPEE, December 2007.

The information availability analysis was developed in two stages. The first stage consisted of analyzing the information available on the institutions' websites, such as surveys,

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online studies and information. The second stage consists of conducting interviews with some of the aforementioned institutions in order to clarify and detail the information.

As a result of both stages of information search, with emphasis on the first, the following information catalogs are presented for each sector, detailing the selected use or subprocess, its relevant variables, the existing specialty, the temporality, source of information, the period of existence of the data, the location of the information and its status among others.

A.13.1. TRANSPORT SECTOR

The following presents the registry of the transport sector and the corresponding analysis of its information.

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Table 42: Cadastre of the Transport Sector

Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibility	Location of the information	Access	State	Comment
Automobiles lightweight and medium	Park Automotive	Regional	Annual	• I HAVE • Plants of Revision Technique - Sect	• 1975 – 2012 • 1998 - 2012	The information of PRT differs with the INE in 1998 and 1999. Since 2001 the information coincide.	• INE Statistics	• Public • Public (upon request)	• PDF, paper and BD • BD Acces	I HAVE and PRT: Information of cars private individuals, taxis, pickup trucks and vans, trucks, buses and others
Automobiles lightweight and medium	Sales of vehicles new	Regional	Annual	ANAC	1998-20012 (detailed), 1989-1993 (added)		www.anac.cl (partial information in pdf)	• Public • Restricted	• PDF • BD ANAC	Details available in several studies
Trucks	Park Automotive	Regional	Annual	• I HAVE • Plants of Revision Technique - Sect	• 1975 – 2012 • 1998 - 2012	The information of PRT differs with the INE in 1998 and 1999. Since 2001 the information coincide.	• INE Statistics • Sect	• Public • Public (upon request)	• PDF, paper and BD • BD Acces	I HAVE and PRT: Information of cars private individuals, taxis, pickup trucks and vans, trucks, buses and others
Trucks	Sales of vehicles new	Regional	Annual	ANAC	1998-2012 (detailed), 1989-1993 (added)		www.anac.cl (partial information in pdf)	• Public • Restricted	• PDF • BD ANAC	Details available in several studies
Buses	Park Automotive	Regional	Annual	• I HAVE • Plants of Revision Technique - Sect • Transantiago (urban buses)	• 1975 – 2012 • 1998 – 2012 • ND	The information of PRT differs with the INE in 1998 and 1999. From 2001 They agree.	• INE Statistics • Sect • Transantiago and operating companies	• Public • Public (upon request) • Restricted	• PDF, paper and BD • BD Acces • ND	I HAVE and PRT: Information of cars private individuals, taxis, pickup trucks and vans, trucks, buses and others
Transport	Antique,	National	Annual	I HAVE	ND	-	INE Statistics	Public	BD	Information by

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibility	Location of the information	Access	State	Comment
passengers	vehicles, traffic, consumption by urban transport, long-distance, school, etc.							(added). Paid (details)		enterprise.
Transport passengers	Vehicles of transport used according to capacity years and of antique	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Transport passengers	Number of services the lines, length of lines or routes, passengers transported, etc.	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Transport passengers	Tariff media per passenger-km. and vehicle-km.	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Passenger transport	Consumption of fuel and other expenses	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
All the modes vehicular	Consumption energy	National	Annual	CNE	1973 - 2012 --.		BNE CNE	Public	Excel, BD	
All the modes vehicular	Yields by vehicle type	National	-	Fuel Economy	1985 - 2012 --		http://www.fuelecono my.gov /feg/findacar.htm	Public	BD	Information of yields new vehicles

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibility	Location of the information	Access	State	Comment
All the modes vehicular	Traffic	common Santiago and toll highways	Diary	Highways Concessionaires	Variable from the start of the service	-	Highways Concessionaires	Restricted	ND	Access was not possible to these sources
All the modes vehicular	Past of vehicles per toll plaza	Monthly	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by toll plaza.
Metro	Consumption electricity	common Santiago	Annual	• CNE • Metro S.A.	1973 - 2012 Ok.		• BNE CNE • Metro Memory Santiago	Public	PDF, Excel, BD	Low consumption. Enough information CNE
Railway	Consumption electricity	Regional	Annual	CNE	1973 - 2012 Ok.		• BNE CNE • Metro Memory Santiago	Public	PDF, Excel, BD	Low consumption. Enough information CNE
Railway	Load and passengers transported per Km.	National	Monthly	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Trucks	and Passenger load per km.	National	Monthly	I HAVE	ND	-	INE Statistics	Public (added). Paid (detail)	BD	Information by enterprise.
Railway	Existence and power of the tractor equipment	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Railway	Existence and capacity to equipment passengers	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.
Railway	Material and combustible used in traction	National	Annual	I HAVE	ND	-	INE Statistics	Public (added). Paid (details)	BD	Information by enterprise.

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibility	Location of the information	Access	State	Comment
Transport maritime	Consumption energy domestic travel and international	National	Annual	CNE	1973 - 2012 --.		BNE CNE	Public	Excel, BD	
Maritime transport	Cargo transported and hours of occupation of spaceships	Regional	Monthly	Subtrans	2002 - 2012 --		http://www.subtrans.cl / subtrans/maritime/ statistics.php	Public	PDF, Excel,	Motion of cargo at state ports
Transport air	Consumption energetic flights national and international	National	Annual	• CNE • Aeronautical companies • System Statistician of Air Traffic (DGAC)	1973 - 2012 --.		• BNE CNE • www.dgac.cl	Public	• Excel, BD • PDF	Statistics of landings and takeoffs by airport

Source: Own elaboration

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A.13.2. ASSESSMENT OF THE STATE OF INFORMATION

The methodology for projecting energy consumption in the transportation sector is based on information about the existing stock of motor vehicles in Chile. This information is primarily obtained from statistics provided by the National Institute of Statistics (INE), which has data on vehicle fleet size and other relevant variables dating back to 1975.

This includes information on private cars, taxis, vans and trucks, buses, and other vehicles. It also includes traffic information and data on vehicles passing through toll plazas.

New vehicle sales data is obtained from information provided by ANAC, which is detailed for the years 1998-2012, and aggregated for the years 1989-1993.

Additionally, information is available from the vehicle inspection stations for the RM for the period 1998-2011. The information from the vehicle inspection stations differs significantly from that of the INE in 1998 and 1999, but by 2011 the information from both sources begins to coincide.

The methodology estimates vehicle sales in a given period as the difference between the current inventory and the previous year's inventory, plus a mortality or obsolescence rate. This figure is not known and must be estimated using national fleet and sales data.
years 1989-2012.

The statistical estimate of this rate (using sales and inventory data) is only valid on average. This is because sales are not related to the calculated inventory differentials. This is due to timing issues, and therefore, there is a measurement error when estimating sales, such as the difference in inventory minus obsolescence.

Other information obtained through assumptions relates to emissions standards. From the PRT data, the proportion of vehicles without catalytic converters is calculated. Additionally, a schedule of standards (emission standards) is available, derived from studies by SECTRA and ANAC, which allows for projecting fuel consumption and emissions from vehicles entering the country in the future. The COPERT IV emission factors are used for this purpose.

It is also necessary to determine vehicle deterioration factors, which allows for the determination of fuel consumption for vehicles of different ages. For this purpose, the deterioration factors of the IVE Model, the US-EPA, and, in Chile, the study “Experimental analysis of emission deterioration factors for light duty catalytic vehicles. Case study: Santiago, Chile,” conducted by the Department of Mechanical Engineering at the University of Chile, are available.

In the case of catalytic converter vehicles sold in the Metropolitan Region, these must pass the Vehicle Homologation and Certification process carried out by the Center of

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Vehicle Control and Certification (3CV)33. To carry out this process, the Vehicle Control and Certification Center uses the deterioration factors that the manufacturer declares for the specific make and model.

In the case of trucks, the deterioration factors recommended by the International Vehicle Emissions Model (IVE)34 are used.

Another important variable that needs to be defined is the activity level of the different modes of Transportation. In the case of the total kilometers traveled by private vehicles in Santiago, this is equal to the number of car trips made during a year multiplied by the average trip distance. The number of annual trips can be obtained from origin-destination surveys (the last one conducted in 2011). The number of car trips is not the same as the number of car journeys, since on many trips the driver is accompanied by another person. A good observed value for car occupancy rates in the city of Santiago is 1.4 people per

vehicle.

Therefore, the number of daily car trips is the total number of trips in the city multiplied by the car modal split and divided by the average occupancy rate of light vehicles:

$$THAT = \frac{Trips \; P \; Km \; F_{Auto} \; d\,ya}{Park}$$

Where:

THAT : These are the kilometers traveled annually by each vehicle.

Trips : This is the number of daily motorized trips in Santiago (includes trips on public transport)

Pact : It is the modal partition of the driver auto mode.

Km These are the average kilometers of each trip.

Fd-a Factor in transitioning from daily trips to annual trips.

Park: It is the car park in the city.

The total number of daily trips can be assumed to be independent of the vehicle fleet and can be obtained from a run of the ESTRAUS transportation model for the years 2005 and 2010. The average distance traveled by car is also obtained from a transportation model, as this allows for incorporating changes in population and estimated location, which results in an increase in the average distance traveled on each trip.

[33] DS N°54/97 of the Ministry of Transport and Telecommunications.

[34] IVE User Manual 2.1 - Attachment A, http://www.issrc.org

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In the case of commercial vehicles, their use is different since they are on the road most of the time. Other studies are available for this purpose, such as “Analysis of ex-post evaluations and re-evaluations, Stage VI” (1997) and studies by CNE35. and SEREMITT36.

For its part, in the truck mode, there is the study "Determination of Emission Factors for Heavy Vehicles in the Metropolitan Region", commissioned by CONAMA to the Department of Mechanical Engineering of the University of Chile.

Finally, Transantiago's activity estimates use data from the ESTRAUS37 (SECTRA) model runs. This model allows us to evaluate, for example, the impact of new urban highways, which enable faster traffic and reduce congestion on nearby roads. It also allows us to predict how many people will stop using public transportation now that cars are becoming more attractive due to reduced travel times.

Regarding public transportation in Santiago, the evolution of the bus fleet is obtained from the study “Evaluation of New Emission Control Measures for the Transportation Sector in the Metropolitan Region,” conducted by DICTUC at the request of CONAMA. The study presents the evolution of the fleet up to the year 2015.

Speed is one of the main factors needed to calculate fuel consumption and emissions for different vehicle categories. During the latest origin-destination survey, the average speed was measured for the city of Santiago. Additionally, traffic speeds were measured separately for the city's main thoroughfares and other major cities in the country.

Speed depends primarily on the traffic flow on each street; in fact, travel time is generally considered to increase exponentially with traffic flow. In recent years, especially over the last decade, there has been a significant increase in transportation demand, leading to substantial increases in congestion, particularly during rush hour.

Finally, other data are available that are useful for projecting fuel consumption and other related variables. The measured data for Chile will be available in the future.

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[35] “Market study of the type of gasoline used”. Prepared for the National Energy Commission by BST Consulting Group, Santiago, May 1998.

[36] “Diagnostic study and recommendations for the taxi and shared taxi market”, 1993; “Analysis and Study of transport services in Chile”, INTRAT SA, Progress Report July 1998

[37] ESTRAUS is a four-stage transportation model designed to represent the functioning of the city of Santiago. In this model, Santiago is considered to be composed of zones, each with an associated number of outbound (generation) and inbound (attraction) trips. ESTRAUS determines, for each type of user, which zones the trips take, which mode of transport is used, and which route. The chosen route is the one that minimizes travel time for each individual. Travel times are calculated taking congestion into account.

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CO2 and fuel consumption data, as well as an energy efficiency label, are provided by the 3CV laboratory. International consumption data is also available in the database www.fueleconomy.org.

A.13.3. COPPER INDUSTRY SECTOR

The following presents the inventory of the copper sector and the corresponding analysis of its information.

A.13.4. ASSESSMENT OF THE STATE OF INFORMATION

The methodology for projecting energy consumption in the copper industry is based on the annual variation in production of companies, the type of process, and in particular the large, energy-intensive equipment in certain processes, such as mills, concentration plants, among others.

Projected production information can be obtained through mining companies, and COCHILCO also develops studies of projected mining investments with
horizons of 5 to 10 years.

Regarding the processes, these are represented in the following figures and include those specific to oxidized and sulfide minerals (Mining, Concentrator, Smelting, Electrolytic Refining, Leaching, Solvent Extraction, Electrowinning). However, this is not the most relevant classification, since the highest consumption occurs in the grinding and concentration stages.

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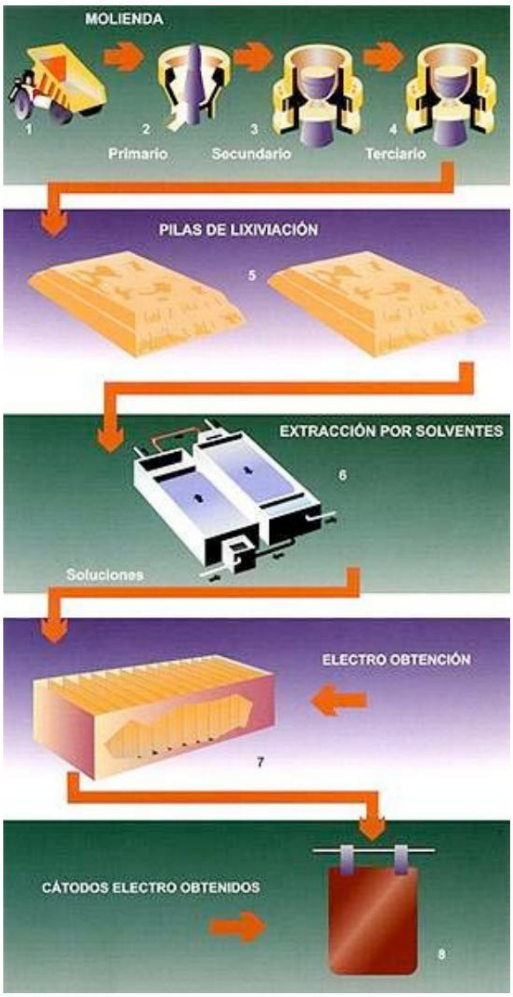
Table 43: Copper Industry Cadastre

Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibility	Location of the information	Access	State	Comment
Copper production	Electricity consumption	• SIC/SING, national • By companies	• Annual • Monthly	COCHILCO and CNE	1973 – 2012		• www.cochilco.cl/ • Cochilco private database and CNE survey	• Public • Restrings the	• PDF • BD	The information from COCHILCO Includes Mine, Concentrator, Smelting, Refining Electrolytic leaching Extraction by solvents, Electro obtaining (by company and aggregate)
Copper production	Consumption fuel at level aggregate and by fuel type	• SIC/SING, national • By companies	Annual	COCHILCO and CNE	2001-2011		• www.cochilco.cl/ • Cochilco private database and CNE survey	• Public • Restricted the	• PDF • BD	Includes Mine, Concentrator, Smelting, Electrolytic refining, Leaching, Extraction by solvents, Electro obtaining (by company and aggregate)
Activity level	Hours worked, shifts, etc.	By company	ND	COCHILCO and companies mining	ND		BD companies	Restricted ND		
Production	TMF mineral	By companies	Annual and monthly	INE, COCHILCO and companies mining				Public		

Source: Own elaboration

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Figure 50: Production Process of Oxidized Minerals



Source: Codelco - Chile

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Figure 51: Production Process of Sulfide Minerals



Source: Codelco - Chile

Based on this, the most important fuel and electricity consumption is related to the projected production of each company, thus obtaining the projection of
consumption in each case.

In the case of information on energy consumption by unit processes and companies, COCHILCO's statistics, according to its executives, cover 95% of the country's copper production.

From an energy efficiency perspective, it is necessary to have consumption data for each type of process in order to apply efficiency measures to specific processes. Similarly, it would be beneficial to establish an "efficiency standard" that allows for comparison between companies (both established and new), thus enabling differentiated efficiency improvements based on each company's potential.

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A qualitative survey of mining companies should be developed, along with studies that allow for estimate this standard.

On the other hand, it's not possible to work with motors at a stock level given the vast number of pieces of equipment, which are difficult to catalog, that exist in companies. Therefore, it would be possible to conduct a survey to characterize the motors and main energy-consuming equipment, but this would be a very complex task. In this regard, there is a study by the Ministry of Mining that estimates energy consumption in the mining sector, with electric motors accounting for almost 75% of that consumption.

A.13.5. INFORMATION ON THE INDUSTRIAL SECTOR

The following presents the inventory of the industrial sector and the corresponding analysis of its information.

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Table 44: Industrial Sector Cadastre

Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of information	Period Existence data	Compatibility	Location information	Access	State	Comment
Activity level Physical units / monetary		Data by sectors, subsectors, and destination sectors in the industry(1)	Monthly, annual	SOFA	1970 - 2011	Department of SOFOFA studies	http://200.27.194.5/sofofa/index.aspx?channel=3548&appintanceid=9116&pubid=5325	Public	Excel / .pdf	
Activity level Physical units	By company		quarterly	I HAVE	1998-2011	ENIA, form 1	http://www.ine.cl/canales/form_eleelectronics/buildingon_energy_researchbut/pdf_fishingo/eoisp.pdf	Public	.pdf	
Activity level Weights		By ISIC category	annual	Banco Central	1960-2011	National Accounts	www.bcentral.cl	Public	Excel / pdf	
Consumption fully electric	Percentage of variation (base year …)	By system interconnected, by sector	annual	SOFA	2000 - 2011	Department of SOFOFA studies	http://www.sofofa.cl/sofofa/index.aspx?channel=3552	public	.pdf	
Electricity purchased	kWh / pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 1	http://www.ine.cl/canals/eastern_chileadistico/estadisticas_economicas/industria/enia/pdf/enia2004.pdf	public	.pdf	
Electricity sold	kWh / pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 1	http://www.ine.cl/canals/eastern_chileadistico/estadisticas_economicas/industria/enia/pdf/enia2004.pdf	public	.pdf	
Electricity generated	kWh / pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 1	http://www.ine.cl/canals/eastern_chileadistico/estadisticas_economicas/in	public	.pdf	

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Use	Variables related	Spatiality (SIC/SING, municipality, province, region)	Temporality	Source of information	Period Existence data	Compatibility	Location information	Access	State	Comment
							dustria/enia/pdf/ enia2004.pdf			
Consumption of fuels for power and heat production	physical units / pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 1	http://espino.ine. cl/cgibin/rpweben gine.exe/portalact ion?&MODE=MAI N&BASE=CLENIA& MAIN=webserver main.inl	public	.pdf	
Consumption of fuels as matter Before	physical units / pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 4	-	restricted		With Requires access key to access the form
Consumption total energy	Teracalories	Sector (paper, steel industry,…) and by companies	Annual	CNE	1991-2011	CNE Statistics	http://www.cne.cl /statistics/f_ba lance.html	public	.pdf	
Intensity index energy	percentage	sector	Annual	CNE	1990 - 1999 CNE	Archives	http://www.cne.cl /download_files/in dices_sector_indu strial.pdf	Public	.pdf	
Cost of the demand	pesos	By company	Annual	I HAVE	1998-2011	ENIA, form 1	http://espino.ine. cl/cgibin/rpweben gine.exe/portalact ion?&MODE=MAI N&BASE=CLENIA& MAIN=webserver main.inl	public	.pdf	
Consumption of fuel the generator sets with superior power and 20 kW	Physical units by company/sector / RM / regions		Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.cona ma.cl/retc/1279/p ropertyvalue-15095.html	Restricted to users registered		With requires access key to access the form
Consumption of fuel boilers industrial	Physical units by company/sector / RM / regions		Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.cona ma.cl/retc/1279/p ropertyvalue-15095.html	Restricted to users registered		With Requires access key to access the form

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Use	Variables related	Spatiality (SIC/SING, municipality, province, region)	Temporality	Source of information	Period Existence data	Compatibility	Location information	Access	State	Comment
(steam generators) with consumption energy combustible greater than 1 Mega Joule per time										
Consumption of fuel boilers of heating with consumption energy combustible greater than 1 Mega Joule per time	Physical units	by company/sector / RM / regions	Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.conama.cl/retc/1279/propertyvalue-15095.html	Restricted to users registered		With requires access key to access the form
Consumption of combustible	Physical units	Through processes with/without combustion combustion	Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.conama.cl/retc/1279/propertyvalue-15095.html	Restricted to users registered		With requires access key to access the form
Activity level	Physical units / monetary	By establishment belonging to any[a] of the sectors determined by DS 138/05 / by[(2)] processes with combustion or without combustion / by devices(3)	Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.conama.cl/retc/1279/propertyvalue-15095.html	Restricted to users registered		With Requires access key to access the form
Number of days of each month of	days	By defined sources by DS 138/05 [(2)] and	Annual	MINSAL, RETC, Authority	2006-2011	Form electronic 138	http://www.conama.cl/retc/1279/p	Restricted to users		With requires access key

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of information	Period Existence data	Compatibility	Location information	Access	State	Comment
year in which the operate sources		by artifacts(3)		Health RM			ropertyvalue-15095.html	registered		to access the form
Cycle characteristic of operation on weekdays		By defined sources by DS 138/05 for artifacts(3) [(2) and]	Annual	MINSAL, RETC, Authority Health RM	2006-2011	Form electronic 138	http://www.cona ma.cl/retc/1279/p ropertyvalue-15095.html	Restricted to users registered		With requires access key to access the form

Source: Own elaboration

The following information has also been detected, which will be available in the future, and will be obtained from the Energy Demand and Efficiency Study Questionnaire for the Manufacturing Industrial Sector (CNE) that is about to be implemented.

Table 45: Industrial Sector Cadastre. Information for future surveys.

Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibilidad	Location information	Access	State	Comment
Cost of energy consumed	pesos	By companies	annual	CNE	...	Questionnaire YES CNE	...	....	...	This information is finds in a questionnaire that has not yet been implemented
Total energy consumption	percentage	By companies	annual	CNE	...	Questionnaire YES CNE	...	....	...	
Disaggregation by uses of consumption total, by energy source:				CNE	...	Questionnaire YES CNE	...	....	...	
Transport	Percentage	By companies	Annual							
Industrial processes	Percentage	By companies	Annual							
Self-generation of electricity	Percentage	By companies	Annual							
Uses	Percentage	By companies	Annual							

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibilidad	Location information	Access	State	Comment
complementary	Percentage	By companies	Annual							
Disaggregation of of the consumption of industrial processes, by energy source:				CNE	...	Questionnaire YES CNE	...	....	...	
Motor use	Percentage	By companies	Annual							
Thermal use	Percentage	By companies	Annual							
Electrical use	Percentage	By companies	Annual							
Other uses	Percentage	By companies	Annual							
Disaggregation of of the consumption of industrial processes for thermal use by energetic:				CNE	...	Questionnaire YES CNE	...	....	...	
Steam generation	Percentage	By companies	Annual							
Use in ovens	Percentage	By companies	Annual							
Heating	Percentage	By companies	Annual							
Performance(4) of the industrial processes of thermal use, by energetic:										
Steam generation	Percentage									
Use in ovens	Percentage									
Heating	Percentage									

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibilidad	Location information	Access	State	Comment
Energy total generated	kWh	By companies	Annual	CNE	...	Questionnaire YES CNE	...	....	...	
Participation of the Additional energy generated:		By companies	Annual	CNE	...	Questionnaire YES CNE	...	....	...	
Self-generation of participation	Percentage									
Cogeneration of participation	Percentage									
Electricity generation by fuel:		By companies	Annual	CNE	...	Questionnaire YES CNE	...	....	...	
Gross generation	Kwh									
Consumption	Physical units									
Total capacity of storage for self-generation	Physical units									
Number of days of normal operation	Days									
Steam generation in boilers, by combustible:		By companies	Annual	CNE	...	Questionnaire YES CNE	...	....	...	
Gross generation	Kj									
Consumption of combustible	Physical units									
Utilization of the heat generated in:		By companies	Annual	CNE	...	Questionnaire YES CNE	...	....	...	
Steam generation	Percentage									

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Use	Variables related	Spatiality (SIC/SING, commune, province, region)	Temporality	Source of the information	Period Existence data	Compatibilidad	Location information	Access	State	Comment
Heating the heating of water	Percentage									
(Pre) warm-up of media of production	Percentage									
Ovens	Percentage									
Waste no opportunists	Percentage									
Others	Percentage									
Annual purchases and sales of energy	Physical units by company		Annual							

Source: Own elaboration

(1) The sectors considered by SOFOFA in the manufacturing industry are:

- Food, beverages and tobacco
- Textiles, clothing and leather
- Wood and furniture
- Cellulose, paper and printing
- Oil, rubber and plastic
- Non-metallic minerals and basic metals
- Metal products, machinery and equipment

(2) The sectors determined by DS 138/05 on the declaration of emissions of atmospheric pollutants are the following:

- Cellulose production
- Primary and secondary foundries
- Thermal power plants

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• Production of cement, lime or gypsum • Production of glass
• Ceramic production
• Steelmaking • Petrochemicals • Asphalt

(3)The devices determined by DS 138 / 05 on the declaration of atmospheric pollutant emissions are the following:

• Generator sets with a power output exceeding 20 kW • Industrial boilers (steam generators) with fuel energy consumption greater than 1 Mega Joule per hour • Fuel consumption of heating boilers with fuel energy consumption greater than 1 Mega Joule per hour

(4) Performance is defined as the ratio of energy generated / energy supplied.

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A.13.6. ASSESSMENT OF THE STATE OF INFORMATION

The first chart presented above shows the current state of information concerning the industrial sector. It reveals three levels of information.

The first level corresponds to aggregated sector information, including the overall activity level of the sector and its subsectors, as defined according to the ISIC classification, in both physical and monetary units. Additionally, there is information on the total energy consumption of each subsector, defined according to the Energy Balance (EB) classification.

Regarding the cost of demand, the INE provides information at the company level.

The second level corresponds to information on energy consumption by use, including data provided by the INE (National Institute of Statistics), which establishes fuel consumption by fuel type for each surveyed company, distinguishing between power generation processes, heat production, and its use as raw material (ENIA, Form 1). Regarding electricity consumption, there is disaggregated information at the company level, differentiating between the purchase, sale, and generation of electricity by companies (ENIA, Form 1).

The third level corresponds to stock information, including the energy consumption of the devices determined by DS 138/05 on the declaration of atmospheric pollutant emissions, and in the sectors determined by the same decree.

Taking into account the information identified in the industrial sector, it is possible to account for energy demand at the usage level, that is, by distinguishing between power generation processes, heat production, and use as raw material. Since stock-level information is restricted to the equipment regulated by Decree 138/05, its inclusion in the demand projection will depend on the representativeness of these devices in the total energy consumption of each industry subsector. This information will be available once the inventory of sources regulated by said decree has been completed.

Regarding sectoral disaggregation, the industry can be classified by categories (according to the ISIC) or by sectors (those defined in the Energy Balance, CNE). This requires criteria for classifying surveyed companies within each category or sector, including the type of final primary product, the company's production level, and its annual sales. In relation to the latter, classifying companies by their annual sales allows us to determine their size and, therefore, whether they belong to the miscellaneous industries sector (which includes SMEs) or another sector.

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In this sense, the work of General Characterization of SMEs with focus on improving energy efficiency38, provides criteria to differentiate SMEs from the rest of the industry.

Regarding the representativeness of the surveys considered, it is necessary to take into account the sample size of each survey, and to adopt assumptions that allow the results of said surveys to be taken as the universe of companies in the industry.

In the case of the National Annual Industrial Survey (ENIA), the universe considered are manufacturing establishments, whose size corresponds to an occupation of at least 10 people, and the directory of establishments considered in the survey is updated through the industrial patent records that the municipalities provide to the INE and according to the taxpayer records of the SII mainly.

In the case of the information provided by SOFOFA, the universe considered are the 7005 companies registered in the industrial directory, considered as the most relevant and representative of the country's business activity.

The preparation of the National Accounts Yearbook of the Central Bank of Chile, for its part, uses information mainly from tax, fiscal, foreign trade records, and financial statements.

Regarding the information collected by MINSAL through form 138, the process is still in the implementation phase, so the “construction of a national registry of the sources regulated by DS No. 138 is still required, and therefore, it is expected that in the first few years local health authorities will publicize the regulations and provide the necessary support to companies in relation to the declaration form. On the other hand, it is necessary to implement quality assurance systems for the information provided by the companies”39.

Projecting energy demand for the industrial sector requires using assumptions related to the activity level of the subsectors considered, as well as assumptions related to energy consumption. Regarding activity levels, projecting GDP by productive sector could allow for projecting the activity level of industrial subsectors. In this sense, the historical series of regional GDP by economic sector compiled by the Central Bank of Chile (which includes the agricultural and forestry, fishing, mining, manufacturing, and construction sectors) can serve as a basis for projecting sectoral GDP.

In the case of energy consumption by the subsectors of industry and by the different uses of each subsector, it is necessary to develop technological assumptions that offer a

38“General characterization of SMEs with a focus on improving energy efficiency” prepared by ValgestaLtda.. for the National Energy Commission (CNE).

39 Declaration of atmospheric sources. CONAMA. www.conama.cl

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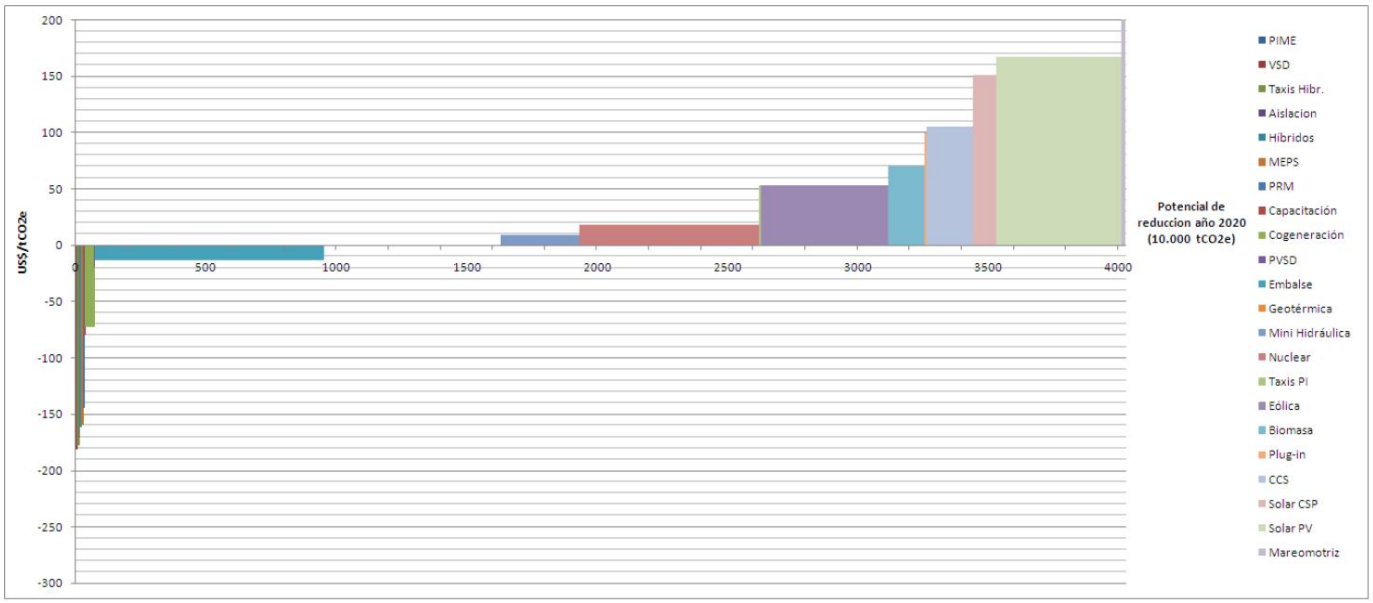
projection of energy efficiency in power and heat production processes, as well as the substitution between them.

The second table in this section shows the information that will be available in the future, once the Energy Demand and Efficiency Study questionnaire for the Manufacturing Industry (CNE) is implemented. This information, along with further development of Form 138, will allow for a more detailed classification of energy uses in each industry subsector, with a particular focus on the thermal use of industrial processes. Currently, only information related to power and heat production processes by companies (without disaggregating heat processes) and the use of fuels as raw materials is available. This disaggregation, however, lacks the specificity needed to evaluate energy efficiency measures targeting specific processes and equipment (such as boiler efficiency in steam generation processes or furnace efficiency in direct heat processes).

Additionally, the Energy Demand and Efficiency Study questionnaire for the Manufacturing Industry will allow us to obtain information about self-generation and cogeneration in the manufacturing industry, thereby enabling more thorough studies on both the energy demand and supply sides.

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A.14. RESULTS WITH A DISCOUNT RATE OF 12%

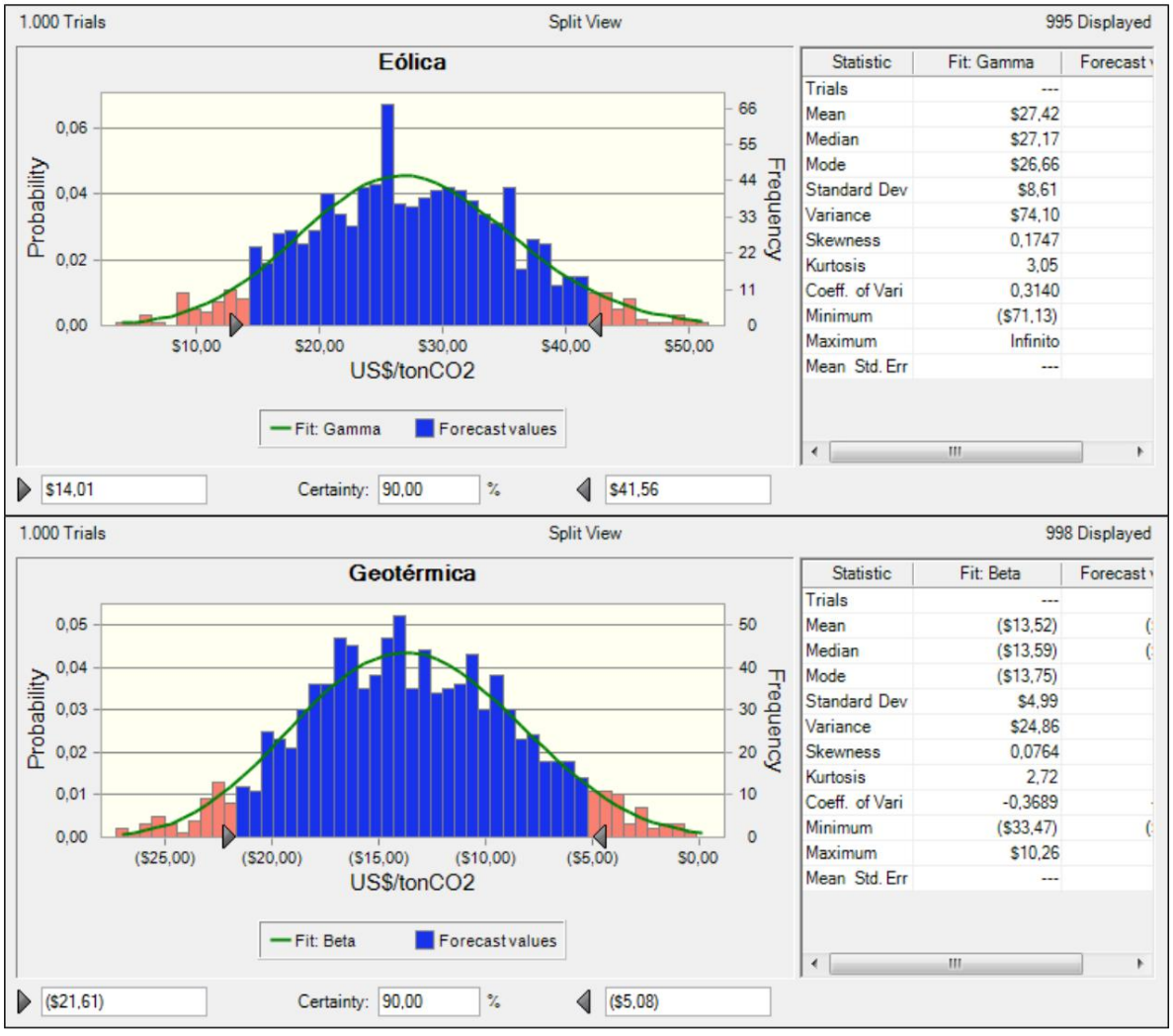


Sector	Measurement	Fashion (US$/tCO2)	5%	95% Potential (tCO2)	
Copper	BLIND	-199,0	-230,7	-166,4	4.016
P&C	VSD	-180,9	-208,6	-151,4	45.599
Transport	Taxis Hibr.	-178,0	-215,5	-124,8	63.093
P&C	Insulation	-170,6	-196,1	-143,2	11.760
Transport	Hybrids	-162,0	-193,6	-112,9	78.096
Copper	MEPS	-159,6	-188,9	-127,7	74.448
Copper	PRM	-144,6	-171,3	-115,8	43.790
Transport	Training	-80,0	-185,2	189,0	37.182
P&C	Cogeneration	-72,8	-103,1	-40,5	347.900
Copper	PVSD	-19,7	-63,7	36,8	21.247
Reservoir **Transformation**		-13,1	-22,3	-4,5	8.760.000
Geothermal **Transformation**		0,0	-10,6	13,1	6.820.000
Mini Hydraulic **Transformation**		8,8	0,7	18,3	3.000.000
Nuclear **Transformation**		17,7	1,0	31,5	6.900.000
Transport	Taxi PI	53,0	8,8	86,0	97.832
Wind **Transformation**		53,4	34,3	73,5	4.866.667
Biomass **Transformation**		69,9	53,4	92,1	1.380.000
Transport	Plug-in	100,0	57,3	130,7	87.193
CCS **Transformation**		104,8	90,6	123,4	1.760.000
Solar CSP **Transformation**		150,7	120,6	188,7	890.000
Solar PV **Transformation**		167,3	142,5	197,3	4.800.000
Tidal **Transformation**		233,2	202,1	274,0	200.000

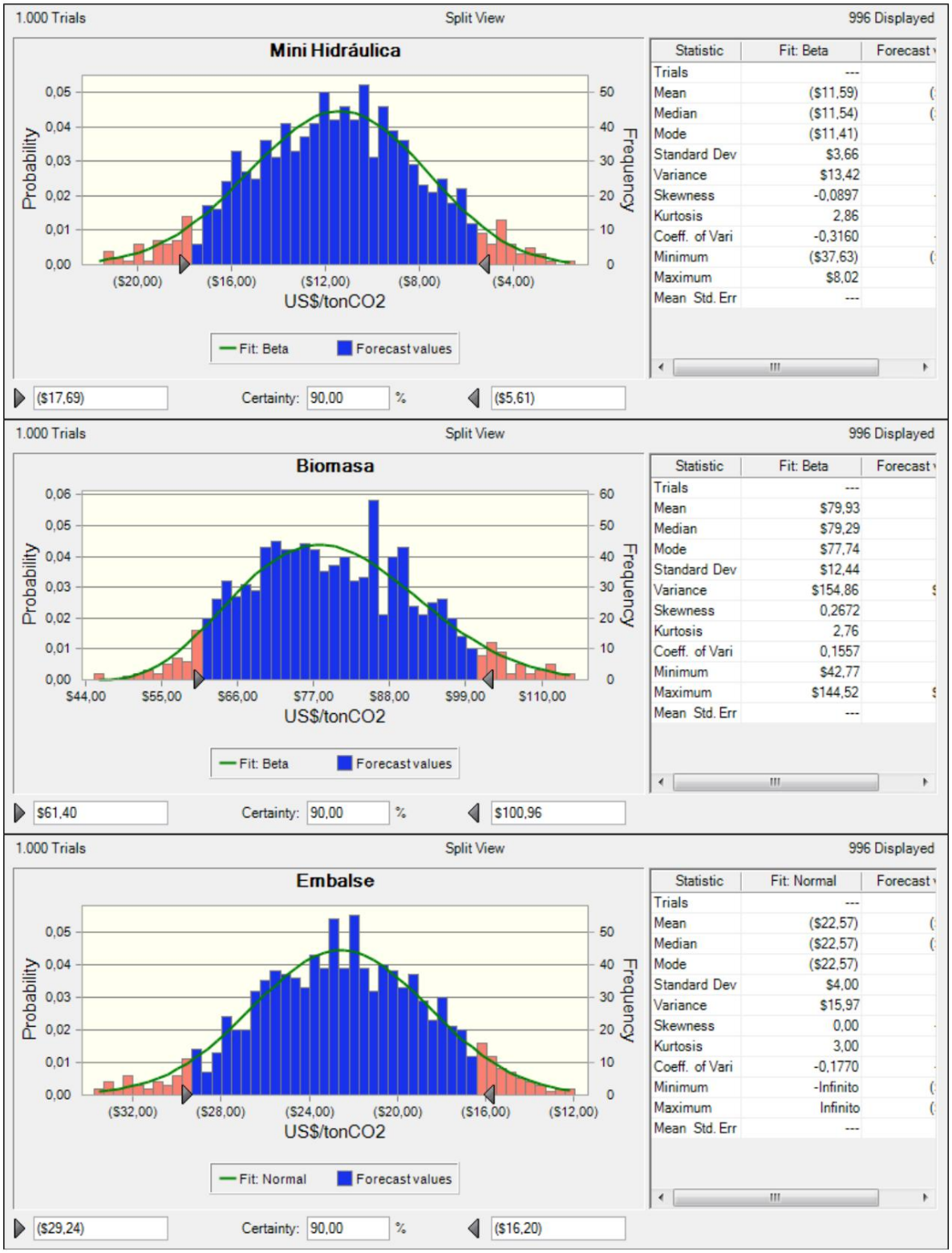
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A.15. STOCHASTIC RESULTS OF THE VARIABLES

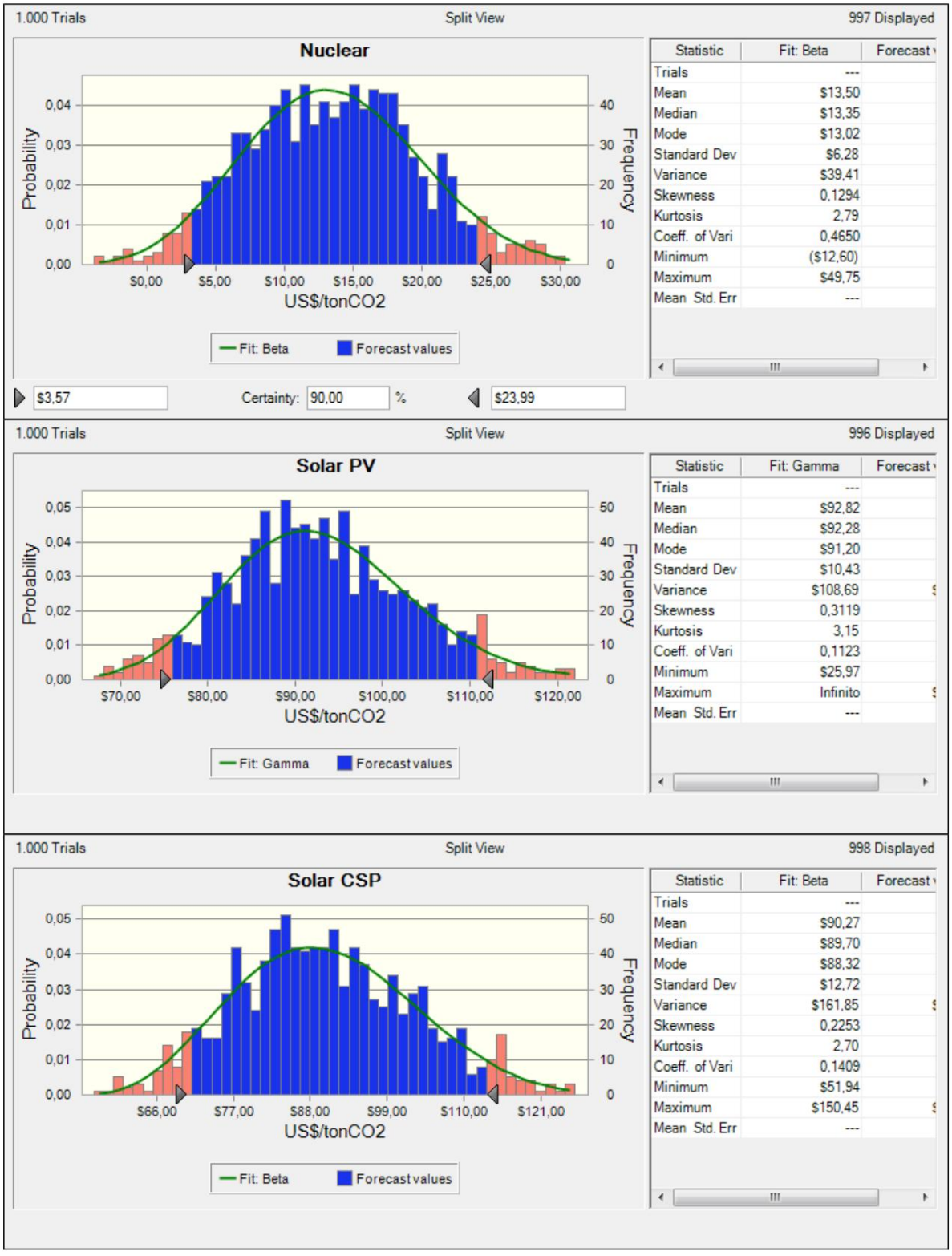
The following figures present the outputs for the Crystal Ball software variables considering a discount rate of 6%. The figures also show the probability distribution that best fits the results obtained.



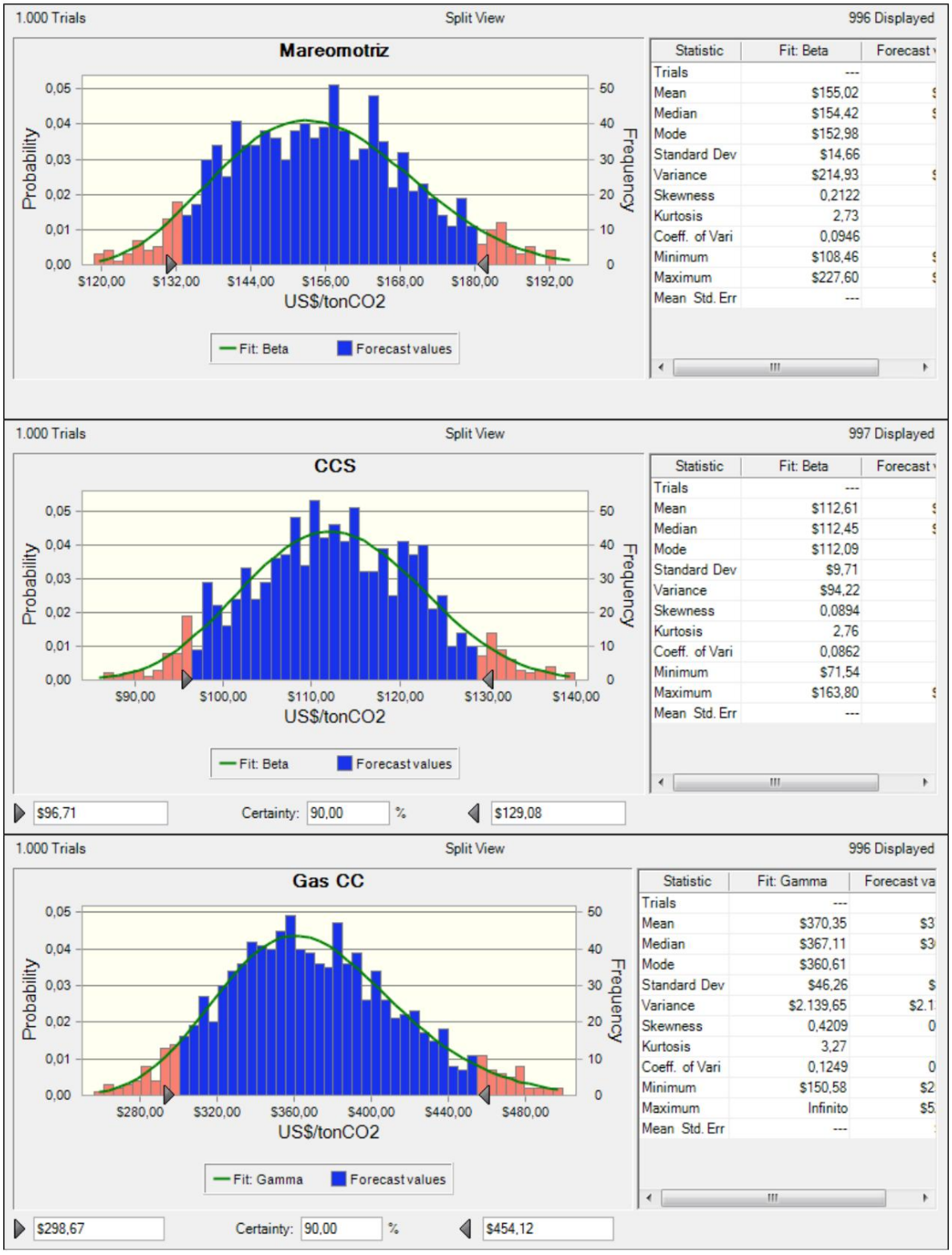
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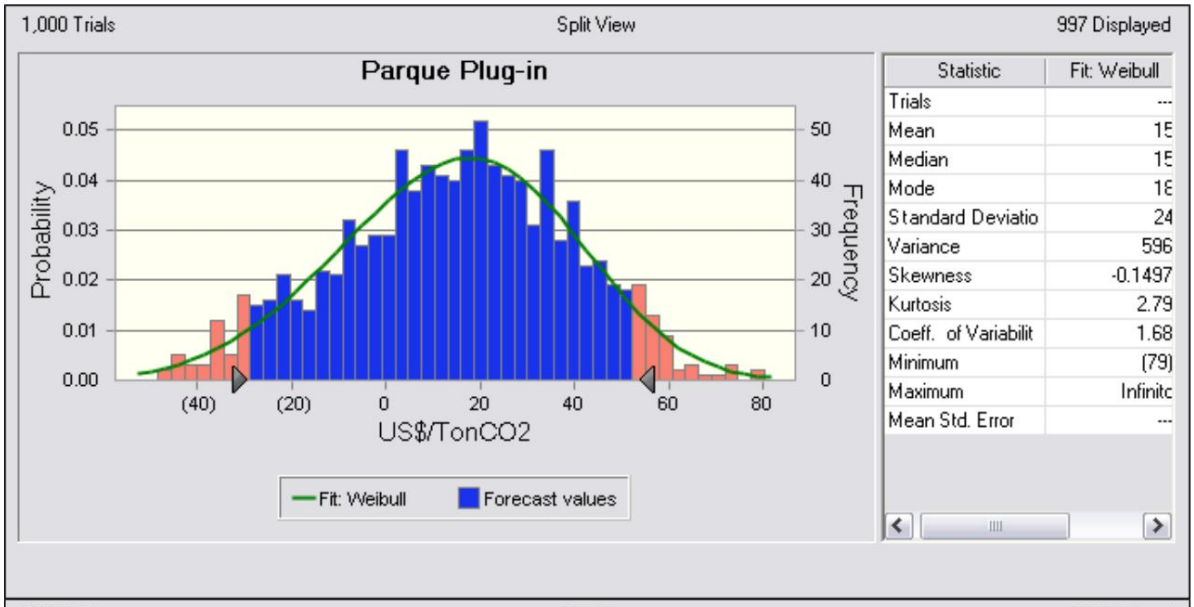
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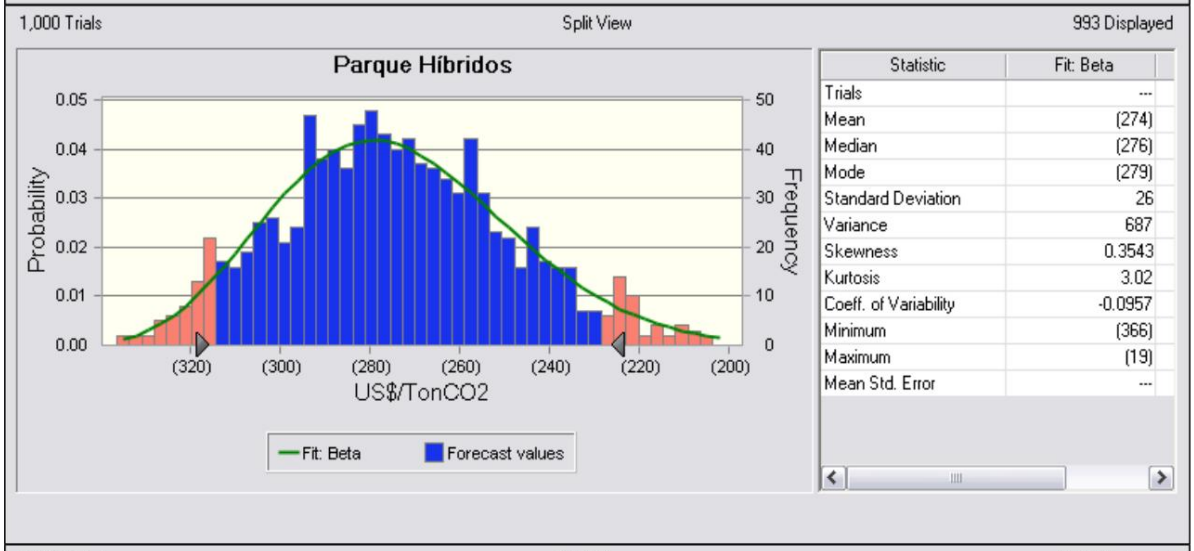


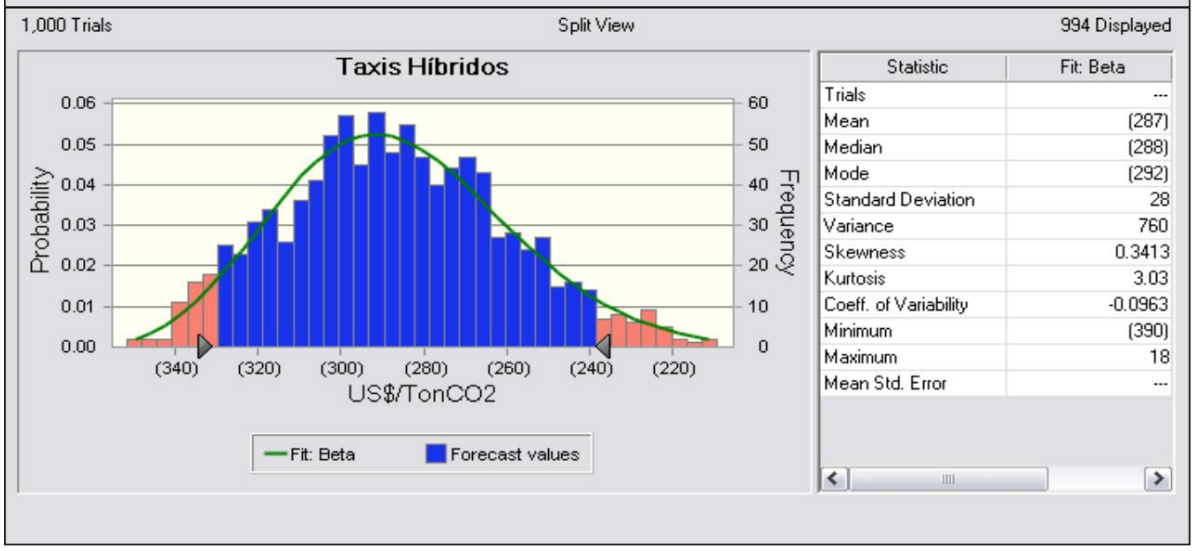
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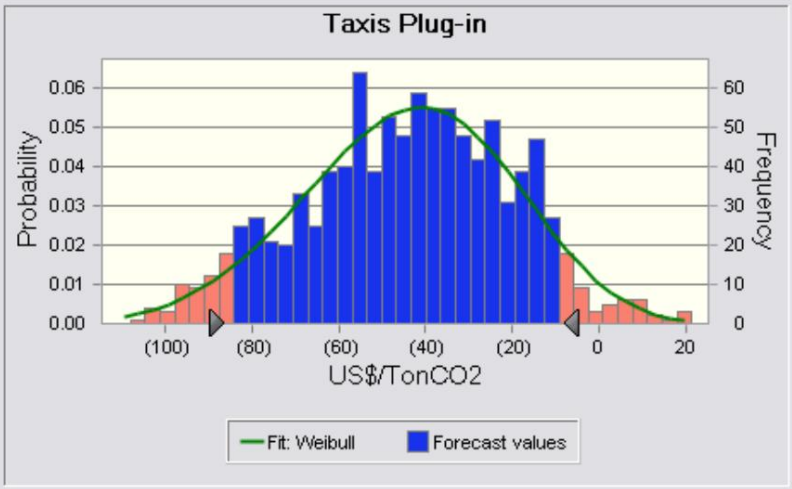






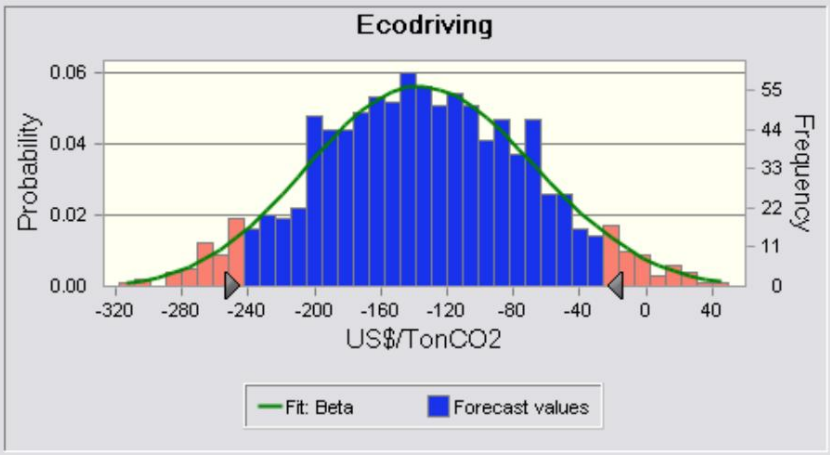
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1,000 Trials Split View 999 Displayed



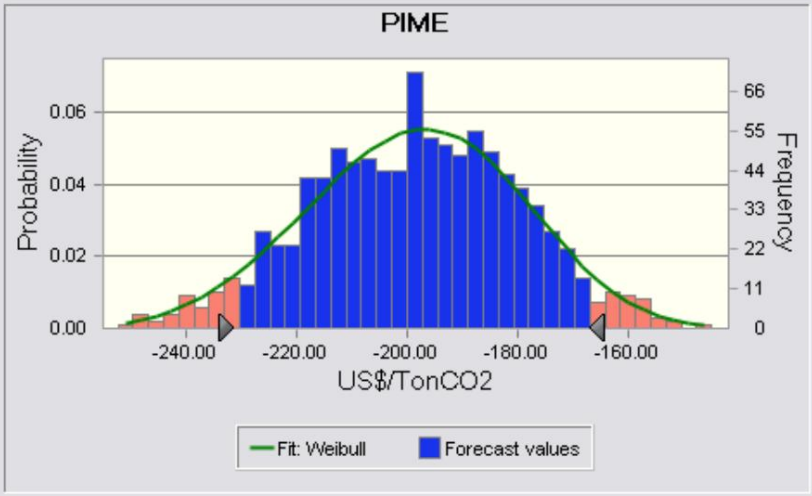
Statistic	Fit: Weibull	Fo
Trials	---	
Mean	(44)	
Median	(43)	
Mode	(41)	
Standard Deviation	24	
Variance	574	
Skewness	-0.1537	
Kurtosis	2.79	
Coeff. of Variability	-0.5432	
Minimum	(136)	
Maximum	Infinito	
Mean Std. Error	---	

1,000 Trials Split View 996 Displayed



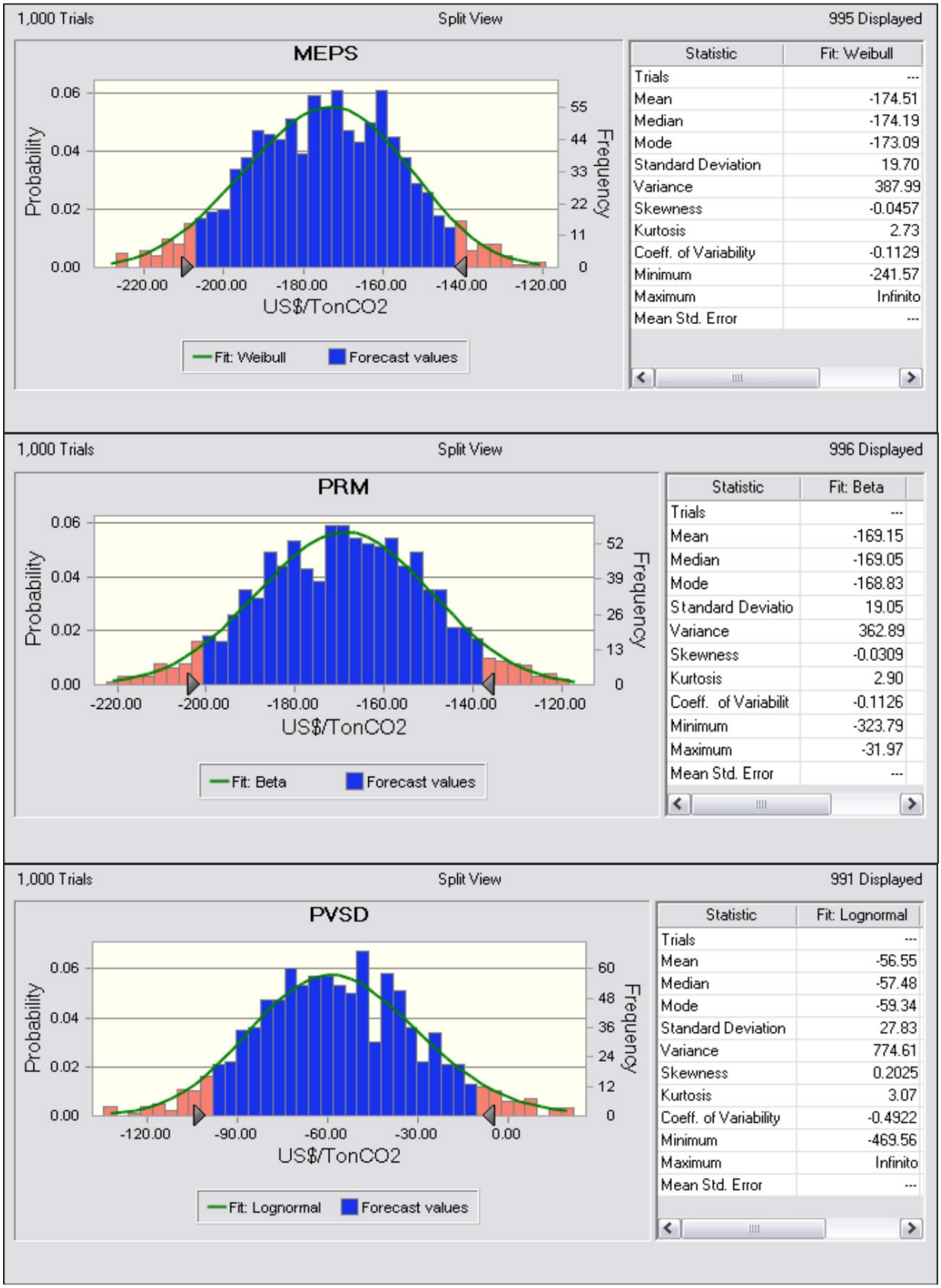
Statistic	Fit: Beta
Trials	---
Mean	-134
Median	-135
Mode	-137
Standard Deviation	66
Variance	4,312
Skewness	0.0733
Kurtosis	2.87
Coeff. of Variability	-0.4897
Minimum	-506
Maximum	339
Mean Std. Error	---

1,000 Trials Split View 996 Displayed



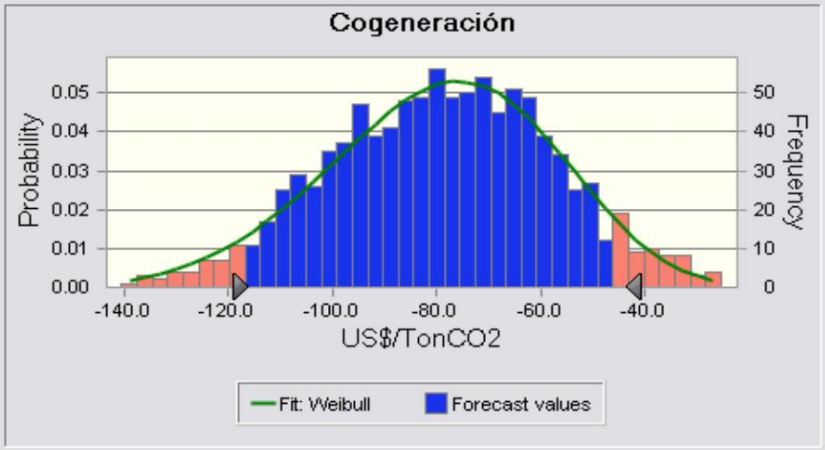
Statistic	Fit: Weibull
Trials	---
Mean	-198.62
Median	-198.18
Mode	-196.85
Standard Deviation	19.18
Variance	367.69
Skewness	-0.0823
Kurtosis	2.75
Coeff. of Variability	-0.0965
Minimum	-266.59
Maximum	Infinito
Mean Std. Error	---

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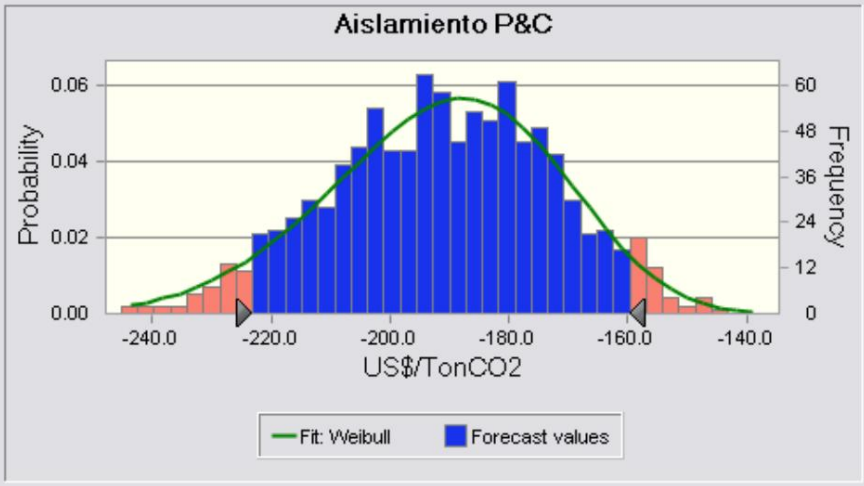
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1,000 Trials Split View 996 Displayed



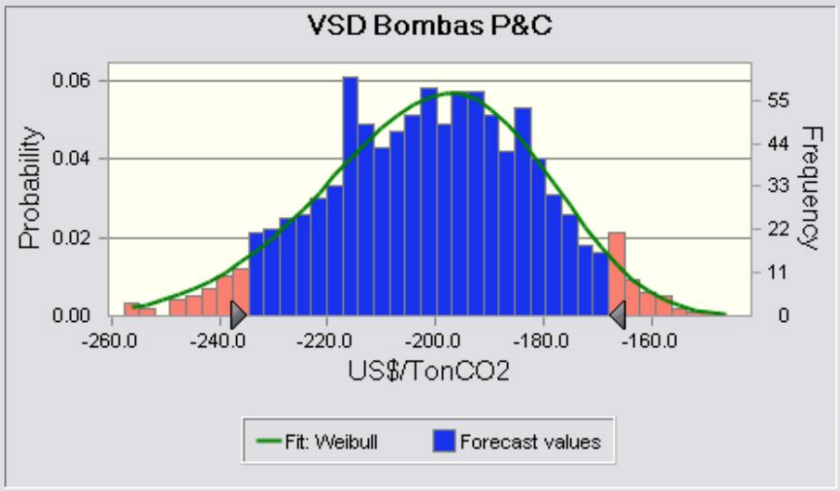
Statistic	Fit: Weibull
Trials	---
Mean	-79.5
Median	-78.5
Mode	-75.9
Standard Deviatio	21.8
Variance	476.1
Skewness	-0.2138
Kurtosis	2.84
Coeff. of Variability	-0.2746
Minimum	-169.9
Maximum	Infinito
Mean Std. Error	---

1,000 Trials Split View 993 Displayed



Statistic	Fit: Weibull
Trials	---
Mean	-191.3
Median	-190.3
Mode	-187.8
Standard Deviatio	19.2
Variance	369.6
Skewness	-0.2621
Kurtosis	2.89
Coeff. of Variabilit	-0.1005
Minimum	-276.2
Maximum	Infinito
Mean Std. Error	---

1,000 Trials Split View 994 Displayed



Statistic	Fit: Weibull
Trials	---
Mean	-201.2
Median	-200.1
Mode	-197.5
Standard Deviatio	20.2
Variance	408.3
Skewness	-0.2622
Kurtosis	2.89
Coeff. of Variabilit	-0.1004
Minimum	-290.4
Maximum	Infinito
Mean Std. Error	---

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A.16. SENSITIVITY **ANALYSIS OF THE VARIABLES**

Sensitivity: Biomass

Variable	Contribution to variance	Correlation ranking
* Biomass price 2035 US$/MWh	36%	0.70
* Biomass price 2020 US$/MWh	33%	0.68
Biomass efficiency %	-8%	-0.34
Biomass TC	-7%	-0.32
* Coal price 2035 US$/Ton	-4%	-0.23
* Coal price 2020 US$/Ton	-3%	-0.21
FE Carbon	-2%	-0.16
Other	6%	

Sensitivity: Wind

Variable	Contribution to variance	Correlation ranking
Wind Energy Investment	55%	0.74
Wind FP %	-22%	-0.47
* Coal price 2035 US$/Ton	-4%	-0.21
* Coal price 2020 US$/Ton	-4%	-0.20
Wind Power TC	-3%	-0.18
TC Carbon	2%	0.14
Coal Investment	-1%	-0.11
Coal efficiency %	1%	0.11
FE Carbon	-1%	-0.10
Other	6%	

Sensitivity: CC Gas

Variable	Contribution to variance	Correlation ranking
FE Gas CC	37%	0.62
FE Carbon	-23%	-0.48
GAS CC TC	-13%	-0.36
* Oil price 2020 US$/Barrel	10%	0.32
CC Gas efficiency %	-4%	-0.21
* Oil price 2035 US$/Barrel	4%	0.20
* Coal price 2035 US$/Ton	-1%	-0.11
* Coal price 2020 US$/Ton	-1%	-0.10
Coal Investment	-1%	-0.09
Other	6%	

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Sensitivity: Tidal

Variable	Contribution to variance	Correlation ranking
Tidal Inversion	54%	0.72
Tidal Power FP %	-21%	-0.46
FE Carbon	-9%	-0.29
Tidal Power Plant TC	-7%	-0.25
Mareomotor O&M	2%	0.12
Coal Investment	-1%	-0.10
TC Carbon	1%	0.09
* Coal price 2035 US$/Ton	-1%	-0.07
* Coal price 2020 US$/Ton	-1%	-0.07
Other	5%	

Sensitivity: Nuclear

Variable	Contribution to variance	Correlation ranking
Nuclear TC	-36%	-0.60
Nuclear Investment	21%	0.46
* Coal price 2020 US$/Ton	-9%	-0.29
Nuclear O&M	6%	0.25
* Coal price 2035 US$/Ton	-6%	-0.24
Uranium Price US$/MWh	5%	0.23
Coal Investment	-3%	-0.18
Coal efficiency %	3%	0.18
TC Carbon	3%	0.16
Other	8%	

Sensitivity: Solar PV

Variable	Contribution to variance	Correlation ranking
Solar PV Investment	42%	0.69
Solar PV FP %	-23%	-0.51
Solar PV O&M	7%	0.27
FE Carbon	-5%	-0.23
* Coal price 2020 US$/Ton	-4%	-0.21
* Coal price 2035 US$/Ton	-4%	-0.21
Solar PV TC	-3%	-0.18
Coal Investment	-2%	-0.17
O&M Coal	-1%	-0.12
Other	10%	

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Sensitivity: CCS

Variable	Contribution to variance	Correlation ranking
CCS O&M	61%	0.82
FE Carbon	-16%	-0.42
CCS TC	-12%	-0.37
CCS Investment*	1%	0.11
Other	9%	

Sensitivity: Reservoir

Variable	Contribution to variance	Correlation ranking
* Coal price 2035 US$/Ton	-22%	-0.50
* Coal price 2020 US$/Ton	-21%	-0.49
Reservoir FP %	-20%	-0.48
Coal Investment	-7%	-0.29
TC Carbon	6%	0.26
Coal efficiency %	4%	0.21
O&M Coal	-4%	-0.20
TC Reservoir	-4%	-0.20
Reservoir O&M	3%	0.18
FE Carbon	3%	0.18
Other	6%	

Sensitivity: Geothermal

Variable	Contribution to variance	Correlation ranking
Geothermal Investment	45%	0.72
* Coal price 2020 US$/Ton	14%	-0.41
* Coal price 2035 US$/Ton	13%	-0.38
Geothermal O&M	7%	0.29
TC Carbon	5%	0.24
Coal Investment	4%	-0.22
O&M Coal	3%	-0.17
Coal efficiency %	2%	0.17
Geothermal TC	1%	-0.12
Other	5%	

Sensitivity: Mini Hydraulics

Variable	Contribution to variance	Correlation ranking
* Coal price 2020 US$/Ton	-26%	-0.56
* Coal price 2035 US$/Ton	-24%	-0.54
MiniHidro FP%	-13%	-0.40

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Coal Investment	-8%	-0.31
MiniHidro TC	-7%	-0.30
TC Carbon	7%	0.29
O&M Coal	-5%	-0.26
Coal efficiency %	3%	0.18
MiniHidro O&M	1%	0.12
Other	5%	

Sensitivity: Solar CSP

Variable	Contribution to variance	Correlation ranking
Solar CSP FP %	-42%	-0.67
Solar CSP	29%	0.55
FE Carbon	-6%	-0.24
Solar CSP O&M	4%	0.21
Solar CSP TC	-4%	-0.20
* Coal price 2035 US$/Ton	-3%	-0.18
* Coal price 2020 US$/Ton	-3%	-0.18
Coal Investment	-2%	-0.13
TC Carbon	1%	0.12
Other	7%	

Sensitivity: Plug-in Park

Variable	Contribution to variance	Correlation ranking
Conventional yield KM/lt	63%	0,83
Additional cost Plug-in 2020 US$	12%	0,37
Hybrid Performance KM/lt	6%	-0,25
* Oil price 2020 US$/Barrel	4%	-0,20
* Oil price 2035 US$/Barrel	2%	-0,16
Other	13%	

Sensitivity: Taxis Plug-in

Variable	Contribution to variance	Correlation ranking
Conventional yield KM/lt	61%	0,83
Additional cost Plug-in 2020 US$	12%	0,37
Hybrid Performance KM/lt	5%	-0,24
* Oil price 2020 US$/Barrel	4%	-0,21
Activity Level Taxis KM/year	4%	-0,21
Other	13%	

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Sensitivity: Hybrid Park

Variable	Contribution to variance	Correlation ranking
Conventional yield KM/lt	46%	0,68
Hybrid Performance KM/lt	20%	-0,45
Additional cost Hybrid 2020 US$	14%	0,37
* Oil price 2020 US$/Barrel	6%	-0,25
* Oil price 2035 US$/Barrel	4%	-0,20
Other	10%	

Sensitivity: Hybrid Taxis

Variable	Contribution to variance	Correlation ranking
Conventional yield KM/lt	51%	0,72
Hybrid Performance KM/lt	22%	-0,47
Additional cost Hybrid 2020 US$	8%	0,29
* Oil price 2020 US$/Barrel	5%	-0,23
* Oil price 2035 US$/Barrel	3%	-0,19
Other	10%	

Sensitivity: Ecodriving

Variable	Contribution to variance	Correlation ranking
Reduction in consumption due to training %	26%	-0,55
* Oil price 2020 US$/Barrel	25%	-0,54
Truck Performance (KM/lt)	23%	0,51
* Oil price 2035 US$/Barrel	9%	-0,33
Activity Level Trucks KM/year	7%	-0,28
Other	10%	

Sensitivity: PIME

Variable	Contribution to variance	Correlation ranking
SING Electricity 2020 $/kWh	87%	-0,84
Efficiency dispersion 5%	2%	-0,12
Engine price dispersion %	1%	0,09
Efficiency dispersion 20-50%	1%	-0,07
Usage time (hrs) Copper	1%	-0,07
Other	9%	

Sensitivity: PRM

Variable	Contribution to variance	Correlation ranking
SING Electricity 2020 $/kWh	97%	-0,89
Efficiency dispersion 5%	1%	-0,10

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Engine price dispersion %	1%	0,08
Efficiency dispersion 20-50%	0%	-0,06
Usage time (hrs) Copper	0%	-0,06
Other	0%	

Sensitivity: MEPS

Variable	Contribution to variance	Correlation ranking
SING Electricity 2020 $/kWh	94%	-0,86
Engine price dispersion %	2%	0,12
Usage time (hrs) Copper	1%	-0,09
Efficiency dispersion 20-50%	1%	-0,09
Efficiency dispersion 5%	1%	-0,08
Other	1%	

Sensitivity: PVSD

Variable	Contribution to variance	Correlation ranking
SING Electricity 2020 $/kWh	45%	-0,62
Engine price dispersion %	15%	0,36
Efficiency dispersion 5%	13%	-0,34
Efficiency dispersion 20-50%	10%	-0,29
Usage time (hrs) Copper	9%	-0,28
Efficiency dispersion 5 - 20%	7%	-0,24
Efficiency dispersion 50 - 100%	0%	-0,04
Penetration replacement %	0%	0,04
Other	0%	

Sensitivity: Cogeneration

Variable	Contribution to variance	Correlation ranking
Electricity SIC 2020 $/KWh	81%	-0,83
* Biomass price 2020 US$/MWh	10%	0,29
* Biomass price 2035 US$/MWh	8%	0,26
O&M US$/KWh	0%	0,06
Investment MMUS$/MW	0%	0,01
Cogeneration time (hrs/year)	0%	0,01

Sensitivity: P&C Insulation

Variable	Contribution to variance	Correlation ranking
Electricity SIC 2020 $/KWh	99%	-0,88
Insulation savings	0%	-0,04

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Sensitivity: VSD Pumps

P&C

Variable	Contribution to variance	Correlation ranking
Electricity SIC 2020 $/KWh	93%	-0,87
Savings from variable frequency drives	2%	-0,12
Cost of variable frequency drives	0%	0,05
Other	5%	

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A.17. PRICE REGRESSIONS

1) Diesel Price Regression

The following model is run to estimate the price of diesel (Pt) based on WTI (West Texas Intermediate) and a restatement.

$$= \quad 0 + \quad _1 \, ' \qquad + \quad _2 \ddot{y}1$$

The results are shown in the following table:

Dependent Variable: DIESEL				
Method: Least Squares				
Date: 01/26/12 Time: 14:40				
Sample(adjusted): 2000:02 2011:04				
Included observations: 135 after adjusting endpoints				
Variable Coefficient	Std. Error	t-Statistic		Prob.
C	0.062349	0.014037	4.441766	0.0000
WTI	0.000701	6.04E-05	11.59223	0.0000
DIESEL(-1)	0.564872	0.038288	14.75321	0.0000
R-squared	0.962249	Mean dependent var	0.961677	0.786322
Adjusted R-squared	S.E.	S.D. dependent var	0.048327	Akaike info 0.246867
of regression	Sum	criterion		-3.199676
squared resid	Log	0.308287	Schwarz criterion	-3.135114
likelihood	Durbin-	218.9781	F-statistic	1682.316
Watson stat		2.217612	Prob(F-statistic)	0.000000

2) Regression of Natural Gas Prices

The following model is run to estimate the price of natural gas (Pt) as a function of Intermediate) and a restatement of WTI and the price of Natural Gas.

$$= \quad _1 \qquad + \quad _2 \, ' \qquad \ddot{y}1 \quad + \quad _3 \ddot{y}1$$

The results are shown in the following table:

Method: Least Squares				
Date: 01/26/12 Time: 14:42				
Sample(adjusted): 2000:02 2011:04				
Included observations: 135 after adjusting endpoints				
Variable Coefficient	Std. Error	t-Statistic		Prob.
WTI	0.000895	9.78E-05	9.148117	0.0000
WTI(-1)	-0.000722	0.000108	-6.705996	0.0000
GN(-1)	0.029716	31.25910	0.973789	Mean dependent 0.0000
R-squared	variable	0.973392	SD dependent variable	0.942875
Adjusted R-squared				0.319577

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S.E. of regression	0.052129	Akaike info criterion	-3.048219
Sum squared resid	0.358701	Schwarz criterion	-2.983657
Log likelihood	208.7548	Durbin-Watson stat	2.168208

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A.18. MODELING SOFTWARE

A number of software packages are available to incorporate uncertainty into analyses. The most widely recognized and used are the commercial packages @Risk40 and Crystal Ball41.
Both are packages designed to operate fully integrated within MS-Excel, with their own toolbar and menus. These packages can be used with minimal knowledge of sampling and calculation techniques, simplifying uncertainty analysis. Another well-known commercial package is Analytica, a quantitative modeling environment with algorithmic functions that integrate and model uncertainty.

If your model is not based on Excel, you can use the SimLab package, which is available. SimLab can also interact with Excel, free programming[42] but this requires some additional steps.
knowledge.

Furthermore, most of these software programs offer the possibility of determining the relative contribution of each parameter's uncertainty to the uncertainty in the model's output, for example, through sensitivity graphs. These can be used for sophisticated trend analysis in
Presence of uncertainty. The main software currently identified in the market is described below.

Crystal Ball

Crystal Ball is a set of applications developed commercially by Oracle, based on the application of predictive, simulation and optimization models, which operate in the MS Excel environment with their own menus and functions.

The software allows for simulation using the Monte Carlo method or the Latin Hypercube method. It can perform time series predictions and also incorporates tools for analyzing and designing probability distributions, sensitivity analyses of results for variables, correlation matrices, decision tables, and optimization.

The software allows for analysis with a wide range of statistical and graphical results, along with the export of cases and details for each iteration, enabling the user to expand their analysis. This complements Crystal Ball's most outstanding feature: its ease of model design and its extensive resources (guides, models, examples). [Sujiyama S. 2007]

Crystal Ball is used in multiple industrial and service sectors, including petrochemical, pharmaceutical, manufacturing, aerospace, logistics, transportation, investment and portfolio management companies, and marketing firms. It has also been used in process improvement methodologies such as Six Sigma.

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[40] http://www.palisade.com
[41] http://www.oracle.com/us/products/applications/crystalball
[42] http://simlab.jrc.ec.europa.eu/

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The study “The Fifth Northwest Electric Power and Conservation Plan” by the Northwest Energy and Conservation Council (USA 2005)43 stands out, as it models uncertainties related to energy and environmental planning. Another interesting application is the RAM tool, which allows for modeling and assessing risks associated with water pollution44.

@Risk

@Risk is a tool developed by Palisade that integrates with MS Excel, adding a series of new functions, each of which allows you to specify a different type of distribution for the values in a cell. Distribution functions can be added to as many cells and formulas as you like in a spreadsheet and can include arguments that reference other cells or expressions, enabling sophisticated uncertainty specifications. To help you assign distributions to uncertain values, @RISK features a graphical window where you can view the distributions and add them to formulas.

The software uses Monte Carlo and Latin Hypercube simulation techniques and can generate distributions of possible outcomes from any cell or range of cells in the spreadsheet model.

The software allows you to view a wide range of statistical and graphical results, along with export capabilities and operational flexibility, making it a simple, yet complete and user-friendly tool for modeling, simulating, and presenting the effects of uncertainty.

@Risk can interact with other Palisade tools to complement the analysis, always in an Excel environment, with sensitivity analysis (TopRank), decision trees (PrecisionTree), distribution calculation (BestFit) and definition of optimization variables (RISKOptimizer).

The software has been used for analysis of mitigation options under uncertainty in the energy sector45 or by the International Energy Agency (IEA) as in the study “Price caps and price floors in climate policy” of December 2008 where the medium-term economic and environmental effects of using maximum and minimum price policies on CO2 emission rights are studied.

Existing applications of @Risk include finance and financial instruments, analysis of
decisions and risk analysis in health, energy, environment, transportation, aerospace, government and defense, among others.

43 http://www.nwcouncil.org/energy/powerplan/5/

44 http://www.esinternational.com/RAM-Water-Risk-Assessment-Software

45 JANSEN J.C. 2006, Social cost-benefit analysis of climate change mitigation option in a European context.

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Risk Solver46

The most recent application for simulation under uncertainty is Risk Solver, a commercial software from Frontline Systems that operates as an add-in for MS Excel. It is designed to solve univariate, multivariate, and nonlinear linear optimization problems. An expansion of the product line incorporated uncertainty analysis, enabling Monte Carlo simulation using complex methods such as Latin Hypercube or Sobol47 numbers, including sensitivity analysis, distribution fitting, and correlation between variables.

This application has become a major player in the competition for uncertainty simulation software in MS Excel, primarily due to its speed and advanced optimization features.

Analytica48

Analytica is a commercial software developed by the company Lumina and is presented as a quantitative decision support environment, which helps to visualize modeling problems clearly and far beyond what is possible in a spreadsheet.

Analytica's "smart arrays" allow for the simple creation and management of multidimensional tables, offering significant advantages over spreadsheets. Analytica's construction employs array abstraction, enabling the modeler to operate from a purely mathematical perspective without writing formulas that reference each element of a table, and without the need for iterative processing routines, unless specific references to individual cells are required. The result is a significant reduction in coding time.

compared to “developer” style models49, along with copy errors and debugging time.

From its inception, Analytica was designed to analyze risk and uncertainty, unlike spreadsheets that require special add-ons. Monte Carlo simulation and Latin HyperCube are fully integrated features in Analytica, making it easy to incorporate uncertainty management much faster than standard spreadsheets. Analytica's influence diagrams allow you to specify uncertain variables simply by selecting an oval node to represent a random variable. You can then choose from a variety of options.

set of probability distributions, simple or multivariate, presenting a wide range of results visualization options such as density functions, cumulative distributions, probability bands, basic statistics, or even the random values obtained in the sample.

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[46] http://www.solver.com/platform/risk-solver-pro.htm
[47] Method for generating pseudo-random numbers with low mathematical discrepancy. http://
[48] www.lumina.com
[49] Refer to models that require development through programming/design codes such as Matlab, and in the particular case of this comparative analysis, to SimLab.

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This allows modelers to quickly assess risk and uncertainty by studying the sensitivity and relevance of the variables.

Existing applications of Analytica include strategic planning, R&D management, and analytics.
decision-making and risk analysis in finance, health, energy, environment, aerospace and telecommunications.

Its use at the national level stands out in the paper on risk analysis of toxic substances in mining “The Use of Probabilistic Analysis to Improve Decision-Making in Environmental Regulation in a Developing Context: The Case of Arsenic Regulation in Chile” [O'Ryan and Díaz 2008], and the study “Preliminary Estimation Of The Potential Ancillary Benefits For Chile” [Cifuentes et al. 2000]

A number of Analytica application cases can be reviewed at http://www.lumina.com/case-studies/ . Among them, the use in the creation of ICAM (Integrated Climate Assessment Model) stands out. ICAM is an integrated climate analysis model that helps to understand and explore the interactions between human activities, the atmosphere, and natural ecosystems, and that considers the uncertainties linked to climate variables.

SimulAr50

SimulAr is free software developed in Argentina that requires users to commit to sharing their analyses in order to be authorized to use it. SimulAr allows users to perform
Monte Carlo simulation software is primarily designed for business analysis and evaluation, as well as decision-making involving risk. It operates as an MS Excel add-in, incorporating its own menu and a series of functions that enable Monte Carlo simulation and sensitivity analysis.

The software allows for Monte Carlo simulation, sensitivity analysis, correlation assignment between variables, and the definition of optimization variables. It does not include the Latin Hypercube algorithm.

SimLab

SimLab is an environment developed as APIs without a graphical interface that requires a hosting environment to run. This environment can be Matlab, Fortran, or C/C++, so knowledge of these applications is required to use SimLab's features. The software is capable of performing uncertainty and sensitivity analyses using various mathematical techniques such as Monte Carlo simulation, Latin Hypercube, sensitivity analysis, and the Sobol method, among others.

This software was used for uncertainty analysis in the research “Risk assessment of nitrate contamination of rural drinking water supply wells in Chile”51

[50] http://www.simularsoft.com.ar
51http://www.scielosp.org/scielo.php?pid=S1020-49892006001100004&script=sci_arttext

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Application Management

Software packages that operate within MS Excel, such as Crystal Ball, @Risk, or Risk Solver, are very easy to learn. They are generally intuitive, due to the user's familiarity with MS Excel, and allow for the quick and easy generation of simulation models.

On the other hand, Analytica, as a pure modeling software, allows the development of more complex, flexible, and dynamic models, with a response speed superior to those applications that operate as an MS Excel add-in. However, the effort required to build simulation models is much greater than that required for models running in Excel.

SimLab, on the other hand, requires advanced programming skills and knowledge of other technical tools, which complicates the design process. Furthermore, a clear deficiency compared to the other tools mentioned is its inability to deliver results that are easily transferable to decision-making. This positions it as a transitional tool for the use of other applications.

Given that the problem this study aims to solve is not particularly complex in terms of its scope and processing time, a tool is required that can easily perform the simulation while incorporating sensitivity analysis tools and providing a flexible, user-friendly, and intuitive set of graphical results. Under these premises, using an application within the MS Excel environment is suitable.

Selection

Of the four software programs that operate on MS Excel (Crystal Ball, @Risk, Risk Solver, and SimulAr), SimulAr was excluded from the final analysis due to a lack of extensive application history and acceptance as a reliable simulation tool. While the three remaining software programs that also operate on MS Excel offer similar features for similar purposes, they possess distinct advantages that should be noted.

Crystal Ball stands out for its ease of use in building models, along with a wide variety of guides, reference manuals, numerous illustrative models, and extensive text. It is also notable for its comprehensive capabilities in developing Monte Carlo simulation and optimization sub-models.

@RISK has the best procedure for identifying the key variables that affect the scope and the Shape of the results. This is your regression sensitivity option. @RISK is a close second Crystal Ball takes the lead in terms of resources. Many texts and examples are available, as well as demos. and the tutorials are simple and well-designed.

The main advantages of Risk Solver are its exceptional response speed and the fact that it offers the most advanced optimization functionalities. An important additional feature is that Risk Solver allows you to define the results of a simulation as an input distribution (SIP, Stochastic Information Package) and as a sub-uncertainty model for use in another simulation. This

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It is a powerful means of transferring results within an organization while maintaining consistency in the key components of the analysis.

Considering the original problem, of incorporating uncertainty into the cost-effective evaluation of the cost of mitigation options in the energy sector, the requirements, given the reduced dimension of the problem, are basically an intuitive and simple development of the incorporation of uncertainty into the model, and a broad and flexible set of results, which allows for detailed analysis of the information from the simulation process.